Rewiring Immune Regulation Through Acute Exercise

A Systematic Review and Meta-Analysis on TH1 and TH2 Cell Responses - Documenation

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2026-06-01

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Background

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This site documents the complete analysis pipeline behind our systematic review and meta-analysis of T-helper-cell responses to acute exercise. For each outcome — **Th1**, **Th2**, and the **Th1/Th2 ratio** — the chapters reproduce the random-effects meta-analysis across three post-exercise time windows (immediate, ~2 h, and ~17 h), together with outlier and influence diagnostics, small-study/publication-bias checks, and moderator and meta-regression analyses. Study quality is appraised with ROBINS-I, and the underlying effect-size computations are documented throughout. The raw data and the full analysis script are available below for reproducibility.

Downloads

Resource	File
Raw data — effect-size computation	`effect_computation_Th1Th2.xlsx`
Analysis script	`meta_Th1Th2.R`
Summary Formulae	`formulae.pdf`

1. Summary of Formulae

1.1. Introduction

Only formulas that were actually used in the calculations are highlighted by placing them in a box to improve readability and to make the key equations easy to identify. Other formulas are included for completeness and conceptual clarity.

Notation

Per study and time point

- n: sample size
- $\tilde{x}, \tilde{y}$: estimated means (approximated from published summary statistics)
- $\tilde{s}_x, \tilde{s}_y$: estimated standard deviations (approximated from published summary statistics)
- m: median
- q_1, q_3: 1st and 3rd quartile
- a: minimum, b: maximum
- Φ^{-1}: inverse standard normal cumulative distribution function (e.g., R: `qnorm`)
- ρ: Correlation between Th1 and Th2 measurements

1.2. Data Extraction and Computation Strategy

1.2.1. Overview

This analysis requires extracting or computing the mean Th1/Th2 ratio and its associated standard deviation for each study and time point. However, the included studies reported their descriptive statistics in heterogeneous formats. To enable uniform computation of effect sizes, we developed a systematic approach:

1. **Classify how each study reported data** (see Section 1.2.2)
2. **Estimate means and SDs** when not directly reported (see Section 1.3)
3. **Compute the Th1/Th2 ratio** and its variance (see Section 1.4)
4. **Calculate standardized effect sizes** with small-sample correction (see Section 1.5)

The **computation table** (see Supplementary Materials) documents which approach (A1, A2, C1, C2, or C3) was applied to each study, providing full methodological transparency.

1.2.2. Data Reporting Types

Studies fell into two categories based on how Th1 and Th2 values were reported:

A1: Summary Statistics Directly Reported - Article reported means and standard deviations (or sufficient summary statistics to estimate them) - No access to individual participant data

A2: Raw Data Available - Individual participant-level data were available (either directly from the article or upon request from authors) - We computed means, SDs, and participant-level correlations directly from the raw data

1.2.3. Summary Statistics Estimation Scenarios

When articles reported summary statistics other than means and SDs, we estimated means and SDs using sample-size-adjusted formulas (Wan et al. 2014) depending on which statistics were available:

Table 1.1.: Summary Statistics Estimation Scenarios and Corresponding Formulas

Scenario	Available Data	$\bar{x}_{estimated}$	$s_{estimated}$
C1	Min, Max, Med., n	(1)	(2)
C2	Q1, Med., Q3, n	(3)	(4)
C3	Mean, SE, n	Direct	(5)

1.3. Estimating mean and SD from median/IQR/range

Table 1.1 above summarizes the estimation scenarios.

The detailed formulas for each scenario are presented below, following the methodology of Wan et al. (2014).

1.3.1. Scenario C1: Min + Max + Median + n (a, m, b, n)

$$\boxed{\bar{x} \approx \frac{a + 2m + b}{4}} \tag{1}$$

$$\boxed{s \approx \frac{b - a}{2\Phi^{-1}\left(\frac{n-0.375}{n+0.25}\right)}} \tag{2}$$

1.3.2. Scenario C2: Median + IQR + n (q_1, m, q_3, n)

$$\boxed{\bar{x} \approx \frac{q_1 + m + q_3}{3}} \tag{3}$$

$$\boxed{s \approx \frac{q_3 - q_1}{2\Phi^{-1}\left(\frac{0.75n-0.125}{n+0.25}\right)}} \tag{4}$$

1.3.3. Scenario C3: Mean, SE

$$\boxed{s = SE\sqrt{n}} \tag{5}$$

1.4. Mean and Variance of a Ratio

Because several studies did not report raw data, the Th1/Th2 ratio could not be calculated directly. For these studies, we first approximated the mean and standard deviation from the published summary statistics (see Section 1.3). Using the estimated mean and SD, we then computed the Th1/Th2 ratio for studies that did not publish the ratio but provided sufficient information to derive it.

1.4.1. Calculation of the mean ratio

Let $\tilde{x}_R$ denote the mean Th1/Th2 ratio, let $\tilde{x}_{\mathrm{Th1}}$ and $\tilde{y}_{\mathrm{Th2}}$ denote the means of Th1 and Th2, respectively, obtained either by estimation from published summary statistics or by calculation using raw data.

$$\boxed{\tilde{x}_R = \frac{\tilde{x}_{\mathrm{Th1}}}{\tilde{y}_{\mathrm{Th2}}} \quad (\tilde{y}_{\mathrm{Th2}} \neq 0)} \tag{6}$$

1.4.2. Calculation of the variance (and standard deviation) of a ratio

It is important to understand that Th1 and Th2 are two dependent variables. Taking this into account, we have used the following formula, published by Van Kempen and Van Vliet (2000):

Their Formula 7 approximates the variance of a ratio using a first-order Taylor expansion (Van Kempen and Van Vliet 2000, 301).

$$var(\frac{\tilde{x}}{\tilde{y}}) \approx \frac{Var(\tilde{x})}{\tilde{y}^2} + \frac{\tilde{x}^2 Var(\tilde{y})}{\tilde{y}^4} - 2\,\frac{\tilde{x}}{\tilde{y}^3} cov(\tilde{x}, \tilde{y}) \tag{7}$$

Since $var() = s^2$ and $cov(\tilde{x}, \tilde{y}) = \rho\, s_{\tilde{x}}\, s_{\tilde{y}}$ (by rearranging) this can be written as:

$$var(\frac{\tilde{x}}{\tilde{y}}) \approx \frac{s_{\tilde{x}}^2}{\tilde{y}^2} + \frac{\tilde{x}^2 s_{\tilde{y}}^2}{\tilde{y}^4} - 2\,\frac{\tilde{x}\rho\, s_{\tilde{x}}\, s_{\tilde{y}}}{\tilde{y}^3} \tag{8}$$

which can be rewritten as:

$$var(\frac{\tilde{x}}{\tilde{y}}) \approx \left(\frac{s_{\tilde{x}}}{\tilde{y}}\right)^2 + \left(\frac{\tilde{x} s_{\tilde{y}}}{\tilde{y}^2}\right)^2 - 2\,\rho \frac{\tilde{x}\, s_{\tilde{x}}\, s_{\tilde{y}}}{\tilde{y}^3} \tag{9}$$

The corresponding standard deviation is obtained by

$$\boxed{\begin{array}{l} SD\left(\dfrac{\tilde{x}}{\tilde{y}}\right) \approx \sqrt{\left(\dfrac{s_{\tilde{x}}}{\tilde{y}}\right)^2 + \left(\dfrac{\tilde{x}\, s_{\tilde{y}}}{\tilde{y}^2}\right)^2 - 2\rho \dfrac{\tilde{x}\, s_{\tilde{x}}\, s_{\tilde{y}}}{\tilde{y}^3}} \\ \qquad \rho = 0.5 \end{array}} \tag{1.1}$$

1.5. Effect Size Calculation

1.5.1. Unstandardized mean difference

- $\tilde{x}_{\text{pre}}$, s_{pre}
- $\tilde{x}_{\text{post}}$, s_{post}
- r: Pre–Post correlation (outcome-specific)
- n: is the number of pairs

Note: A fixed pre–post correlation of $r = 0.5$ was applied uniformly across all effect size calculations.

Mean Change

$$\boxed{D = \tilde{x}_{\text{post}} - \tilde{x}_{\text{pre}}} \tag{11}$$

(Borenstein 2013, 22)

Standard deviation of D

$$\boxed{SD_{\text{diff}} = \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2 - 2r\, s_{\text{pre}}\, s_{\text{post}}}} \tag{12}$$

(Borenstein 2013, 24)

Variance of D

$$\boxed{V_D = \frac{SD_{\text{diff}}^2}{n}} \tag{13}$$

(Borenstein 2013, 24)

Standard error of D

$$\boxed{SE_D = \sqrt{V_D}} \tag{14}$$

(Borenstein 2013, 24)

Within-subject SD

$$\boxed{S_{\text{within}} = \frac{SD_{\text{diff}}}{\sqrt{2(1-r)}}} \tag{15}$$

(Borenstein 2013, 29)

1.5.2. Standardized mean difference

$$\boxed{d = \frac{D}{S_{\text{within}}}} \tag{16}$$

(Borenstein 2013, 29)

Variance and standard error of d

$$\boxed{V_d = \left(\frac{1}{n} + \frac{d^2}{2n}\right) 2(1-r)} \tag{17}$$

(Borenstein 2013, 29)

$$\boxed{SE_d = \sqrt{V_d}} \tag{18}$$

(Borenstein 2013, 29)

1.5.3. Small-sample correction (Hedges' g)

Degrees of freedom for pre-post measures (df):

$$\boxed{df = n - 1} \tag{19}$$

(Borenstein 2013, 29)

Correction factor (J)

$$\boxed{J = 1 - \frac{3}{4df - 1}} \tag{20}$$

(Borenstein 2013, 27)

Hedges' g:

$$\boxed{g = J\,d} \tag{21}$$

(Borenstein 2013, 27)

Variance and SE of g

$$\boxed{V_g = J^2\,V_d} \tag{22}$$

(Borenstein 2013, 27)

$$\boxed{SE_g = \sqrt{V_g}} \tag{23}$$

(Borenstein 2013, 27)

1.6. References

2. Workspace Setup

All chapters load a shared setup file (`_common.R`) that installs/loads the required packages, sets a consistent global `ggplot2` theme, defines the leave-one-out outlier-detection functions, and exposes small helpers for the standard funnel, forest, and bubble plots used throughout. This keeps every analysis chapter self-contained and reproducible regardless of render order.

2.1. Packages

2.2. Outlier Detection Functions

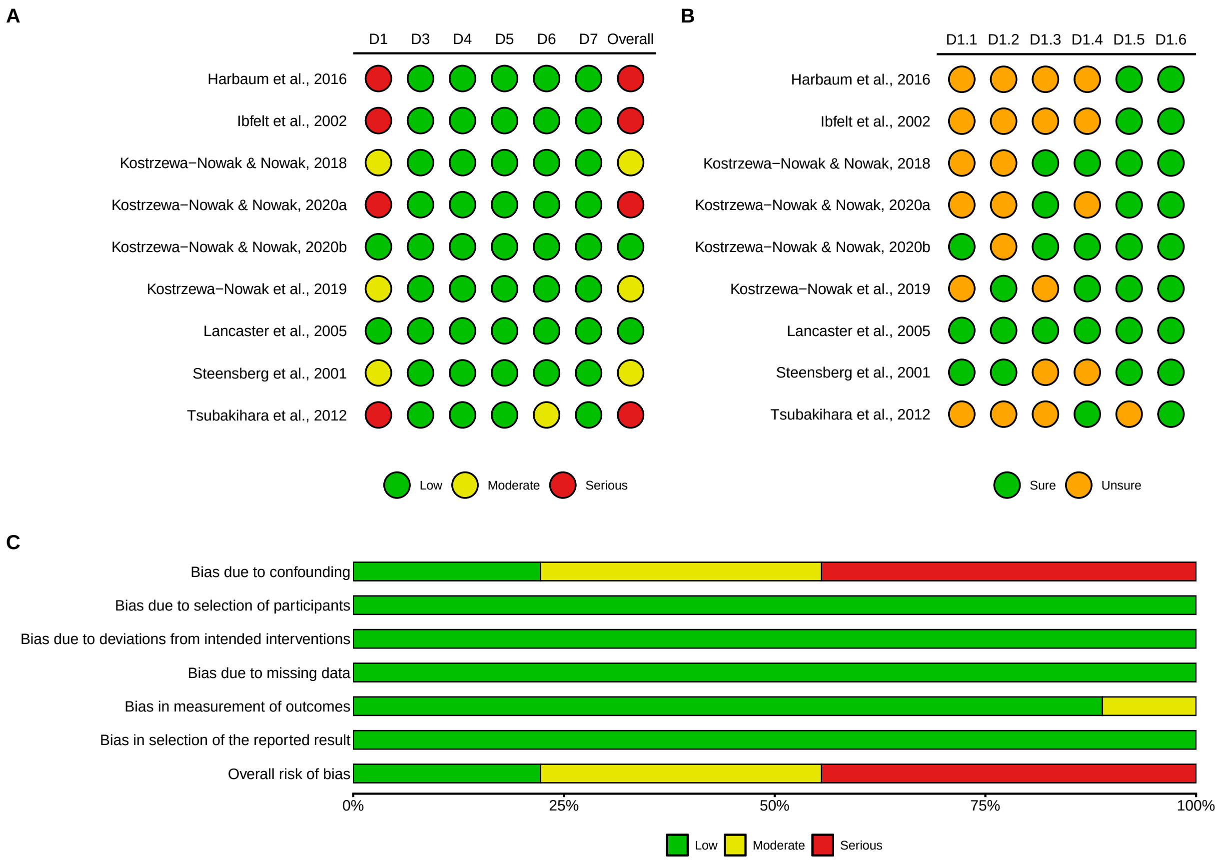
Custom functions — required for all outlier analyses below.

The following four functions implement the leave-one-out standardised residual approach described in Hedges and Olkin (1985) and Viechtbauer and Cheung (2010). A study is classified as an outlier if its standardised residual exceeds |3|; values above |1.96| warrant closer inspection. The model (fixed- vs. random-effects) is selected automatically from the robust heterogeneity index $I\mathrm{r}^2$ ($< 30\% \rightarrow$ fixed-effects, otherwise random-effects). The outlier routine is always passed the **variance** (`seTE^2`) as its `s2` argument.

The full definitions live in `_common.R` and are sourced automatically at the top of every chapter.

3. Risk of Bias Assessment (ROBINS-I)

Downloads: ROBINS-I Data (xlsx) · ROBINS-I R-Script



Confounding-factor items (Panel B).

- **D1.1** — No exercise 24–48 h pre-intervention?
- **D1.2** — Standardized food intake among participants (e.g. all fasted or standardized breakfast)?
- **D1.3** — No caffeine, acute medication, or alcohol 24 h before exercise?
- **D1.4** — Identical warm-up between participants?
- **D1.5** — Good way to assess the controlled intensity (HR control, VO , VT)?
- **D1.6** — Homogeneity of the group (sex, age, BMI, training status)?

Part I.

Th1

4. Th1 — Immediate Post-Exercise

4.1. Data Import

The base analysis uses the pre-computed Hedges' g file for Th1 at $r = 0.5$ (`effect_size/Th1/pre_post/Th1_pre_post_r_050`
Sensitivity analyses can substitute the $r = 0.3$ / 0.4 / 0.6 / 0.7 files in the same folder. The pipeline column names are mapped to the analysis conventions (`VE_g`, `Sex`, `Age`, `BMI`) by `load_es()`.

4.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes (g, SE_g) using the **Hedges–Olkin (HE)** ^{2} estimator and the **Knapp–Hartung** small-sample correction (`hakn = TRUE`). After inspecting the full set, **Kostrzewa-Nowak et al. (2019)** was detected as very different across meta-analysis and while not influential on the overall effect, its values were unrealistic and brought substantial misleading variance.

4.2.1. Full Modell {meta}

```
Number of studies: k = 18

                             SMD            95% CI    t
Random effects model (HK) 0.8982 [ 0.3316; 1.4649] 3.34
Prediction interval              [-1.5159; 3.3124]
                          p-value
Random effects model (HK)  0.0038
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.2328 [0.5533; 2.8692]; tau = 1.1103 [0.7438; 1.6939]
 I^2 = 86.8% [80.7%; 91.0%]; H = 2.76 [2.27; 3.34]

Test of heterogeneity:
      Q d.f.  p-value
 129.23   17 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 17)
- Prediction interval based on t-distribution (df = 17)
```

4.2.2. Adapted Model {meta}

> Caution
>
> Kostrzewa-Nowak et al. (2019) removed (unrealistic values / misleading variance); re-fit

```
Number of studies: k = 17

                              SMD            95% CI    t
Random effects model (HK) 0.9980 [ 0.4524; 1.5436] 3.88
Prediction interval              [-1.2488; 3.2448]
                          p-value
Random effects model (HK)  0.0013
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.0535 [0.4647; 2.5537]; tau = 1.0264 [0.6817; 1.5980]
 I^2 = 86.4% [79.8%; 90.9%]; H = 2.72 [2.22; 3.32]

Test of heterogeneity:
      Q d.f.  p-value
 117.96   16 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)
```

4.2.3. Adapted Model {metafor}

```
Random-Effects Model (k = 17; tau^2 estimator: HE)

tau^2 (estimated amount of total heterogeneity): 1.0535 (SE = 0.4224)
tau (square root of estimated tau^2 value):      1.0264
I^2 (total heterogeneity / total variability):   90.76%
H^2 (total variability / sampling variability):  10.82

Test for Heterogeneity:
Q(df = 16) = 117.9586, p-val < .0001

Model Results:

estimate      se    tval  df    pval   ci.lb   ci.ub
  0.9980  0.2574  3.8774  16  0.0013  0.4524  1.5436  **
```

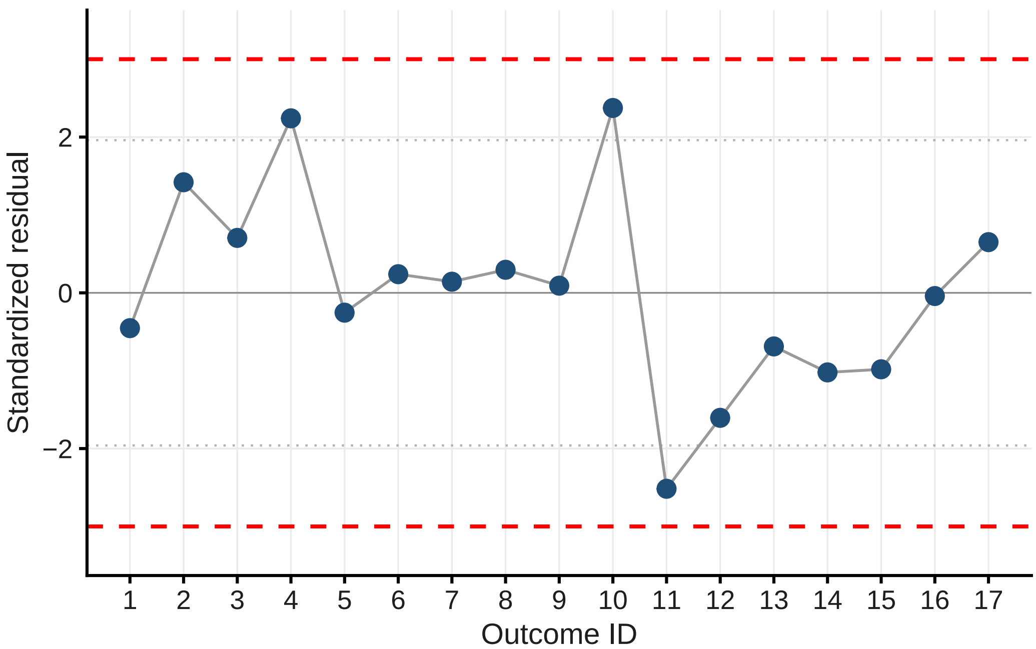
```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m1$seTE^2`). Studies with $|z| > 3$ are flagged as outliers; $|z| > 1.96$ warrants inspection.

Table 4.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	0.577	-0.454
Kostrzewa-Nowak & Nowak (2018)a	2.289	1.420
Kostrzewa-Nowak & Nowak (2018)b	1.613	0.705
Kostrzewa-Nowak & Nowak (2020a)a	2.781	2.241
Kostrzewa-Nowak & Nowak (2020a)b	0.761	-0.255
Kostrzewa-Nowak & Nowak (2020b)a	1.195	0.239
Kostrzewa-Nowak & Nowak (2020b)b	1.111	0.143
Kostrzewa-Nowak & Nowak (2020b)c	1.245	0.296
Kostrzewa-Nowak & Nowak (2020b)d	1.066	0.092
Kostrzewa-Nowak & Nowak (2020b)e	3.282	2.374
Steensberg et al. (2001)	-1.092	-2.516
Tsubakihara et al. (2012)	-0.276	-1.605
Lancaster et al. (2005)a	0.373	-0.688
Lancaster et al. (2005)b	0.091	-1.022
Lancaster et al. (2005)c	0.124	-0.982
Harbaum et al. (2016)a	0.949	-0.041
Harbaum et al. (2016)b	1.584	0.651



> **!** Important
>
> No outcome has standardized residuals above 3 or below -3.

4.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones (= 0.10, 0.05, 0.01). The dfbetas diagnostic from `metafor::rma.mv()` identifies **influential cases** — distinct from statistical outliers and not necessarily excluded.

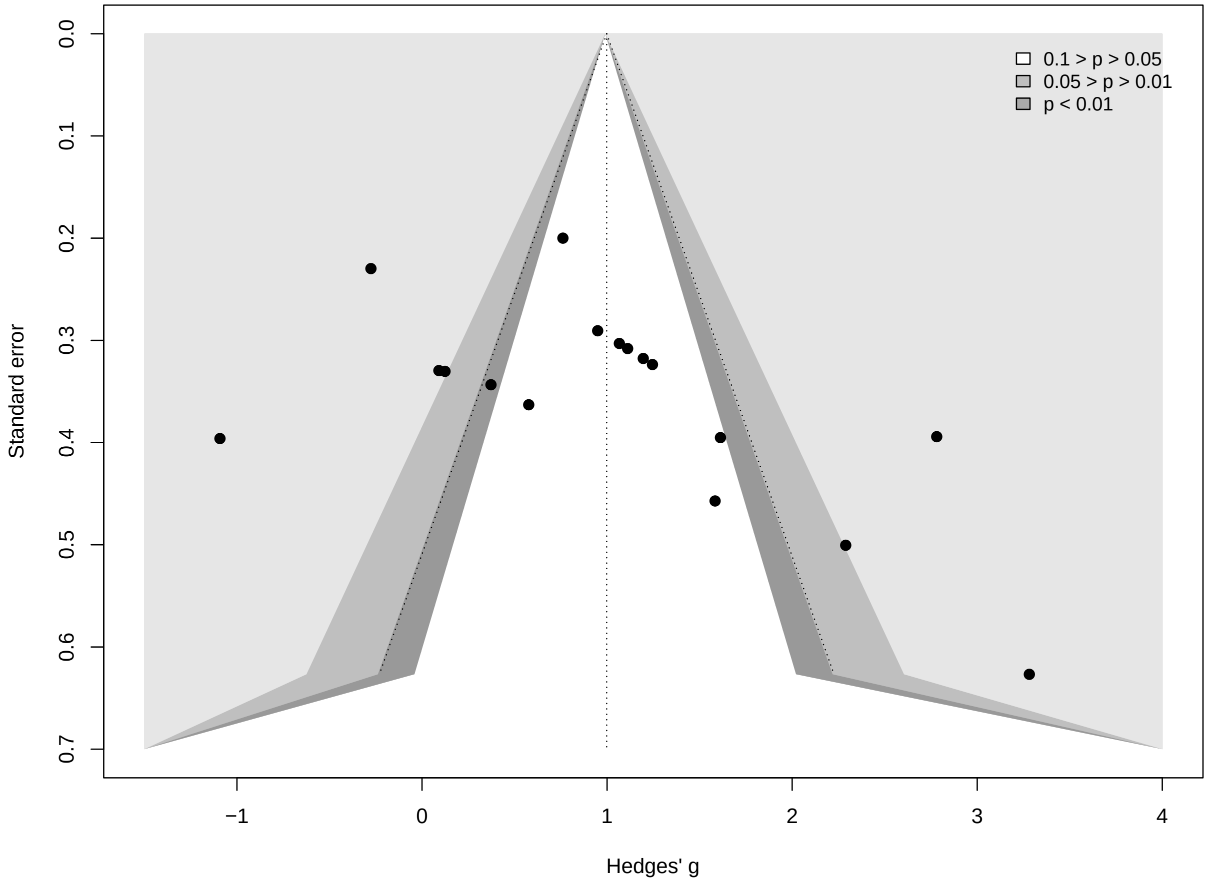


Table 4.2.: DFBETAS influence diagnostics

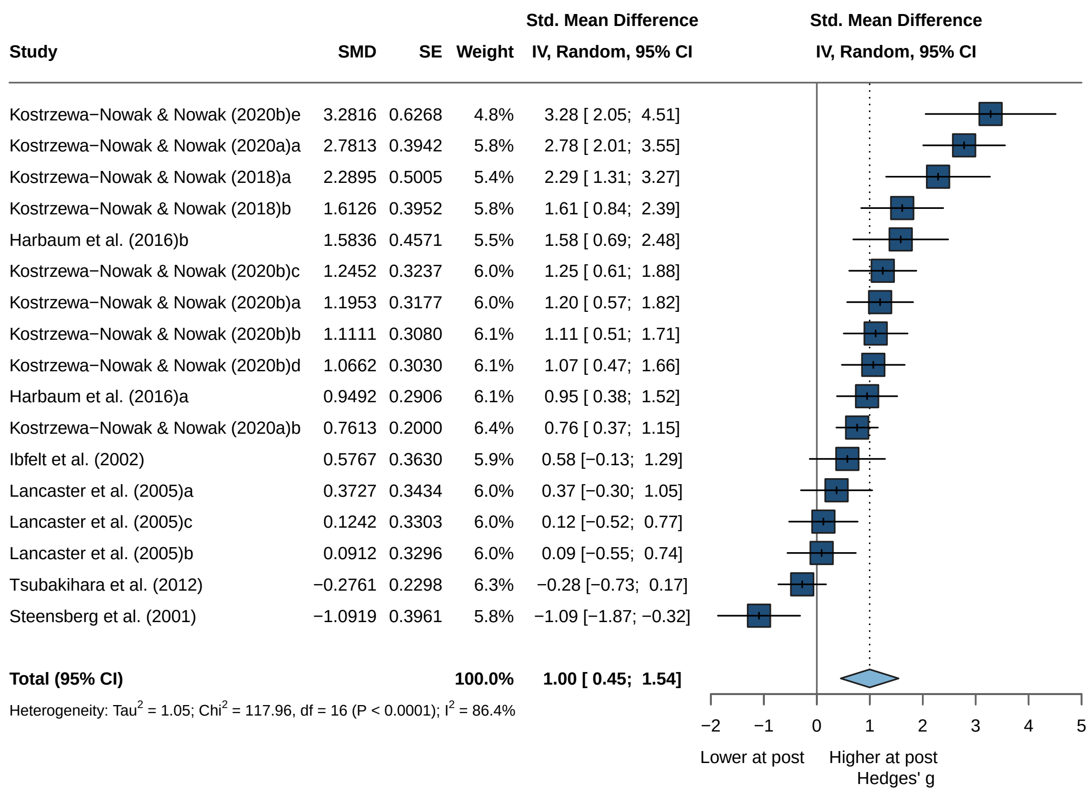
Short_reference	g	dfbetas
Ibfelt et al. (2002)	0.577	-0.13459692
Kostrzewa-Nowak & Nowak (2018)a	2.289	0.48230850
Kostrzewa-Nowak & Nowak (2018)b	1.613	0.43065936
Kostrzewa-Nowak & Nowak (2020a)a	2.781	1.04875284
Kostrzewa-Nowak & Nowak (2020a)b	0.761	-0.07030521
Kostrzewa-Nowak & Nowak (2020b)a	1.195	0.33513405
Kostrzewa-Nowak & Nowak (2020b)b	1.111	0.28334594
Kostrzewa-Nowak & Nowak (2020b)c	1.245	0.36199846
Kostrzewa-Nowak & Nowak (2020b)d	1.066	0.25226569

Table 4.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Kostrzewa-Nowak & Nowak (2020b)e	3.282	0.50656391
Steensberg et al. (2001)	-1.092	-0.98309619
Tsubakihara et al. (2012)	-0.276	-1.80121819
Lancaster et al. (2005)a	0.373	-0.29482591
Lancaster et al. (2005)b	0.091	-0.53785734
Lancaster et al. (2005)c	0.124	-0.50993030
Harbaum et al. (2016)a	0.949	0.15837850
Harbaum et al. (2016)b	1.584	0.30720822

One value was below −1 and one above 1, so both were identified as influential cases. However, neither is an outlier (Kostrzewa-Nowak & Nowak (2020a)a and Tsubakihara et al. (2012)). Accordingly, all outcomes are retained, and the sources of heterogeneity are examined in the moderator analysis.

4.5. Forest Plot



4.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed, $p < .10$). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially missing studies.

4.6.1. Egger´s regression test

```
Linear regression test of funnel plot asymmetry

Test result: t = 2.11, df = 15, p-value = 0.0523
Bias estimate: 4.9260 (SE = 2.3369)

Details:
- multiplicative residual heterogeneity variance (tau^2 = 6.0667)
- predictor: standard error
- weight:    inverse variance
- reference: Egger et al. (1997), BMJ
```

4.6.2. Trim and Fill

```
Estimated number of missing studies on the left side: 0 (SE = 2.5907)

Random-Effects Model (k = 17; tau^2 estimator: HE)

tau^2 (estimated amount of total heterogeneity): 1.0535 (SE = 0.4224)
tau (square root of estimated tau^2 value):      1.0264
I^2 (total heterogeneity / total variability):   90.76%
H^2 (total variability / sampling variability):  10.82

Test for Heterogeneity:
Q(df = 16) = 117.9586, p-val < .0001

Model Results:

estimate      se    zval    pval   ci.lb   ci.ub
  0.9980  0.2643  3.7765  0.0002  0.4800  1.5159  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

No missing studies

4.7. Moderator Analyses

4.7.1. Risk of Bias

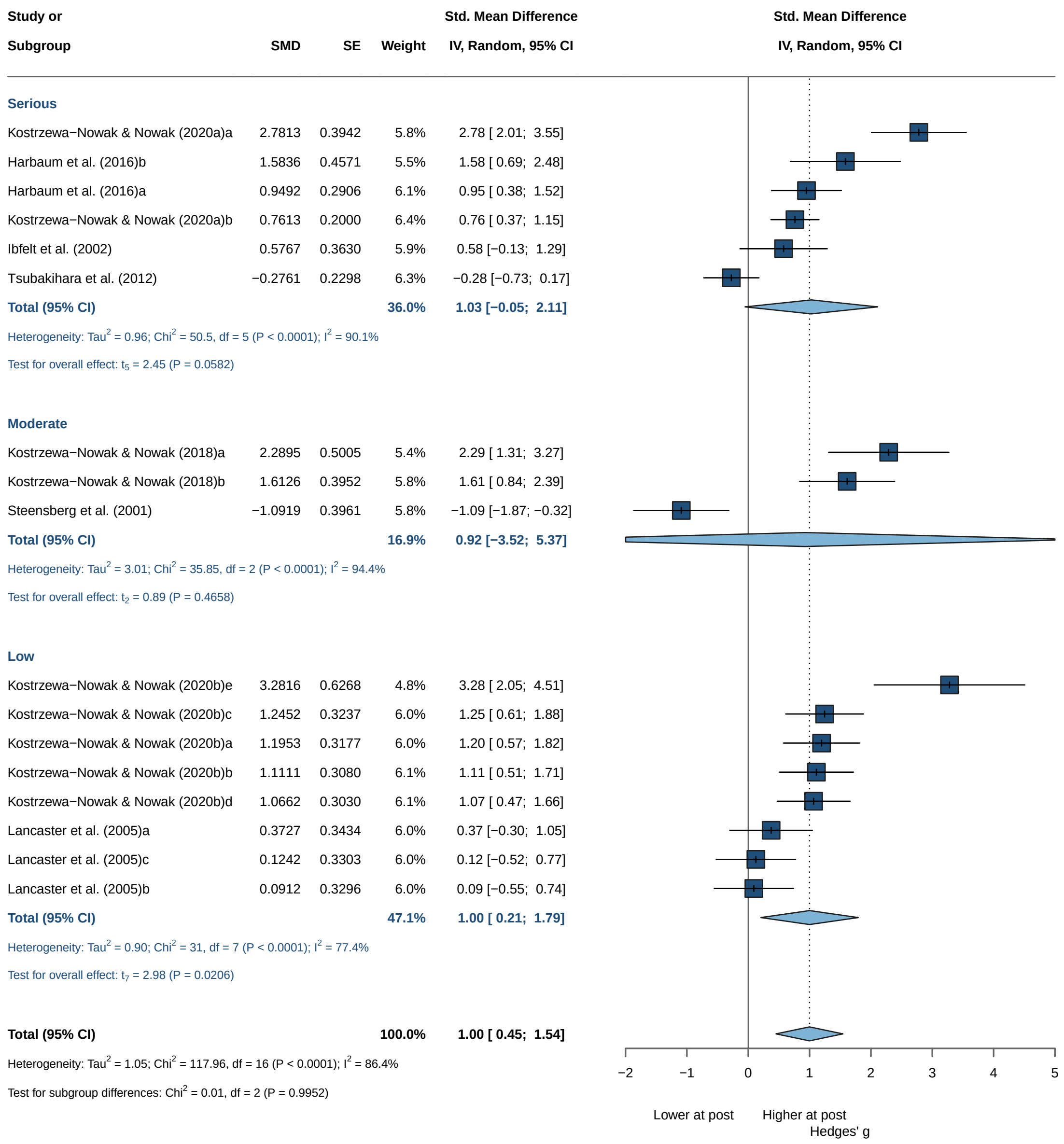
Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic (Q_b) indicates whether subgroup means differ.

4.7.1.1. Model

```
Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.0535 [0.4647; 2.5537]; tau = 1.0264 [0.6817; 1.5980]
 I^2 = 86.4% [79.8%; 90.9%]; H = 2.72 [2.22; 3.32]

Test of heterogeneity:
      Q d.f.  p-value
 117.96   16 < 0.0001

Results for subgroups (random effects model (HK)):
                 k    SMD            95% CI  tau^2    tau
RoB = Serious    6 1.0287 [-0.0523; 2.1097] 0.9592 0.9794
RoB = Moderate   3 0.9237 [-3.5234; 5.3709] 3.0131 1.7358
RoB = Low        8 0.9996 [ 0.2061; 1.7932] 0.8995 0.9484
                   Q   I^2
RoB = Serious  50.50 90.1%
RoB = Moderate 35.85 94.4%
RoB = Low      31.00 77.4%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 0.01    2  0.9952

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **Note**
>
> Not significant results

4.7.1.2. Forrestplot



4.7.2. Staining

Number of studies: k = 17

```
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.0535 [0.4647; 2.5537]; tau = 1.0264 [0.6817; 1.5980]
 I^2 = 86.4% [79.8%; 90.9%]; H = 2.72 [2.22; 3.32]

Test of heterogeneity:
      Q d.f.  p-value
 117.96   16 < 0.0001

Results for subgroups (random effects model (HK)):
                  k    SMD             95% CI  tau^2    tau
staining = intra  14 1.0632 [ 0.4218; 1.7046] 1.1630 1.0784
staining = extra   3 0.7028 [-1.6445; 3.0501] 0.7783 0.8822
                    Q   I^2
staining = intra 91.40 85.8%
staining = extra 18.92 89.4%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 0.34    1  0.5617

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **Note**
>
> Not significant results

4.7.3. Sex

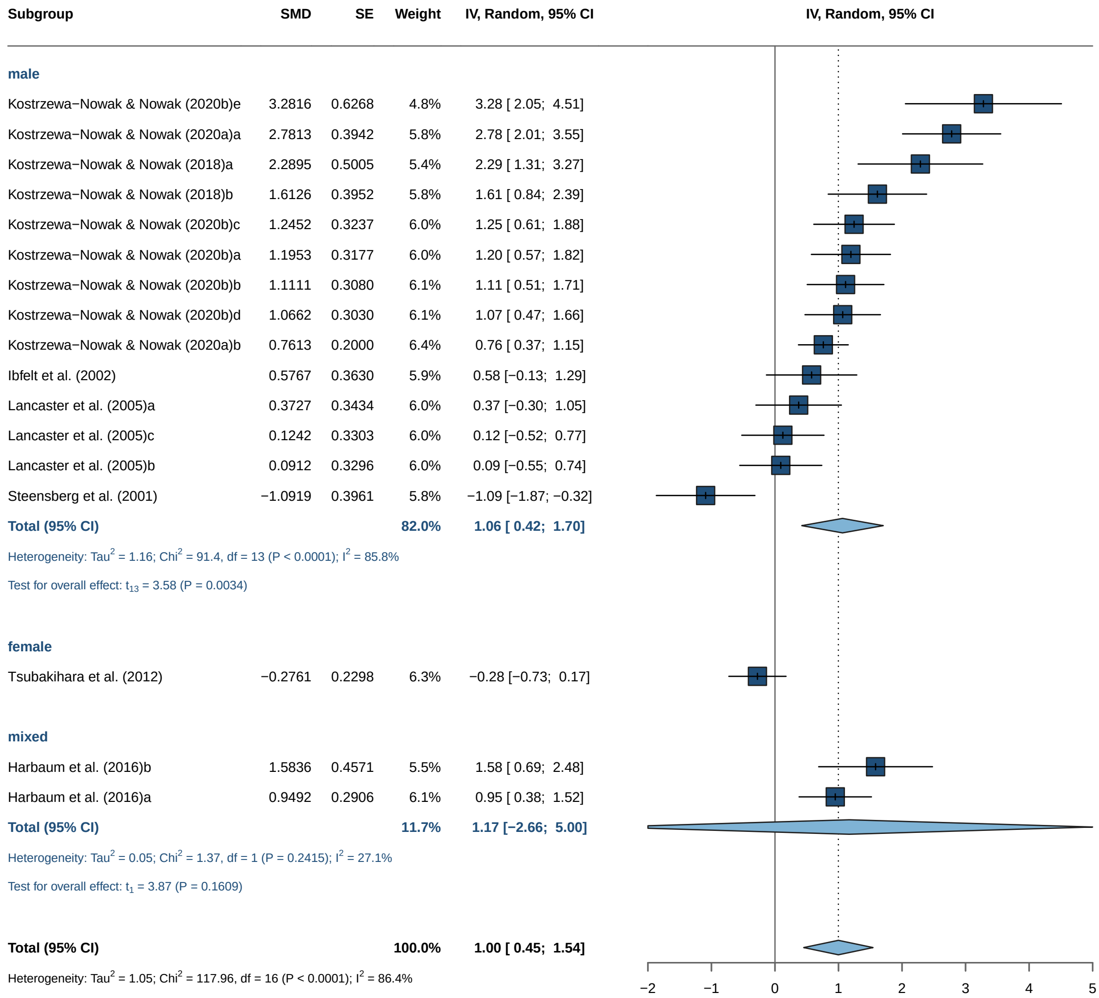
4.7.3.1. Model

```
Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.0535 [0.4647; 2.5537]; tau = 1.0264 [0.6817; 1.5980]
 I^2 = 86.4% [79.8%; 90.9%]; H = 2.72 [2.22; 3.32]

Test of heterogeneity:
      Q d.f.  p-value
 117.96   16 < 0.0001

Results for subgroups (random effects model (HK)):
              k     SMD             95% CI  tau^2    tau
sex = male    14  1.0632 [ 0.4218; 1.7046] 1.1630 1.0784
```

```
sex = female   1 -0.2761 [-0.7265; 0.1743]      --      --
sex = mixed    2  1.1682 [-2.6647; 5.0012] 0.0545 0.2335
                 Q   I^2
sex = male   91.40 85.8%
sex = female  0.00    --
sex = mixed   1.37 27.1%

Test for subgroup differences (random effects model (HK)):
                   Q d.f.  p-value
Between groups 19.89    2 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **! Important**
>
> Significant results but 14 vs 1 vs 2

4.7.3.2. Forrest Plot



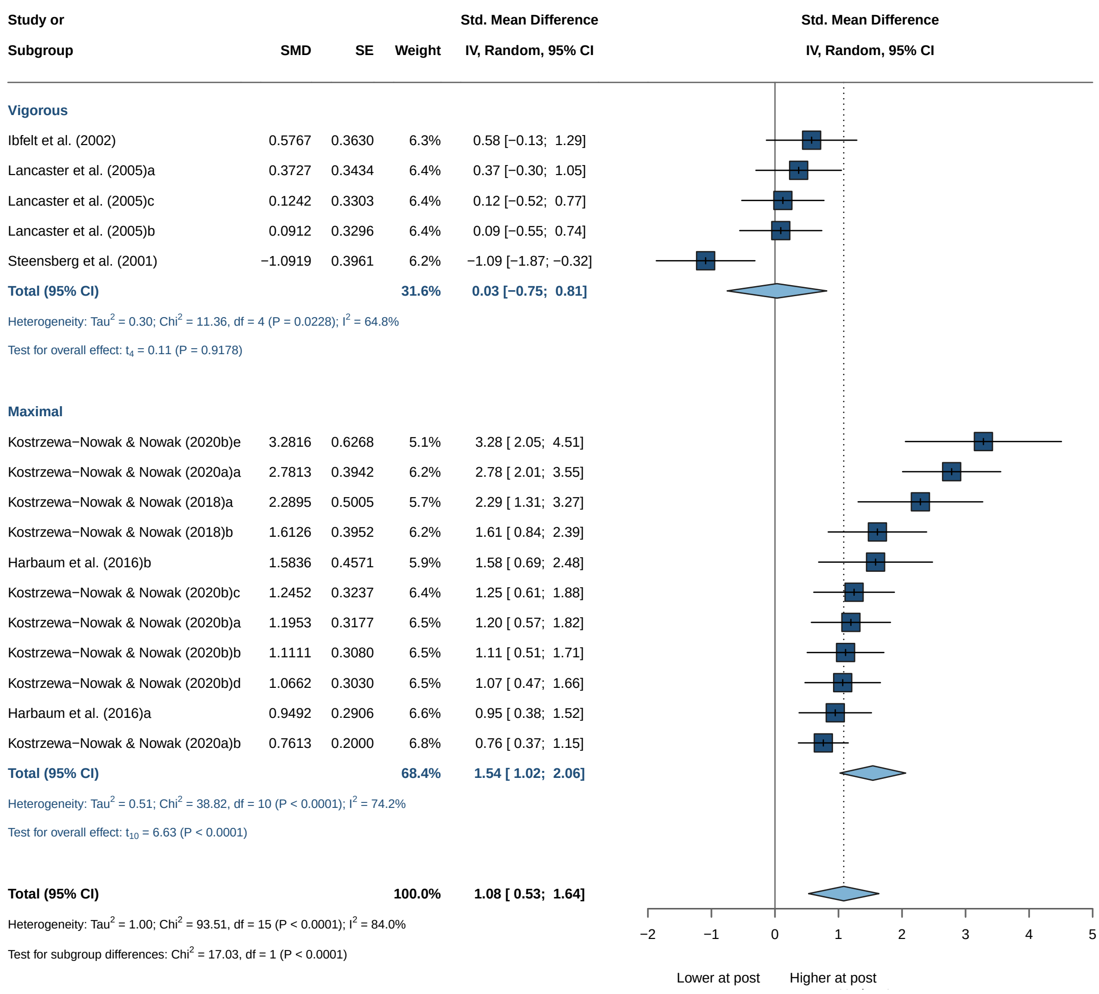
4.7.4. Intensity

```
Number of studies: k = 16

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.0050 [0.4207; 2.5358]; tau = 1.0025 [0.6486; 1.5924]
 I^2 = 84.0% [75.3%; 89.6%]; H = 2.50 [2.01; 3.10]

Test of heterogeneity:
     Q d.f.  p-value
 93.51   15 < 0.0001

Results for subgroups (random effects model (HK)):
                       k    SMD             95% CI  tau^2
intensity = Vigorous   5 0.0310 [-0.7527; 0.8147] 0.2967
intensity = Maximal   11 1.5400 [ 1.0222; 2.0579] 0.5104
                        tau     Q   I^2
intensity = Vigorous 0.5447 11.36 64.8%
intensity = Maximal  0.7145 38.82 74.2%
```

```
Test for subgroup differences (random effects model (HK)):
                   Q d.f.  p-value
Between groups 17.03    1 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **! Important**
>
> Significant results

4.7.4.1. Forrest Plot



4.7.5. Continuous Moderators (Meta-Regression)

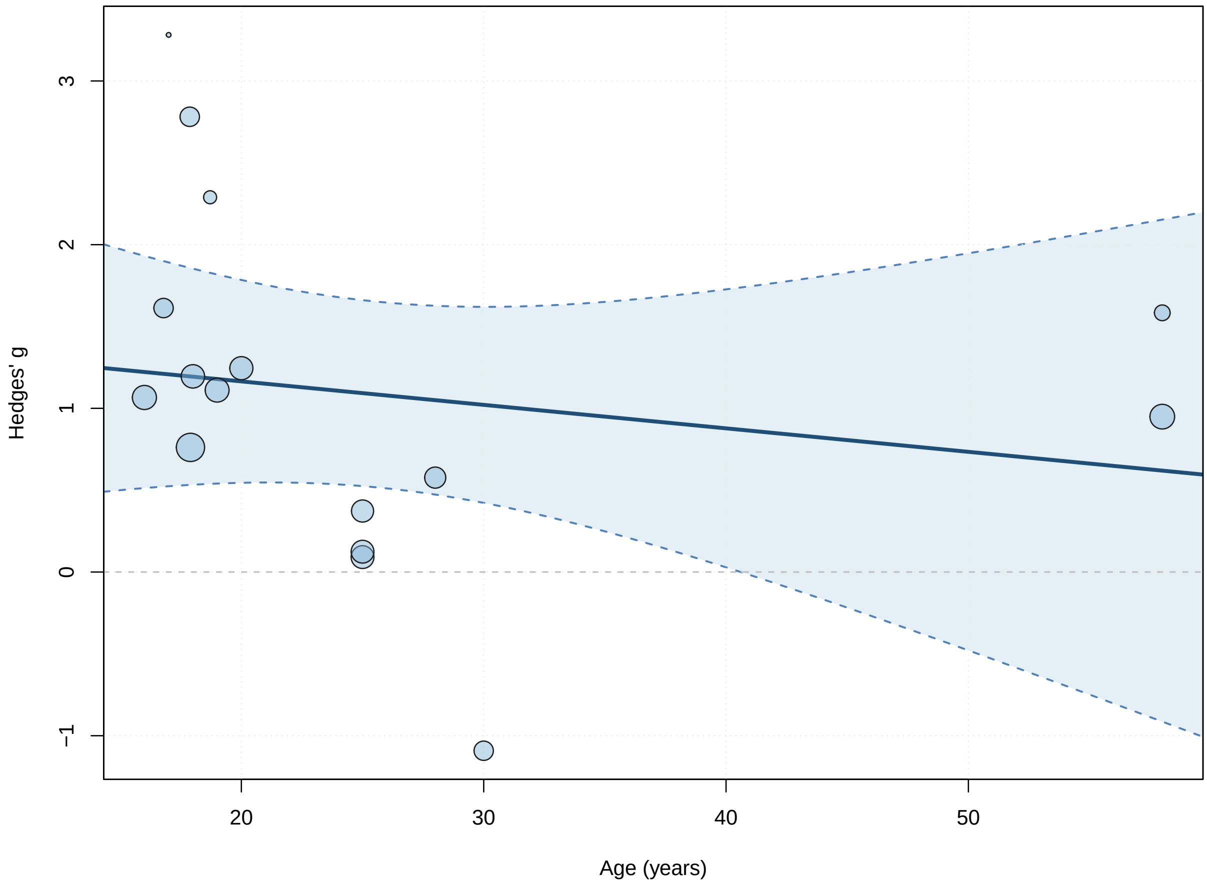
Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity (²); bubble plots show the fitted line with

95% CI.

4.7.5.1. Age

```
Mixed-Effects Model (k = 16; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-22.6865   49.0041   51.3730   53.6908   53.3730

tau^2 (estimated amount of residual heterogeneity):     1.0410 (SE = 0.4487)
tau (square root of estimated tau^2 value):             1.0203
I^2 (residual heterogeneity / unaccounted variability): 89.97%
H^2 (unaccounted variability / sampling variability):   9.97
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 14) = 91.8671, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 14) = 0.4880, p-val = 0.4963

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    1.4515  0.5896   2.4621  14  0.0274   0.1871
age       -0.0143  0.0205  -0.6986  14  0.4963  -0.0584
          ci.ub
intrcpt  2.7160  *
age      0.0297

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant



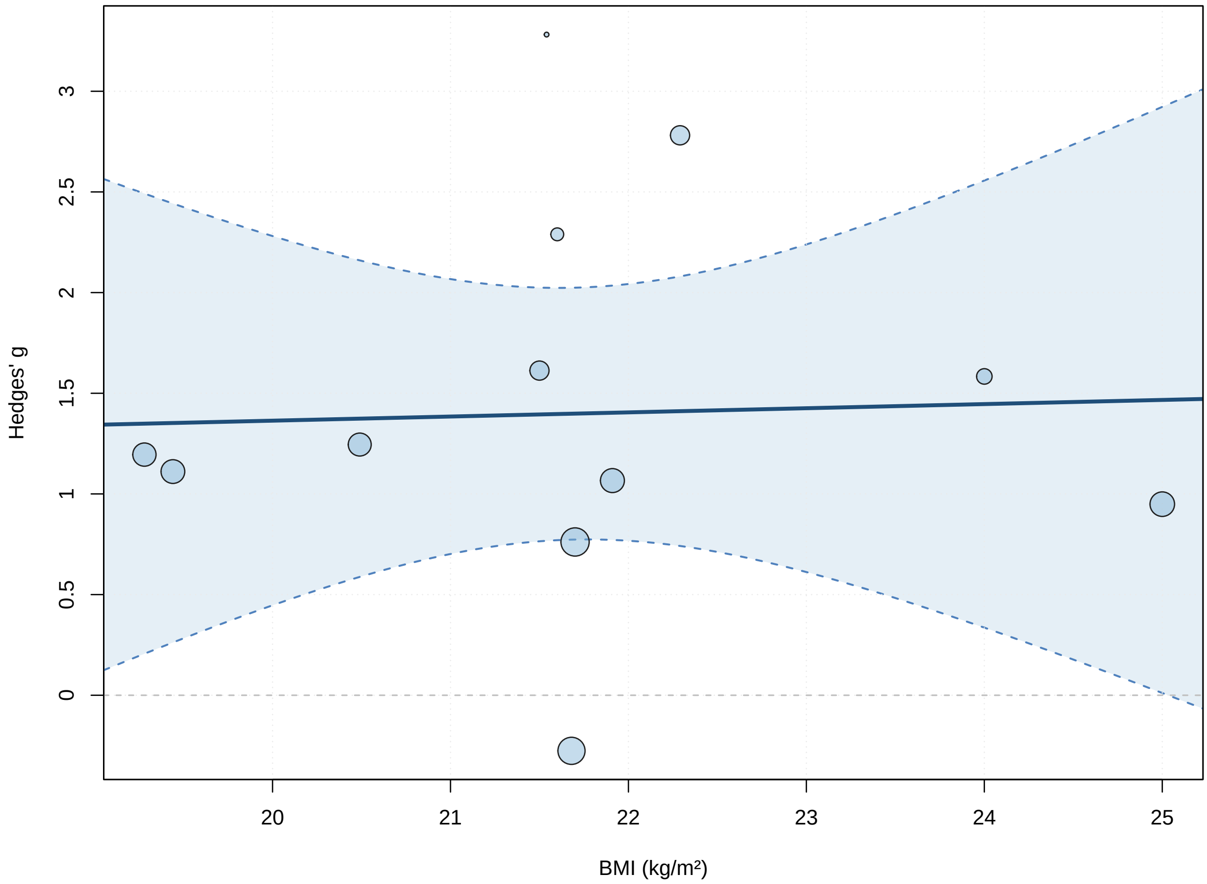
4.7.5.2. BMI

```
Mixed-Effects Model (k = 12; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-15.7587   34.9893   37.5174   38.9721   40.5174

tau^2 (estimated amount of residual heterogeneity):     0.8458 (SE = 0.4461)
tau (square root of estimated tau^2 value):             0.9197
I^2 (residual heterogeneity / unaccounted variability): 89.09%
H^2 (unaccounted variability / sampling variability):   9.17
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 10) = 77.5843, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 10) = 0.0134, p-val = 0.9102

Model Results:
```

```
         estimate      se    tval  df    pval    ci.lb
intrcpt    0.9518  3.8746  0.2457  10  0.8109  -7.6813
bmi        0.0206  0.1782  0.1156  10  0.9102  -0.3764
          ci.ub
intrcpt  9.5849
bmi      0.4176

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant



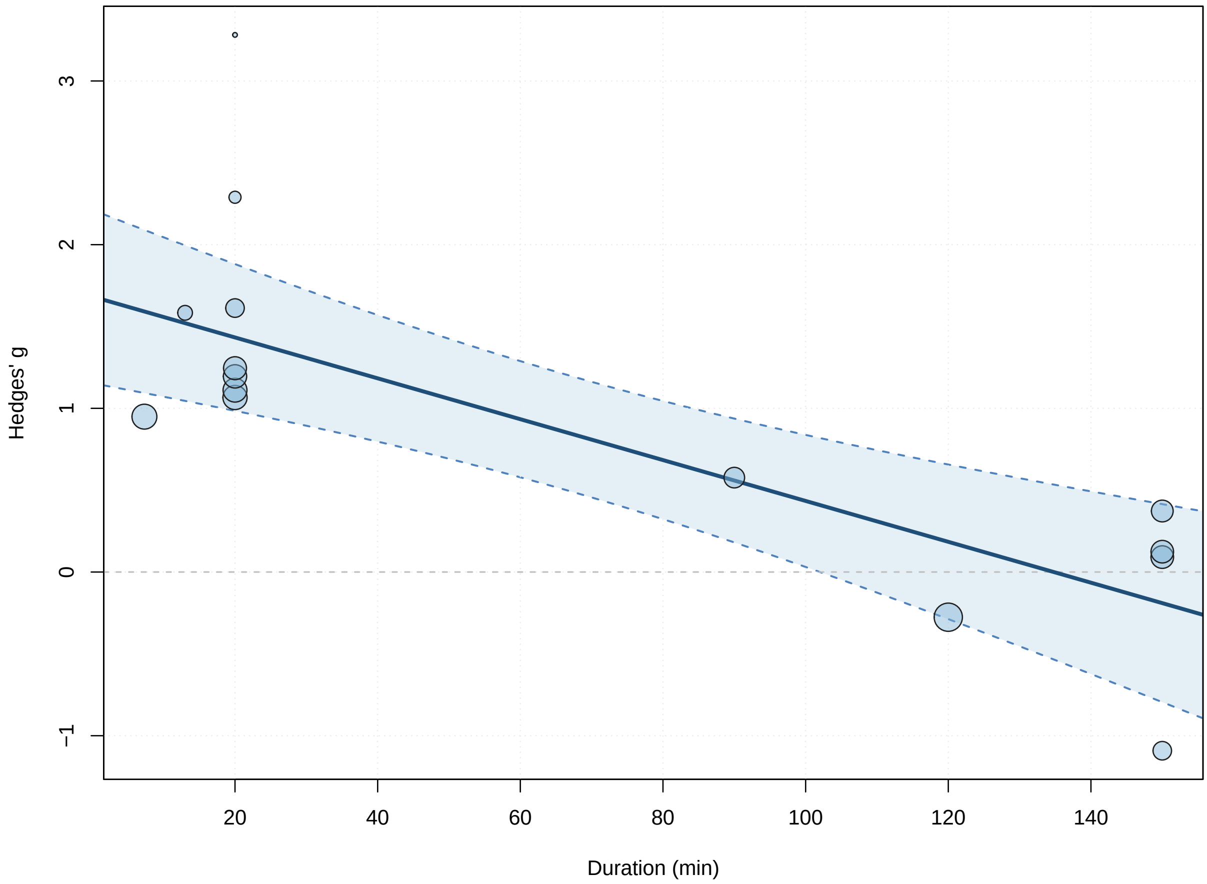
4.7.5.3. Duration

```
Mixed-Effects Model (k = 15; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-13.7530   30.8355   33.5060   35.6301   35.6878
```

```
tau^2 (estimated amount of residual heterogeneity):     0.3425 (SE = 0.1937)
tau (square root of estimated tau^2 value):             0.5853
I^2 (residual heterogeneity / unaccounted variability): 74.35%
H^2 (unaccounted variability / sampling variability):   3.90
R^2 (amount of heterogeneity accounted for):            65.26%

Test for Residual Heterogeneity:
QE(df = 13) = 31.7293, p-val = 0.0026

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 13) = 20.8420, p-val = 0.0005

Model Results:

          estimate      se     tval  df    pval    ci.lb
intrcpt     1.6827  0.2451   6.8662  13  <.0001   1.1532
duration   -0.0125  0.0027  -4.5653  13  0.0005  -0.0184
             ci.ub
intrcpt     2.2121  ***
duration   -0.0066  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

! Important

Significant



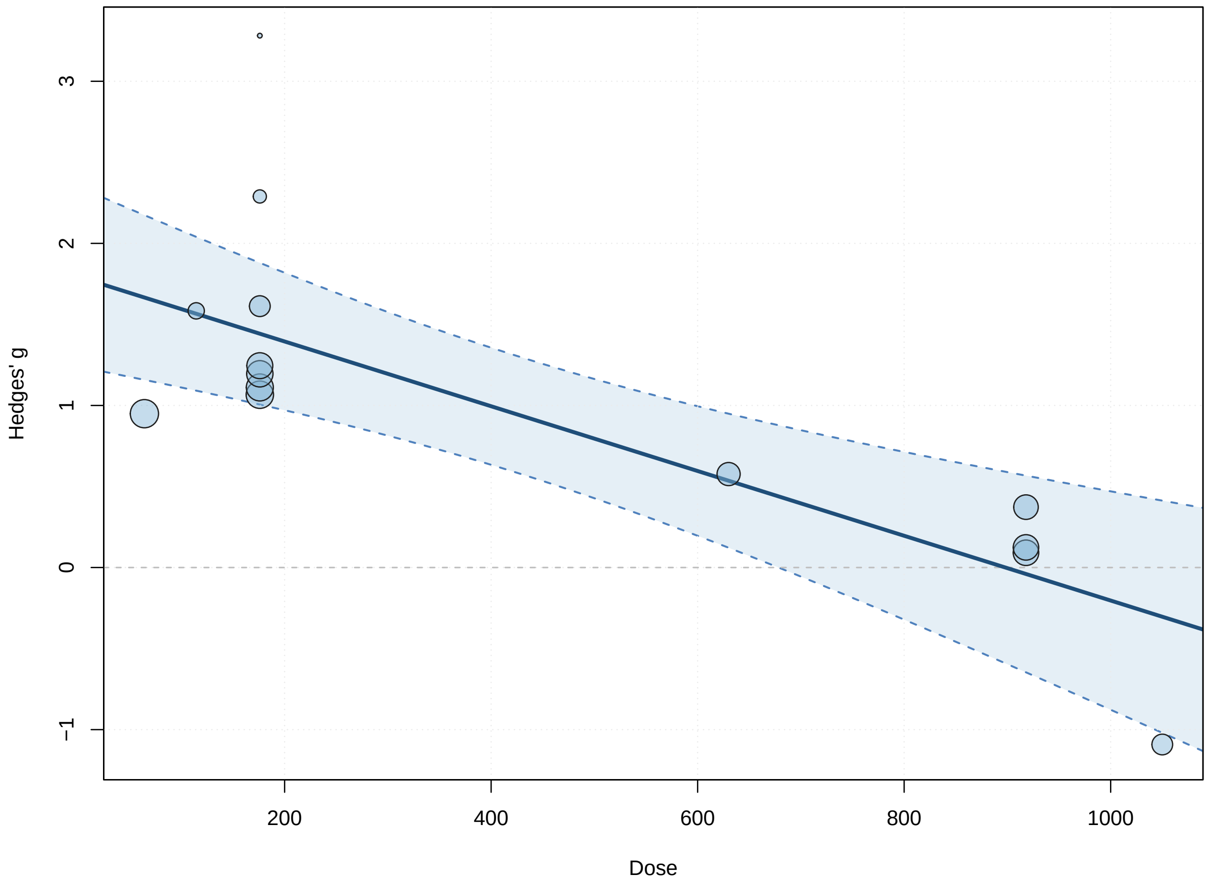
4.7.5.4. Dose

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-12.5188   27.2639   31.0376   32.9547   33.4376

tau^2 (estimated amount of residual heterogeneity):     0.3246 (SE = 0.1970)
tau (square root of estimated tau^2 value):             0.5697
I^2 (residual heterogeneity / unaccounted variability): 71.72%
H^2 (unaccounted variability / sampling variability):   3.54
R^2 (amount of heterogeneity accounted for):            65.62%

Test for Residual Heterogeneity:
QE(df = 12) = 25.4779, p-val = 0.0127

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 12) = 19.2815, p-val = 0.0009

Model Results:
```

```
         estimate      se      tval  df     pval    ci.lb
intrcpt    1.7943  0.2547    7.0457  12  <.0001    1.2394
dose      -0.0020  0.0005   -4.3911  12  0.0009   -0.0030
            ci.ub
intrcpt    2.3491  ***
dose      -0.0010  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

! Important

Significant — collinear with duration



4.7.5.5. IFN

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -9.1465   20.7348   24.2929   24.1307   32.2929
```

```
tau^2 (estimated amount of residual heterogeneity):     0.9848 (SE = 0.7226)
tau (square root of estimated tau^2 value):             0.9924
I^2 (residual heterogeneity / unaccounted variability): 88.95%
H^2 (unaccounted variability / sampling variability):   9.05
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 27.1266, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.0809, p-val = 0.7875

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    2.0560  1.7626   1.1665   5  0.2960  -2.4748
IFN       -0.2718  0.9556  -0.2845   5  0.7875  -2.7282
          ci.ub
intrcpt  6.5869
IFN      2.1846

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

4.7.5.6. IL-4

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-10.3229   23.4488   26.6457   27.2374   31.4457

tau^2 (estimated amount of residual heterogeneity):     0.6578 (SE = 0.4320)
tau (square root of estimated tau^2 value):             0.8111
I^2 (residual heterogeneity / unaccounted variability): 86.45%
H^2 (unaccounted variability / sampling variability):   7.38
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 7) = 30.1320, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 0.8032, p-val = 0.3999

Model Results:
```

```
         estimate      se    tval  df    pval    ci.lb
intrcpt    1.3165  0.3308  3.9793   7  0.0053   0.5342
IL_4       0.2506  0.2796  0.8962   7  0.3999  -0.4105
          ci.ub
intrcpt  2.0988  **
IL_4     0.9117

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

4.7.5.7. IL-6

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-18.0578   39.5434   42.1155   44.0327   44.5155

tau^2 (estimated amount of residual heterogeneity):     0.8055 (SE = 0.3889)
tau (square root of estimated tau^2 value):             0.8975
I^2 (residual heterogeneity / unaccounted variability): 88.00%
H^2 (unaccounted variability / sampling variability):   8.34
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 12) = 63.3467, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 12) = 0.1327, p-val = 0.7220

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    1.2063  0.2945  4.0964  12  0.0015   0.5647
IL_6       0.0604  0.1659  0.3642  12  0.7220  -0.3010
          ci.ub
intrcpt  1.8479  **
IL_6     0.4219

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

4.7.5.8. IL_10

```
Mixed-Effects Model (k = 11; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-12.3204   27.0095   30.6409   31.8345   34.0694

tau^2 (estimated amount of residual heterogeneity):     0.5518 (SE = 0.3375)
tau (square root of estimated tau^2 value):             0.7428
I^2 (residual heterogeneity / unaccounted variability): 81.00%
H^2 (unaccounted variability / sampling variability):   5.26
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 9) = 35.0975, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 9) = 0.2716, p-val = 0.6148

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    1.5574  0.2433   6.3998   9  0.0001   1.0069
IL_10     -0.3495  0.6706  -0.5211   9  0.6148  -1.8664
          ci.ub
intrcpt  2.1078  ***
IL_10    1.1675

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

4.7.5.9. IL_2

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -8.6034   19.6487   23.2069   23.0446   31.2069

tau^2 (estimated amount of residual heterogeneity):     0.8092 (SE = 0.5809)
```

```
tau (square root of estimated tau^2 value):             0.8996
I^2 (residual heterogeneity / unaccounted variability): 89.58%
H^2 (unaccounted variability / sampling variability):   9.59
R^2 (amount of heterogeneity accounted for):            0.73%

Test for Residual Heterogeneity:
QE(df = 5) = 32.1011, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.8054, p-val = 0.4106

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.2553  1.4923  0.1711   5  0.8709  -3.5808
IL_2       0.7369  0.8211  0.8974   5  0.4106  -1.3739
          ci.ub
intrcpt  4.0915
IL_2     2.8477

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

4.8. Multivariate Models

Multivariate meta-regression tests interactions between candidate moderators. Neither model below yields independent predictors.

4.8.1. Intensity * Duration

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.3855 (SE = 0.2452)
tau (square root of estimated tau^2 value):             0.6209
I^2 (residual heterogeneity / unaccounted variability): 74.36%
H^2 (unaccounted variability / sampling variability):   3.90
R^2 (amount of heterogeneity accounted for):            59.17%

Test for Residual Heterogeneity:
QE(df = 10) = 24.8856, p-val = 0.0056

Test of Moderators (coefficients 2:4):
F(df1 = 3, df2 = 10) = 5.5862, p-val = 0.0164
```

```
Model Results:

                            estimate      se     tval  df
intrcpt                       0.7778  0.9262   0.8398  10
intensityVigorous             0.8233  1.9758   0.4167  10
duration                      0.0403  0.0508   0.7938  10
intensityVigorous:duration   -0.0517  0.0523  -0.9887  10
                              pval    ci.lb   ci.ub
intrcpt                     0.4206  -1.2859  2.8415
intensityVigorous           0.6857  -3.5792  5.2257
duration                    0.4457  -0.0728  0.1534
intensityVigorous:duration  0.3461  -0.1681  0.0648

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

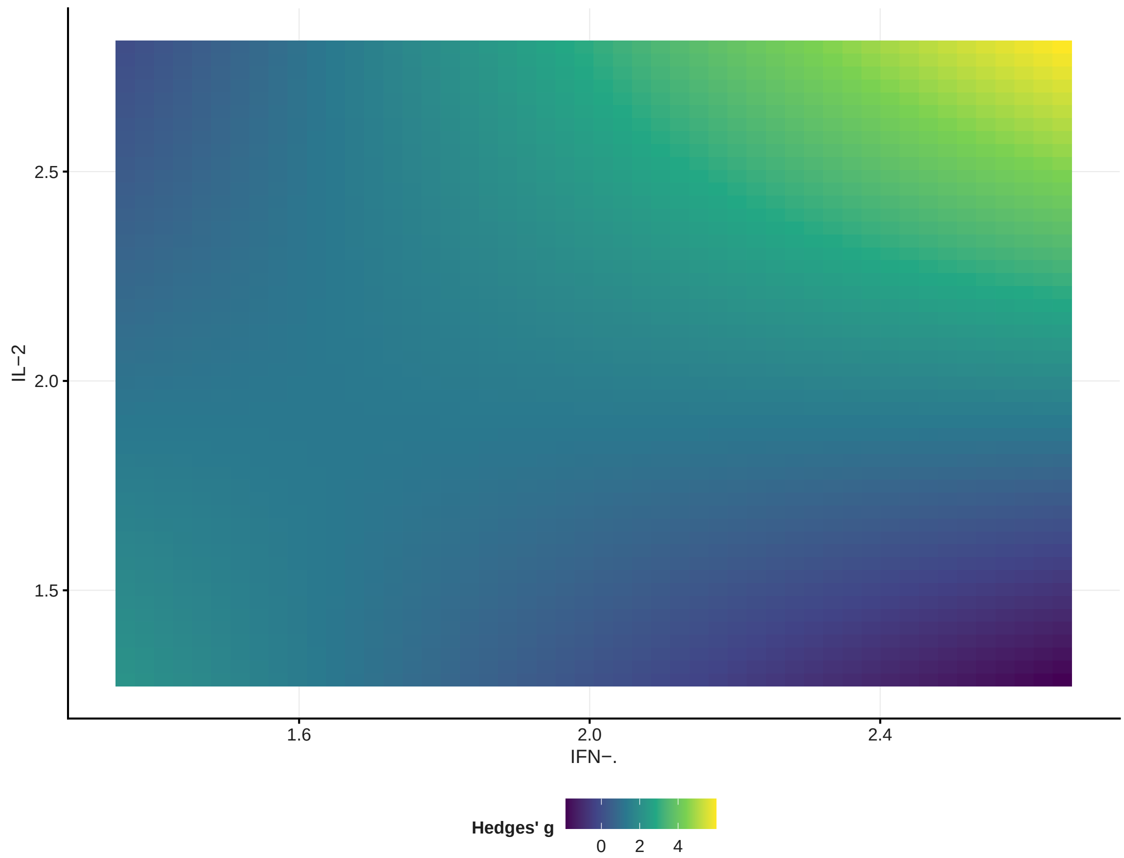
4.8.2. IFN * IL_2

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.8792 (SE = 0.8102)
tau (square root of estimated tau^2 value):             0.9376
I^2 (residual heterogeneity / unaccounted variability): 88.83%
H^2 (unaccounted variability / sampling variability):   8.95
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 3) = 21.9341, p-val < .0001

Test of Moderators (coefficients 2:4):
F(df1 = 3, df2 = 3) = 0.7599, p-val = 0.5866

Model Results:

          estimate       se     tval  df    pval     ci.lb
intrcpt    17.1616  16.2628   1.0553   3  0.3688  -34.5937
IFN        -9.6844   9.0286  -1.0726   3  0.3620  -38.4174
IL_2       -8.3898   9.5134  -0.8819   3  0.4428  -38.6658
IFN:IL_2    5.1176   5.1389   0.9959   3  0.3927  -11.2366
             ci.ub
intrcpt    68.9169
IFN        19.0485
IL_2       21.8862
```

```
IFN:IL_2  21.4719

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

4.9. Interaction Surface (IFN- × IL-2)



5. Th1 — ~2 h Post-Exercise

5.1. Data Import

The ~2 h window (`effect_size/Th1/pre_post+2h/Th1_pre_post+2h_r_050.csv`) contains few arms and no cytokine/BMI coverage, so it is reported for the main analysis, outlier check, funnel, forest, and asymmetry only — no moderator modelling is possible. No study is excluded a priori in this window.

5.2. Primary Meta-Analysis

5.2.1. {meta}

```
Number of studies: k = 5

                              SMD            95% CI     t
Random effects model (HK) -0.4137 [-1.0063; 0.1789] -1.94
Prediction interval               [-1.5190; 0.6917]
                          p-value
Random effects model (HK)  0.1247
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.1110 [0.0000; 1.8375]; tau = 0.3332 [0.0000; 1.3555]
 I^2 = 42.5% [0.0%; 78.9%]; H = 1.32 [1.00; 2.17]

Test of heterogeneity:
    Q d.f. p-value
 6.96    4  0.1382

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 4)
- Prediction interval based on t-distribution (df = 4)
```

5.2.2. {metafor}

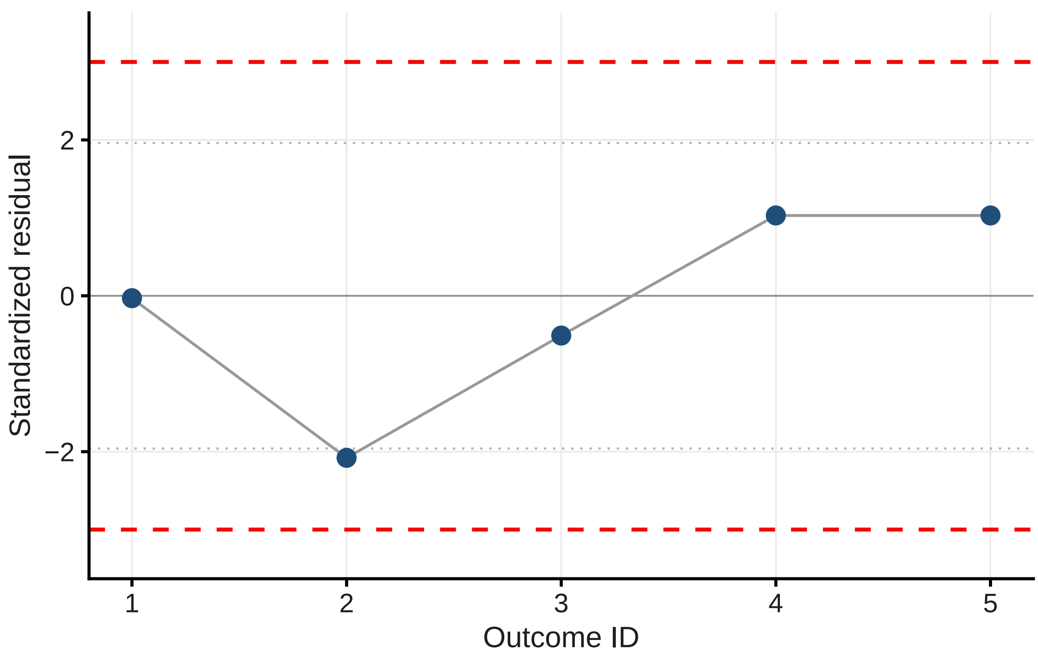
```
Random-Effects Model (k = 5; tau^2 estimator: HE)
```

```
tau^2 (estimated amount of total heterogeneity): 0.1110 (SE = 0.1699)
tau (square root of estimated tau^2 value):      0.3332
I^2 (total heterogeneity / total variability):   46.86%
H^2 (total variability / sampling variability):  1.88

Test for Heterogeneity:
Q(df = 4) = 6.9560, p-val = 0.1382

Model Results:

estimate      se     tval  df    pval    ci.lb   ci.ub
 -0.4137  0.2134  -1.9381   4  0.1247  -1.0063  0.1789

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

5.3. Outlier Detection

Table 5.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	-0.435	-0.031
Steensberg et al. (2001)	-1.164	-2.079
Lancaster et al. (2005)a	-0.657	-0.510
Lancaster et al. (2005)b	0.000	1.031
Lancaster et al. (2005)c	0.000	1.031



ℹ Note

No outcome has standardized residuals above 3 or below -3.

5.4. Funnel Plot & Influential Cases

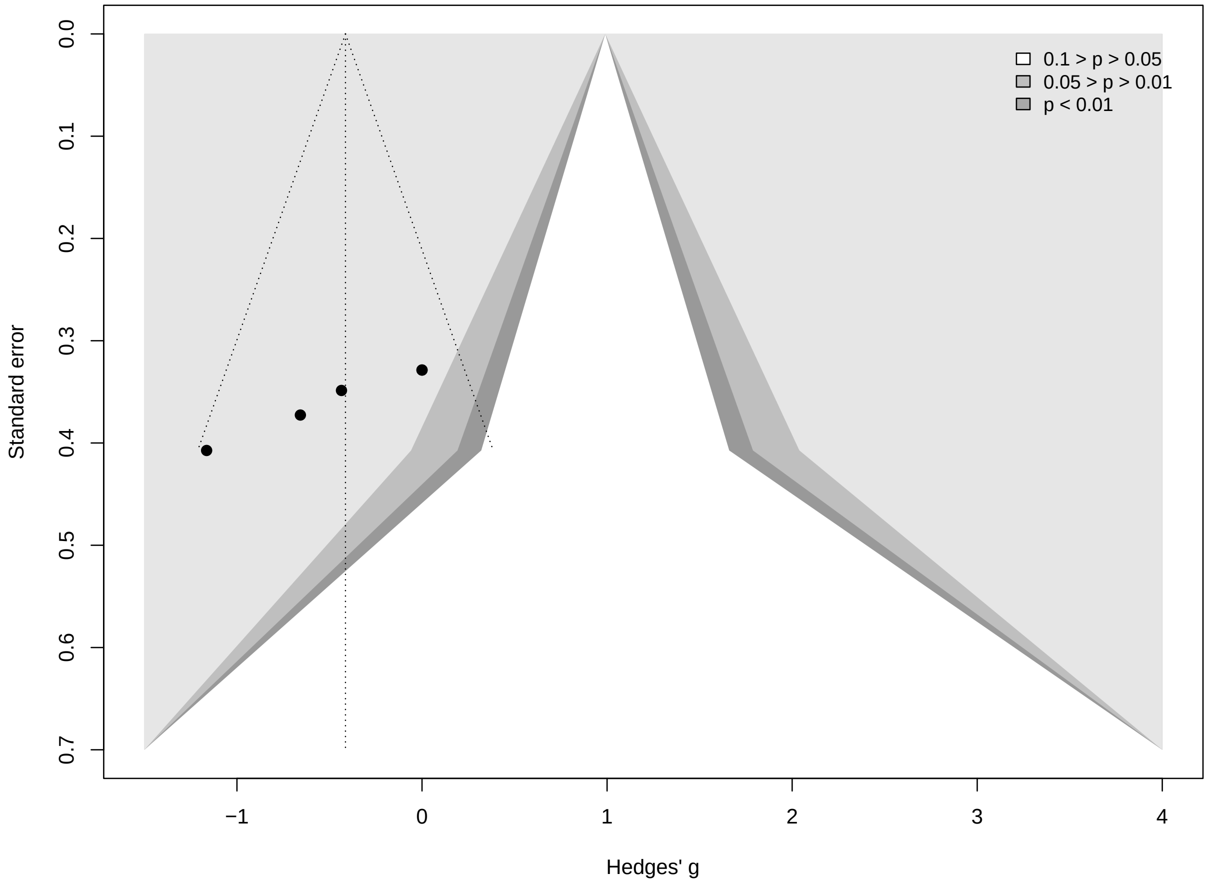
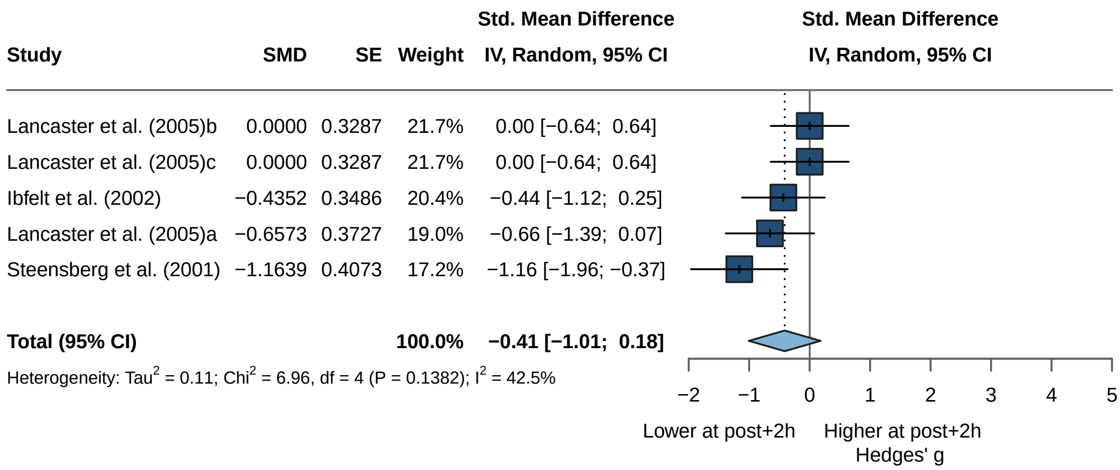


Table 5.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Ibfelt et al. (2002)	-0.435	-0.08474579
Steensberg et al. (2001)	-1.164	-0.87614442
Lancaster et al. (2005)a	-0.657	-0.38036241
Lancaster et al. (2005)b	0.000	0.73086010
Lancaster et al. (2005)c	0.000	0.73086010

One value below -1 and one above 1 were identified as influential cases. Neither is an outlier (Kostrzewa-Nowak & Nowak (2020a)a and Tsubakihara et al. (2012)); all outcomes are retained.

5.5. Forest Plot



5.6. Asymmetry Test

Note: the number of studies ($k = 5$) is too small to reliably test for small-study effects (minimum $k = 10$; Borenstein et al., 2009). No Egger's test or trim-and-fill is performed.

6. Th1 — 17 h Post-Exercise

6.1. Data Import

6.2. Primary Meta-Analysis

As in the immediate window, **Kostrzewa-Nowak et al. (2019)** is removed before pooling. Outlier detection then flags **Steensberg et al. (2001)**, which is both an outlier and an influential case and is therefore withdrawn from the downstream analyses (`m2_1`).

6.2.1. {meta}

```
Number of studies: k = 11

                             SMD            95% CI    t
Random effects model (HK) 1.8184 [ 0.8759; 2.7610] 4.30
Prediction interval              [-2.1567; 5.7935]
                          p-value
Random effects model (HK)  0.0016
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 2.8848 [0.3797; 7.3113]; tau = 1.6985 [0.6162; 2.7039]
 I^2 = 83.2% [71.3%; 90.1%]; H = 2.44 [1.87; 3.18]

Test of heterogeneity:
     Q d.f.  p-value
 59.37   10 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 10)
- Prediction interval based on t-distribution (df = 10)
```

6.2.2. {metafor}

```
Random-Effects Model (k = 11; tau^2 estimator: HE)
```

```
tau^2 (estimated amount of total heterogeneity): 2.8848 (SE = 2.0141)
tau (square root of estimated tau^2 value):      1.6985
I^2 (total heterogeneity / total variability):   95.32%
H^2 (total variability / sampling variability):  21.39

Test for Heterogeneity:
Q(df = 10) = 59.3676, p-val < .0001

Model Results:

estimate      se    tval  df    pval   ci.lb   ci.ub
  1.8184  0.4230  4.2986  10  0.0016  0.8759  2.7610  **

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6.2.3. Adapted Modell {meta}

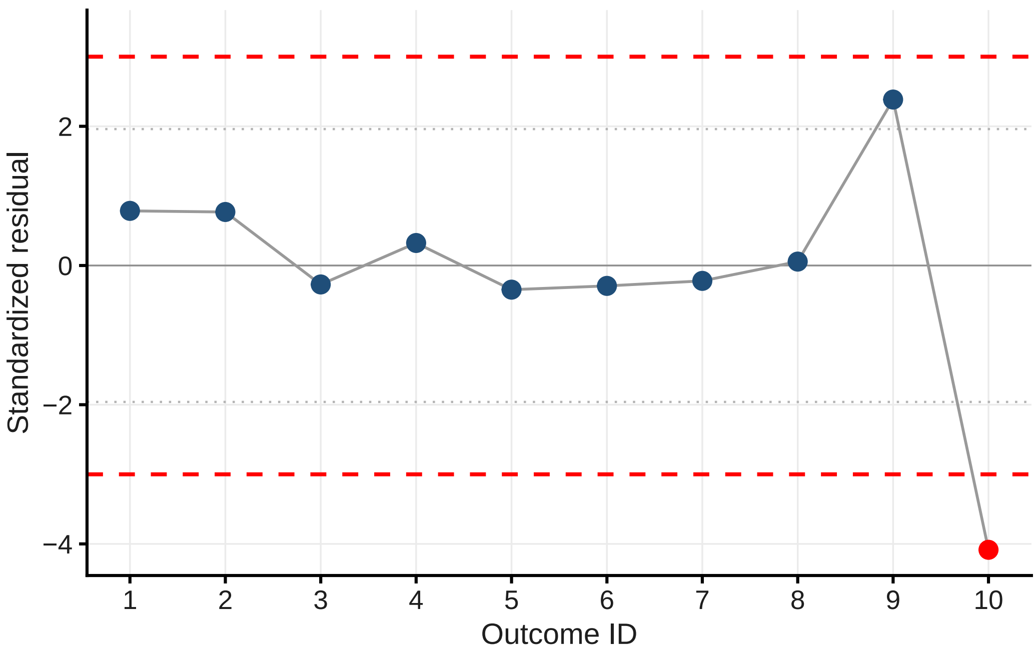
> Caution
>
> Re-Fit: Removed Konstrezw-Nowak et al. (2019)

```
Number of studies: k = 10

                             SMD            95% CI    t
Random effects model (HK) 1.6316 [ 0.8788; 2.3845] 4.90
Prediction interval              [-0.8087; 4.0720]
                          p-value
Random effects model (HK)  0.0008
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.0428 [0.3167; 3.7647]; tau = 1.0212 [0.5628; 1.9403]
 I^2 = 83.8% [71.7%; 90.7%]; H = 2.48 [1.88; 3.28]

Test of heterogeneity:
     Q d.f.  p-value
 55.52    9 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 9)
- Prediction interval based on t-distribution (df = 9)
```

6.3. Outlier Detection

Table 6.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2018)a	2.320	0.785
Kostrzewa-Nowak & Nowak (2018)b	2.305	0.769
Kostrzewa-Nowak & Nowak (2020a)a	1.374	-0.272
Kostrzewa-Nowak & Nowak (2020a)b	1.889	0.323
Kostrzewa-Nowak & Nowak (2020b)a	1.305	-0.347
Kostrzewa-Nowak & Nowak (2020b)b	1.352	-0.293
Kostrzewa-Nowak & Nowak (2020b)c	1.416	-0.221
Kostrzewa-Nowak & Nowak (2020b)d	1.660	0.056
Kostrzewa-Nowak & Nowak (2020b)e	3.959	2.384
Steensberg et al. (2001)	-0.454	-4.085



Note

One outcome has standardized residuals below -3 (Steensberg et al. (2001)).

6.4. Funnel Plot & Influential Cases

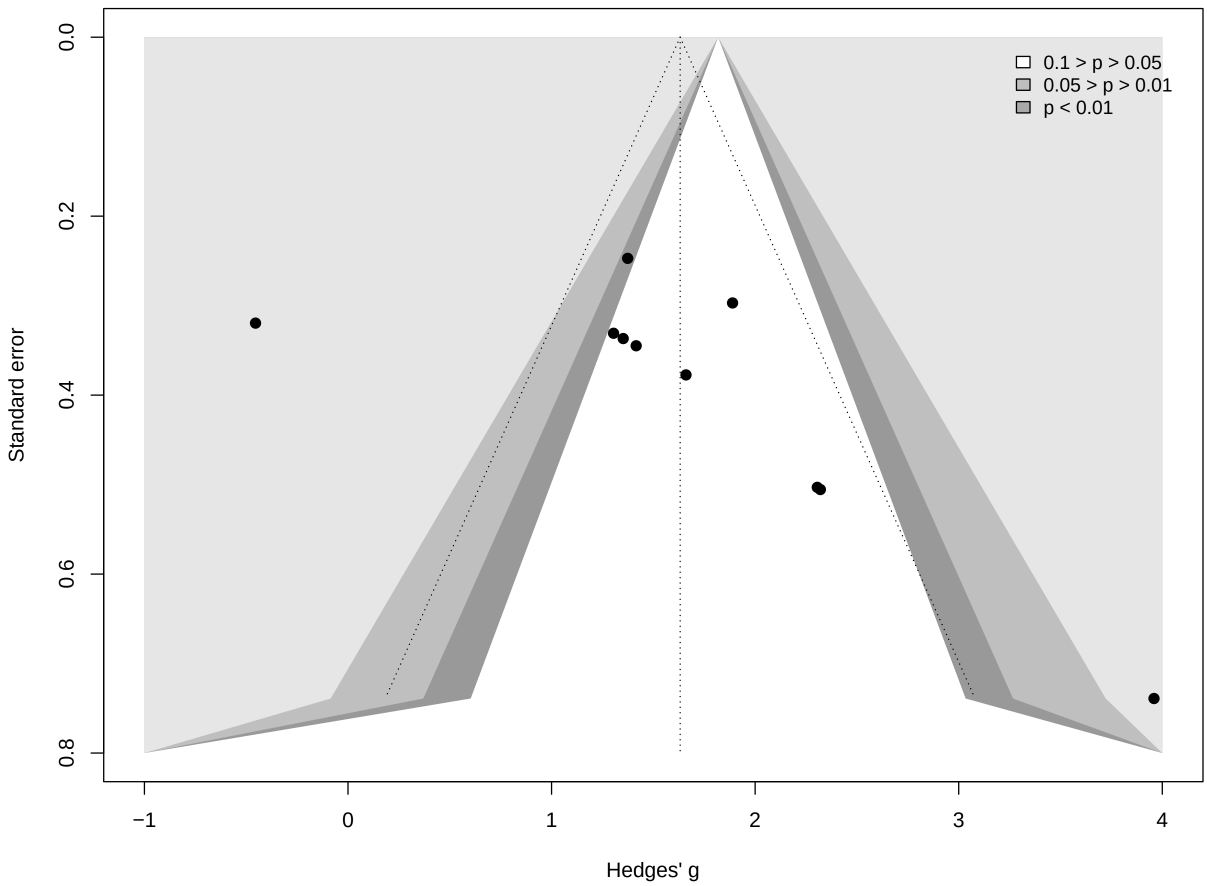


Table 6.2.: DFBETAS influence diagnostics

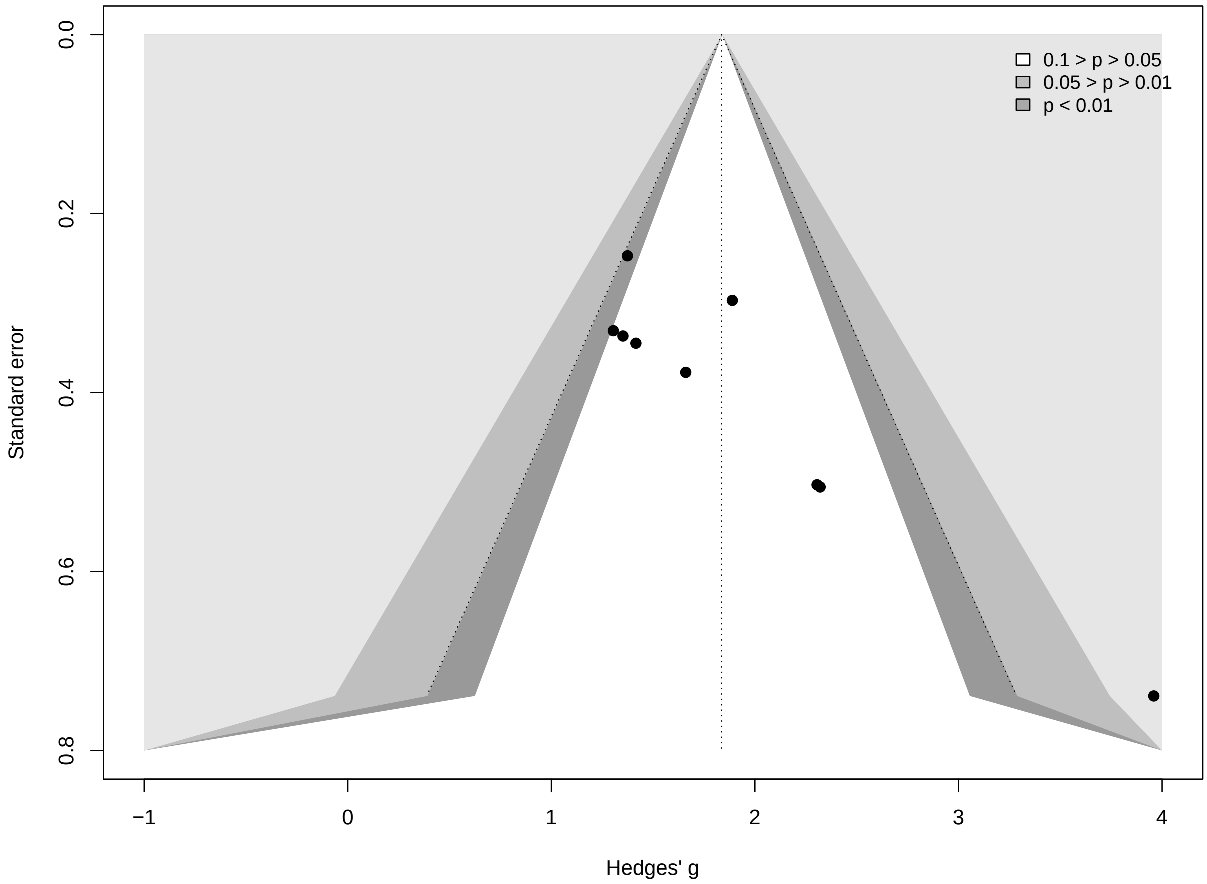
Short_reference	g	dfbetas
Kostrzewa-Nowak & Nowak (2018)a	2.320	0.42461320
Kostrzewa-Nowak & Nowak (2018)b	2.305	0.42202631
Kostrzewa-Nowak & Nowak (2020a)a	1.374	-0.04511648
Kostrzewa-Nowak & Nowak (2020a)b	1.889	0.72827130
Kostrzewa-Nowak & Nowak (2020b)a	1.305	-0.10227485
Kostrzewa-Nowak & Nowak (2020b)b	1.352	-0.04589921
Kostrzewa-Nowak & Nowak (2020b)c	1.416	0.02298307
Kostrzewa-Nowak & Nowak (2020b)d	1.660	0.22848951
Kostrzewa-Nowak & Nowak (2020b)e	3.959	0.53549707
Steensberg et al. (2001)	-0.454	-2.29509141

Caution

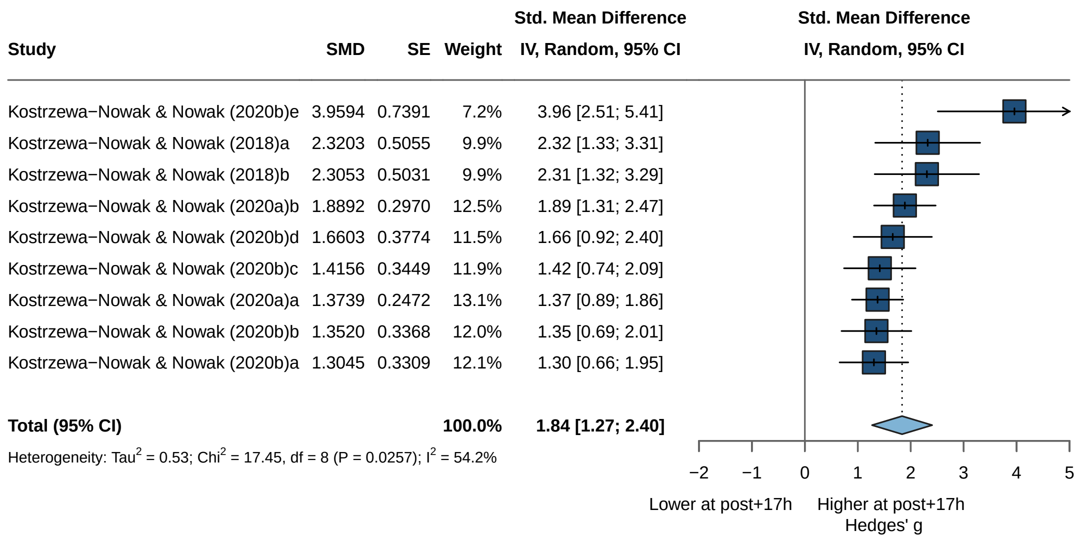
Steensberg et al. (2001) is < -1 (influential) and also an outlier -> withdrawn.

6.4.1. Adapted Model

```
Number of studies: k = 9

                             SMD           95% CI    t
Random effects model (HK) 1.8367 [1.2690; 2.4043] 7.46
Prediction interval              [0.0315; 3.6418]
                           p-value
Random effects model (HK) < 0.0001
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.5349 [0.0000; 2.2412]; tau = 0.7314 [0.0000; 1.4971]
 I^2 = 54.2% [2.8%; 78.4%]; H = 1.48 [1.01; 2.15]

Test of heterogeneity:
     Q d.f. p-value
 17.45    8  0.0257

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 8)
- Prediction interval based on t-distribution (df = 8)
```



6.5. Forest Plot



6.6. Asymmetry Test

6.6.1. Egger´s regression test

6.6.2. Trim and Fill

```
Estimated number of missing studies on the left side: 0 (SE = 1.9045)

Random-Effects Model (k = 9; tau^2 estimator: HE)

tau^2 (estimated amount of total heterogeneity): 0.5349 (SE = 0.3673)
tau (square root of estimated tau^2 value):      0.7314
I^2 (total heterogeneity / total variability):   80.24%
H^2 (total variability / sampling variability):  5.06

Test for Heterogeneity:
Q(df = 8) = 17.4510, p-val = 0.0257

Model Results:

estimate      se    zval    pval   ci.lb   ci.ub
  1.8367  0.2791  6.5818  <.0001  1.2897  2.3836  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ℹ No missing studies

6.7. Moderator Analyses

6.7.1. Risk of Bias

6.7.1.1. Model

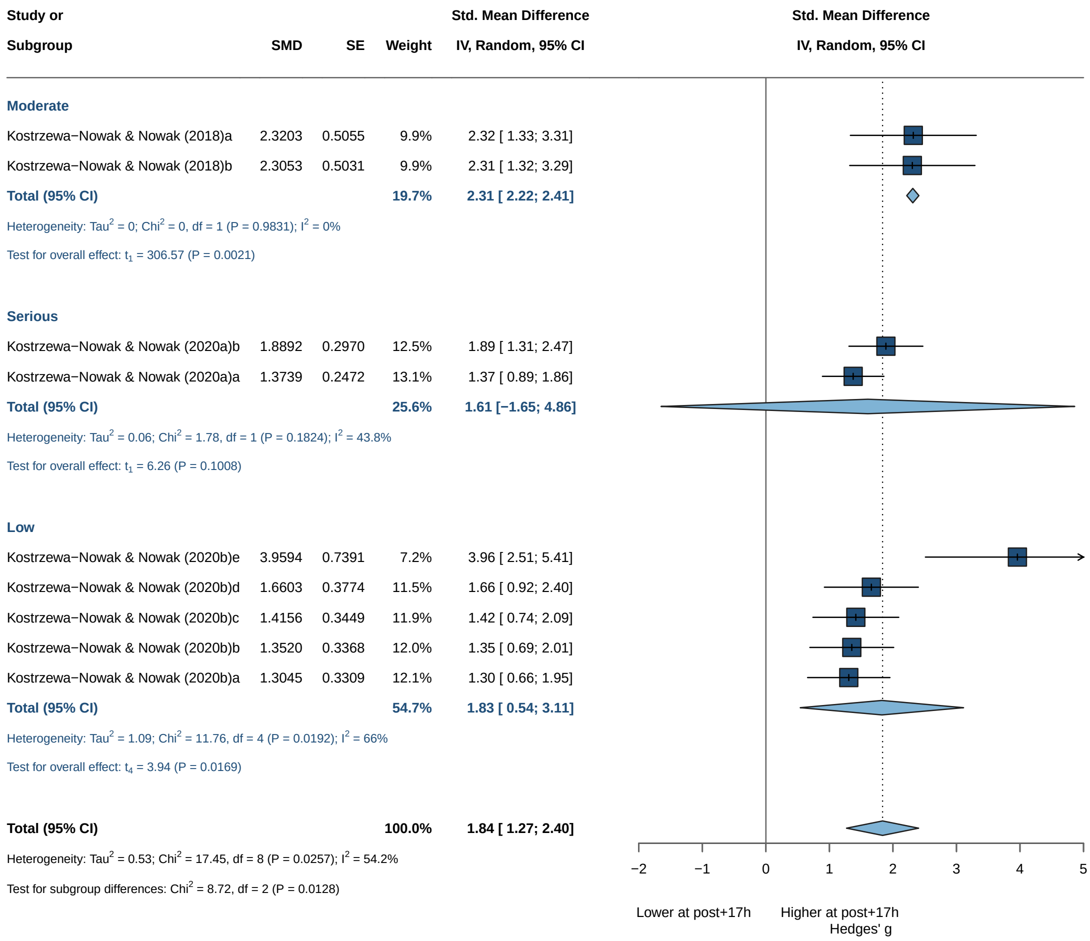
```
Number of studies: k = 9

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.5349 [0.0000; 2.2412]; tau = 0.7314 [0.0000; 1.4971]
 I^2 = 54.2% [2.8%; 78.4%]; H = 1.48 [1.01; 2.15]

Test of heterogeneity:
     Q d.f. p-value
 17.45    8  0.0257

Results for subgroups (random effects model (HK)):
                 k    SMD            95% CI  tau^2    tau
```

```
RoB = Moderate   2 2.3128 [ 2.2169; 2.4086]      0      0
RoB = Serious    2 1.6052 [-1.6511; 4.8615] 0.0581 0.2410
RoB = Low        5 1.8263 [ 0.5408; 3.1117] 1.0891 1.0436
                    Q   I^2
RoB = Moderate  0.00  0.0%
RoB = Serious   1.78 43.8%
RoB = Low      11.76 66.0%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 8.72    2  0.0128

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

Note

Not significant results

6.7.1.2. Forrest Plot



Intensity: all retained arms share the same intensity, so no intensity subgroup analysis is performed.

6.7.2. Continuous Moderators (Meta-Regression)

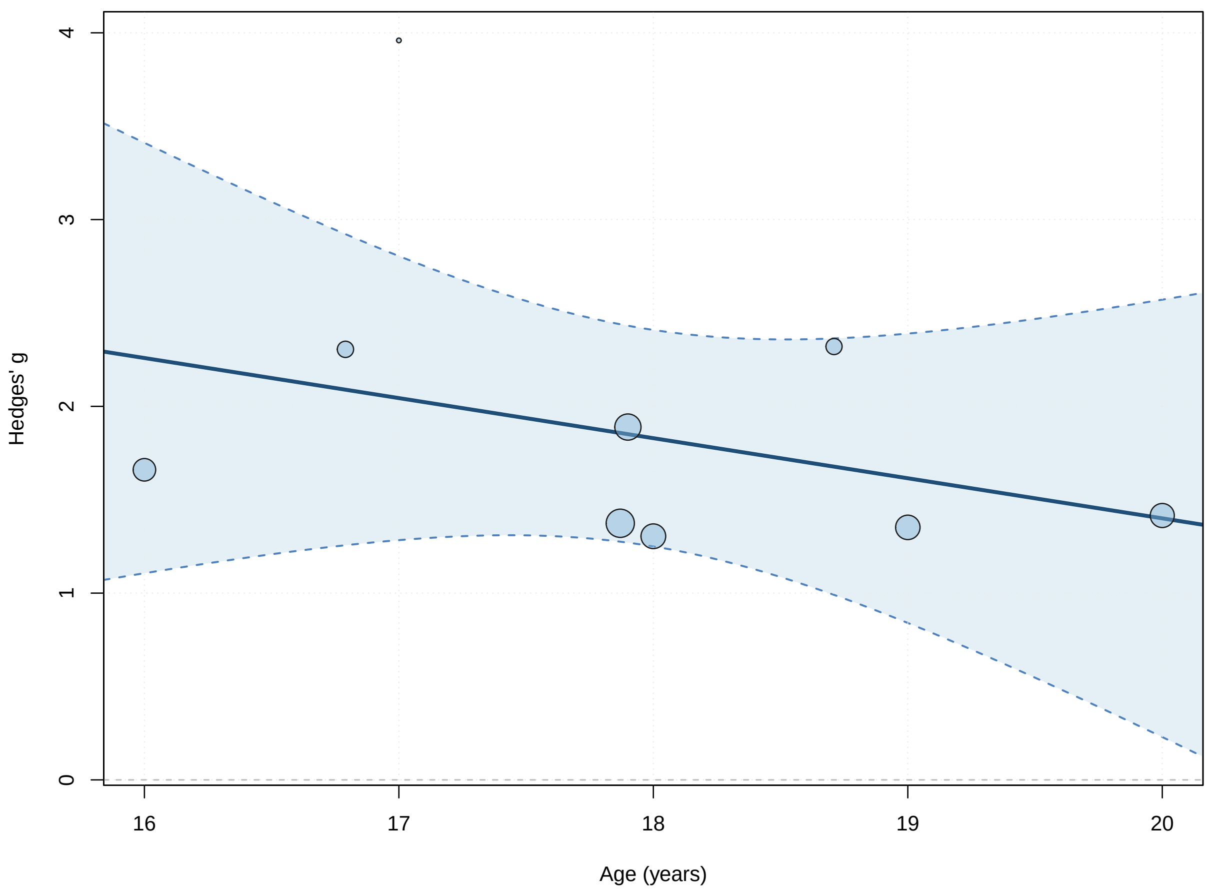
6.7.2.1. Age

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -9.3856   19.2634   24.7713   25.3629   29.5713

tau^2 (estimated amount of residual heterogeneity):     0.5156 (SE = 0.3831)
tau (square root of estimated tau^2 value):             0.7181
I^2 (residual heterogeneity / unaccounted variability): 79.71%
H^2 (unaccounted variability / sampling variability):   4.93
R^2 (amount of heterogeneity accounted for):            3.61%

Test for Residual Heterogeneity:
QE(df = 7) = 15.6877, p-val = 0.0281
```

```
Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 1.0164, p-val = 0.3470

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    5.6926  3.8350   1.4844   7  0.1813  -3.3758
age       -0.2146  0.2129  -1.0082   7  0.3470  -0.7180
            ci.ub
intrcpt  14.7609
age       0.2888

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

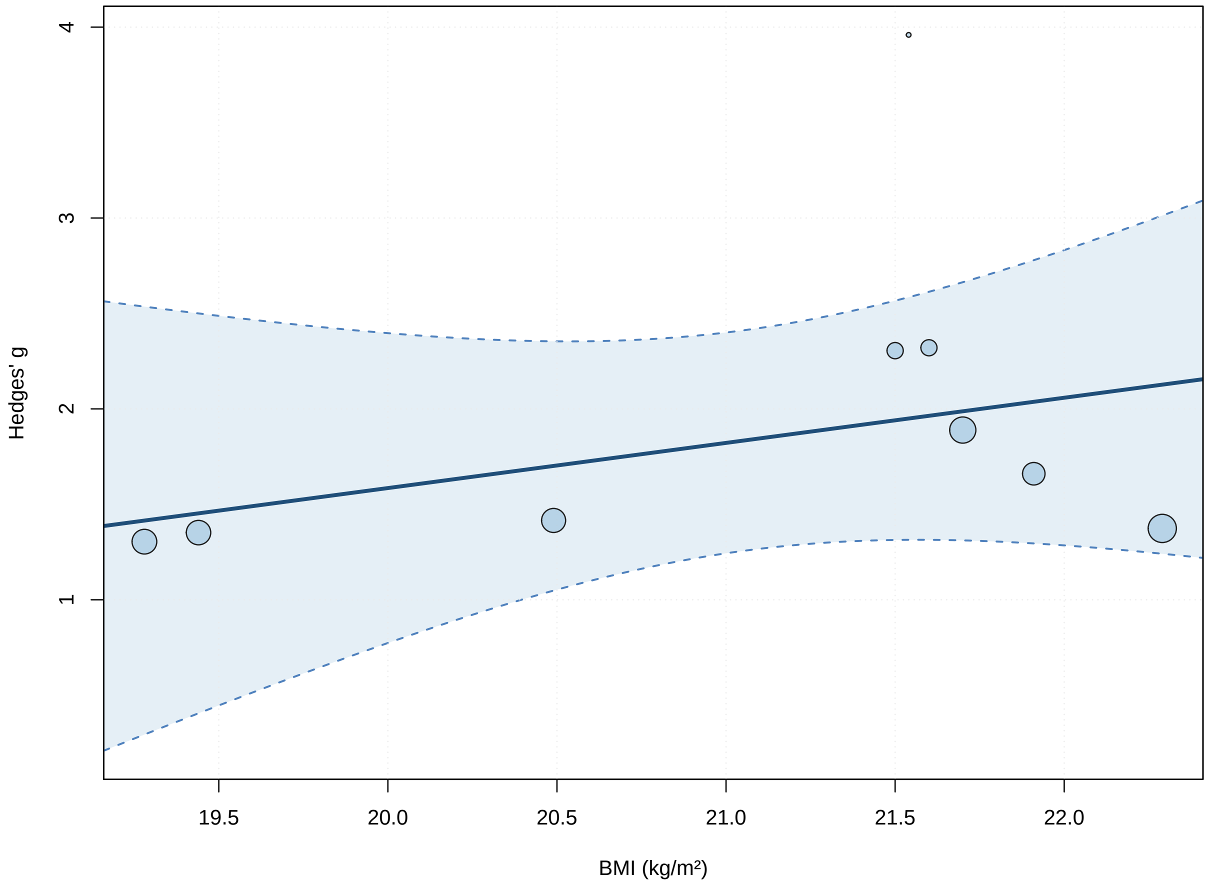


6.7.2.2. BMI

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -9.3661   19.2242   24.7321   25.3238   29.5321

tau^2 (estimated amount of residual heterogeneity):     0.5147 (SE = 0.3872)
tau (square root of estimated tau^2 value):             0.7175
I^2 (residual heterogeneity / unaccounted variability): 78.61%
H^2 (unaccounted variability / sampling variability):   4.68
R^2 (amount of heterogeneity accounted for):            3.77%

Test for Residual Heterogeneity:
QE(df = 7) = 15.9237, p-val = 0.0258

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 1.0645, p-val = 0.3365

Model Results:

         estimate      se     tval  df    pval     ci.lb
intrcpt   -3.1460  4.8331  -0.6509   7  0.5359  -14.5743
bmi        0.2366  0.2293   1.0317   7  0.3365   -0.3056
          ci.ub
intrcpt  8.2824
bmi      0.7787

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant results



6.7.2.3. Duration

```
Random-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -8.5222   15.9889   21.0443   20.9362   24.0443

tau^2 (estimated amount of total heterogeneity): 0.6770 (SE = 0.5235)
tau (square root of estimated tau^2 value):      0.8228
I^2 (total heterogeneity / total variability):   80.13%
H^2 (total variability / sampling variability):  5.03

Test for Heterogeneity:
Q(df = 6) = 15.4832, p-val = 0.0168

Model Results:

estimate      se    tval  df    pval   ci.lb   ci.ub
  1.9274  0.3239  5.9501   6  0.0010  1.1348  2.7200  **

---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant results

6.7.2.4. Dose

```
Random-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -8.5222   15.9889   21.0443   20.9362   24.0443

tau^2 (estimated amount of total heterogeneity): 0.6770 (SE = 0.5235)
tau (square root of estimated tau^2 value):      0.8228
I^2 (total heterogeneity / total variability):   80.13%
H^2 (total variability / sampling variability):  5.03

Test for Heterogeneity:
Q(df = 6) = 15.4832, p-val = 0.0168

Model Results:

estimate      se    tval  df    pval   ci.lb   ci.ub
  1.9274  0.3239  5.9501   6  0.0010  1.1348  2.7200  **

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant results

6.7.2.5. IFN

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -8.6058   18.6411   23.2116   23.0493   31.2116

tau^2 (estimated amount of residual heterogeneity):     0.8890 (SE = 0.6816)
tau (square root of estimated tau^2 value):             0.9429
I^2 (residual heterogeneity / unaccounted variability): 87.53%
H^2 (unaccounted variability / sampling variability):   8.02
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
```

```
QE(df = 5) = 13.1061, p-val = 0.0224

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.0320, p-val = 0.8651

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    1.9455  1.1277   1.7252   5  0.1451  -0.9533
IFN       -0.0645  0.3608  -0.1788   5  0.8651  -0.9920
          ci.ub
intrcpt  4.8442
IFN      0.8630

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ℹ Note

Not significant results

6.7.2.6. IL_4

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -8.6353   18.7000   23.2706   23.1083   31.2706

tau^2 (estimated amount of residual heterogeneity):     0.9003 (SE = 0.6946)
tau (square root of estimated tau^2 value):             0.9488
I^2 (residual heterogeneity / unaccounted variability): 87.21%
H^2 (unaccounted variability / sampling variability):   7.82
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 13.3188, p-val = 0.0206

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.0105, p-val = 0.9225

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    1.7842  0.4532   3.9366   5  0.0110   0.6191
IL_4      -0.0606  0.5927  -0.1023   5  0.9225  -1.5842
          ci.ub
intrcpt  2.9494  *
IL_4     1.4629
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant results

6.7.2.7. IL_6

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -8.2134   16.9190   22.4269   23.0185   27.2269

tau^2 (estimated amount of residual heterogeneity):     0.3862 (SE = 0.3095)
tau (square root of estimated tau^2 value):             0.6214
I^2 (residual heterogeneity / unaccounted variability): 75.64%
H^2 (unaccounted variability / sampling variability):   4.10
R^2 (amount of heterogeneity accounted for):            27.80%

Test for Residual Heterogeneity:
QE(df = 7) = 12.0828, p-val = 0.0979

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 3.2234, p-val = 0.1157

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    1.4941  0.2761  5.4119   7  0.0010   0.8413
IL_6       0.1709  0.0952  1.7954   7  0.1157  -0.0542
          ci.ub
intrcpt  2.1469  ***
IL_6     0.3960

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant results

6.7.2.8. IL_10

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)
```

```
  logLik  deviance       AIC       BIC      AICc  
-10.1243   20.7407   26.2486   26.8403   31.0486  

tau^2 (estimated amount of residual heterogeneity):     0.6430 (SE = 0.4456)
tau (square root of estimated tau^2 value):             0.8019
I^2 (residual heterogeneity / unaccounted variability): 83.76%
H^2 (unaccounted variability / sampling variability):   6.16
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 7) = 17.0898, p-val = 0.0168

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 0.0868, p-val = 0.7768

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    1.7784  0.3578  4.9701   7  0.0016   0.9323
IL_10      0.0714  0.2424  0.2946   7  0.7768  -0.5017
          ci.ub
intrcpt  2.6245  **
IL_10    0.6445

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant results

6.7.2.9. IL_2

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc  
 -8.6267   18.6829   23.2535   23.0912   31.2535  

tau^2 (estimated amount of residual heterogeneity):     0.9041 (SE = 0.6941)
tau (square root of estimated tau^2 value):             0.9509
I^2 (residual heterogeneity / unaccounted variability): 87.75%
H^2 (unaccounted variability / sampling variability):   8.16
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 11.8338, p-val = 0.0371

Test of Moderators (coefficient 2):
```

```
F(df1 = 1, df2 = 5) = 0.0501, p-val = 0.8317

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    1.6362  0.6316  2.5905   5  0.0488   0.0126
IL_2       0.0662  0.2956  0.2239   5  0.8317  -0.6937
          ci.ub
intrcpt  3.2599  *
IL_2     0.8261

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant results

6.8. Multivariate Models

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     1.4592 (SE = 1.3845)
tau (square root of estimated tau^2 value):             1.2080
I^2 (residual heterogeneity / unaccounted variability): 90.49%
H^2 (unaccounted variability / sampling variability):   10.52
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 3) = 11.4927, p-val = 0.0093

Test of Moderators (coefficients 2:4):
F(df1 = 3, df2 = 3) = 0.0705, p-val = 0.9719

Model Results:

          estimate      se     tval  df    pval    ci.lb
intrcpt     1.7103  2.6663   0.6415   3  0.5669  -6.7751
IFN        -0.1372  0.8430  -0.1627   3  0.8811  -2.8200
IL_2        0.5964  1.9677   0.3031   3  0.7816  -5.6658
IFN:IL_2   -0.0966  0.4584  -0.2108   3  0.8466  -1.5554
            ci.ub
intrcpt   10.1958
IFN        2.5457
IL_2       6.8586
IFN:IL_2   1.3621

---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

Part II.

Th2

7. Th2 — Immediate Post-Exercise

7.1. Data Import

The base analysis uses the pre-computed Hedges' g file for Th2 at $r = 0.5$ (`effect_size/Th2/pre_post/Th2_pre_post_r_050` Sensitivity analyses can substitute the $r = 0.3$ / 0.4 / 0.6 / 0.7 files in the same folder. The pipeline column names are mapped to the analysis conventions (`VE_g`, `Sex`, `Age`, `BMI`) by `load_es()`.

7.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes (g, SE_g) using the **Hedges–Olkin (HE)** ^{2} estimator and the **Knapp–Hartung** small-sample correction (`hakn = TRUE`). **Kostrzewa-Nowak et al. (2019)** is removed before pooling and the model re-fitted; `metafor::rma()` cross-validates.

7.2.1. Full Model {meta}

```
Number of studies: k = 18

                              SMD            95% CI    t
Random effects model (HK) 0.3214 [-0.0871; 0.7298] 1.66
Prediction interval              [-1.9144; 2.5571]
                           p-value
Random effects model (HK)  0.1152
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.0569 [0.1238; 1.5639]; tau = 1.0281 [0.3518; 1.2506]
 I^2 = 71.9% [54.9%; 82.5%]; H = 1.89 [1.49; 2.39]

Test of heterogeneity:
     Q d.f.  p-value
 60.51   17 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 17)
- Prediction interval based on t-distribution (df = 17)
```

7.2.2. Full Model {metafor}

```
Random-Effects Model (k = 18; tau^2 estimator: HE)

tau^2 (estimated amount of total heterogeneity): 1.0569 (SE = 0.4699)
tau (square root of estimated tau^2 value):      1.0281
I^2 (total heterogeneity / total variability):   93.23%
H^2 (total variability / sampling variability):  14.78

Test for Heterogeneity:
Q(df = 17) = 60.5052, p-val < .0001

Model Results:

estimate      se    tval  df    pval    ci.lb   ci.ub
  0.3214  0.1936  1.6601  17  0.1152  -0.0871  0.7298

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

7.2.3. Adapted Model {meta}

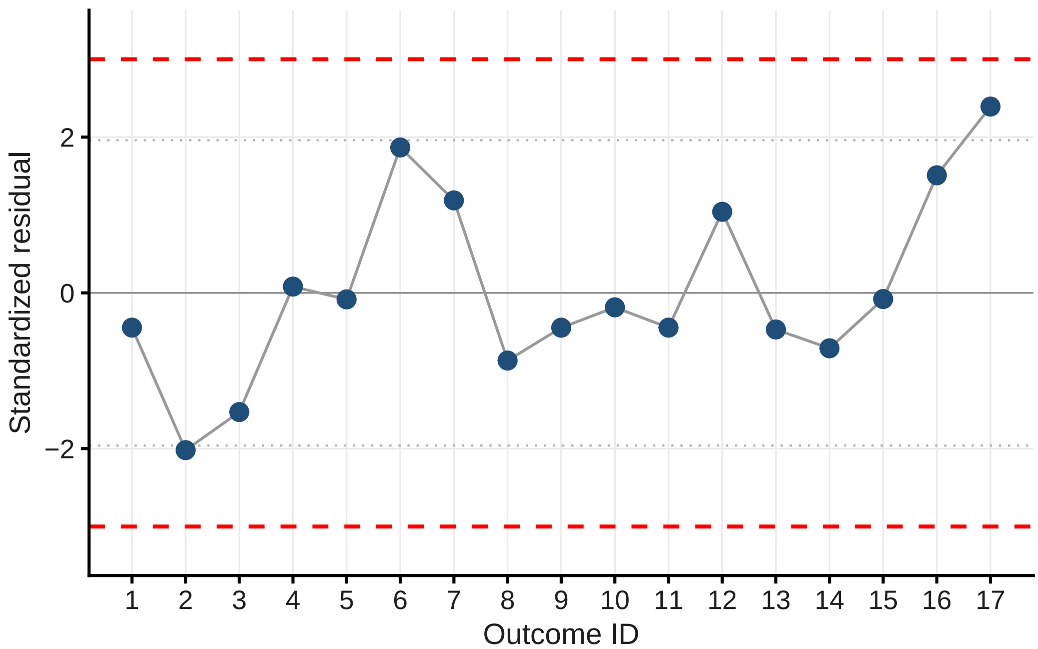
```
Number of studies: k = 17

                             SMD            95% CI    t
Random effects model (HK) 0.3791 [ 0.0865; 0.6718] 2.75
Prediction interval              [-0.7743; 1.5326]
                          p-value
Random effects model (HK)  0.0143
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.2749 [0.0825; 0.7087]; tau = 0.5243 [0.2873; 0.8418]
 I^2 = 70.2% [51.2%; 81.8%]; H = 1.83 [1.43; 2.35]

Test of heterogeneity:
     Q d.f.  p-value
 53.73   16 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)
```

7.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m3$seTE^2`). Studies with $|z| > 3$ are flagged as outliers; $|z| > 1.96$ warrants inspection.

Table 7.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	0.137	-0.446
Kostrzewa-Nowak & Nowak (2018)a	-0.529	-2.019
Kostrzewa-Nowak & Nowak (2018)b	-0.330	-1.530
Kostrzewa-Nowak & Nowak (2020a)a	0.410	0.080
Kostrzewa-Nowak & Nowak (2020a)b	0.334	-0.084
Kostrzewa-Nowak & Nowak (2020b)a	1.272	1.867
Kostrzewa-Nowak & Nowak (2020b)b	0.941	1.187
Kostrzewa-Nowak & Nowak (2020b)c	-0.036	-0.869
Kostrzewa-Nowak & Nowak (2020b)d	0.158	-0.447
Kostrzewa-Nowak & Nowak (2020b)e	0.282	-0.186
Steensberg et al. (2001)	0.144	-0.446
Tsubakihara et al. (2012)	0.861	1.041
Lancaster et al. (2005)a	0.124	-0.470
Lancaster et al. (2005)b	0.000	-0.712
Lancaster et al. (2005)c	0.329	-0.079
Harbaum et al. (2016)a	1.096	1.510
Harbaum et al. (2016)b	1.829	2.392



> Caution
>
> One outcome has standardized residuals around -2 (Kostrzewa-Nowak et al. (2019)).

7.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones (= 0.10, 0.05, 0.01). The dfbetas diagnostic from `metafor::rma.mv()` identifies **influential cases** — distinct from statistical outliers and not necessarily excluded.

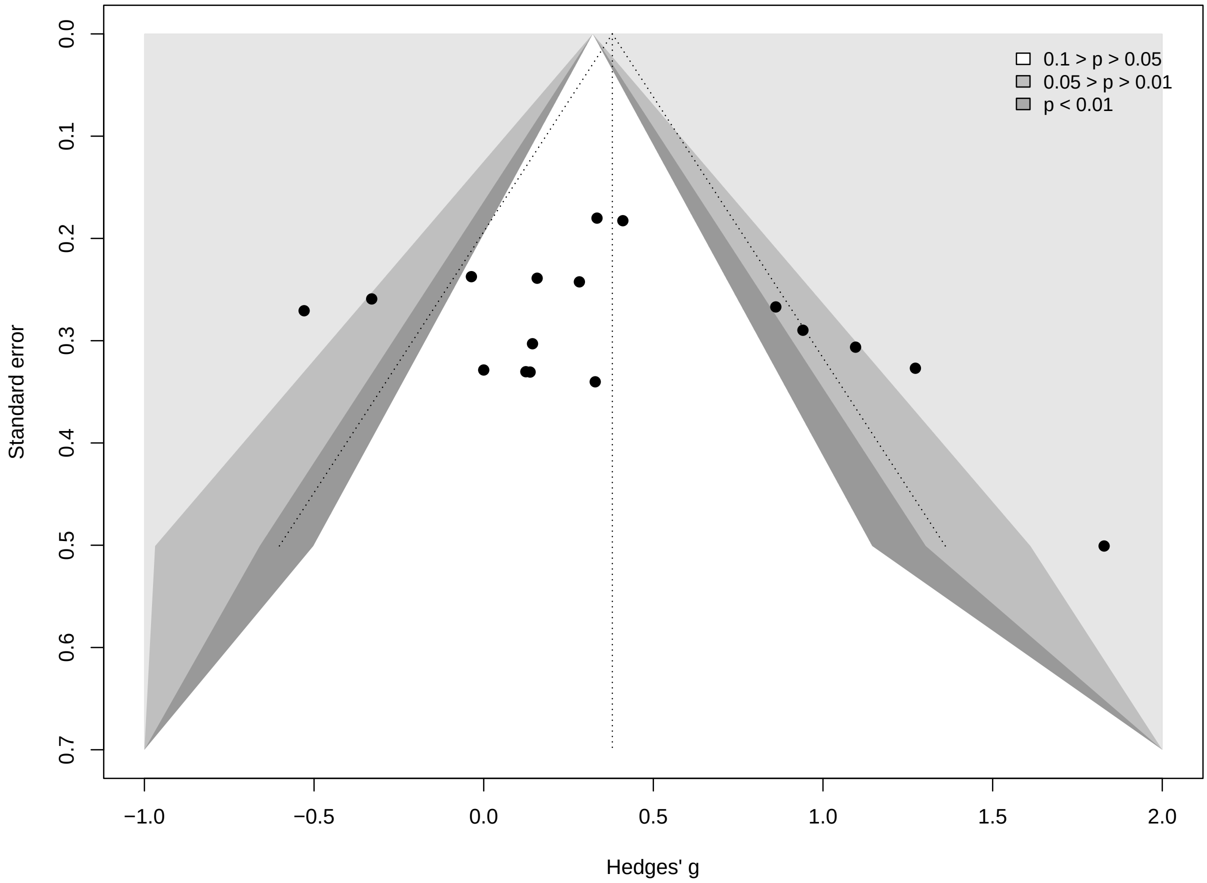


Table 7.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Ibfelt et al. (2002)	0.137	-0.1178026325
Kostrzewa-Nowak & Nowak (2018)a	-0.529	-0.8034118886
Kostrzewa-Nowak & Nowak (2018)b	-0.330	-0.6766632755
Kostrzewa-Nowak & Nowak (2020a)a	0.410	0.1818284116
Kostrzewa-Nowak & Nowak (2020a)b	0.334	0.0132850848
Kostrzewa-Nowak & Nowak (2020b)a	1.272	0.5951183620
Kostrzewa-Nowak & Nowak (2020b)b	0.941	0.4971720392
Kostrzewa-Nowak & Nowak (2020b)c	-0.036	-0.4523407882
Kostrzewa-Nowak & Nowak (2020b)d	0.158	-0.2088082167
Kostrzewa-Nowak & Nowak (2020b)e	0.282	-0.0547346426
Steensberg et al. (2001)	0.144	-0.1362231954
Tsubakihara et al. (2012)	0.861	0.5140815471
Lancaster et al. (2005)a	0.124	-0.1258018657
Lancaster et al. (2005)b	0.000	-0.2045859006

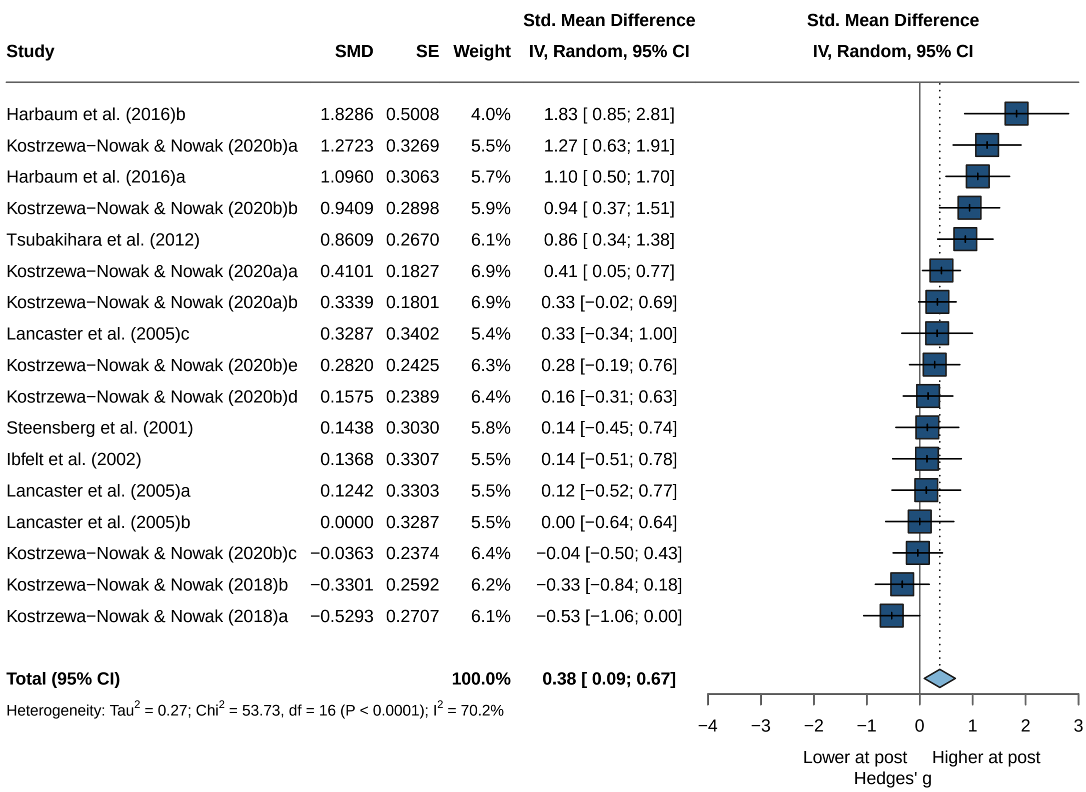
Table 7.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Lancaster et al. (2005)c	0.329	0.0003012762
Harbaum et al. (2016)a	1.096	0.5544242443
Harbaum et al. (2016)b	1.829	0.3938945044

Tip

No influential case.

7.5. Forest Plot



7.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed, $p < .10$). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially missing studies.

```
Linear regression test of funnel plot asymmetry

Test result: t = 1.17, df = 15, p-value = 0.2595
Bias estimate: 2.2167 (SE = 1.8916)

Details:
- multiplicative residual heterogeneity variance (tau^2 = 3.2814)
- predictor: standard error
- weight:    inverse variance
- reference: Egger et al. (1997), BMJ
```

> **Note**
>
> No asymmetry -> no trim-and-fill

7.7. Moderator Analyses

7.7.1. Risk of Bias

Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic (Q_b) indicates whether subgroup means differ.

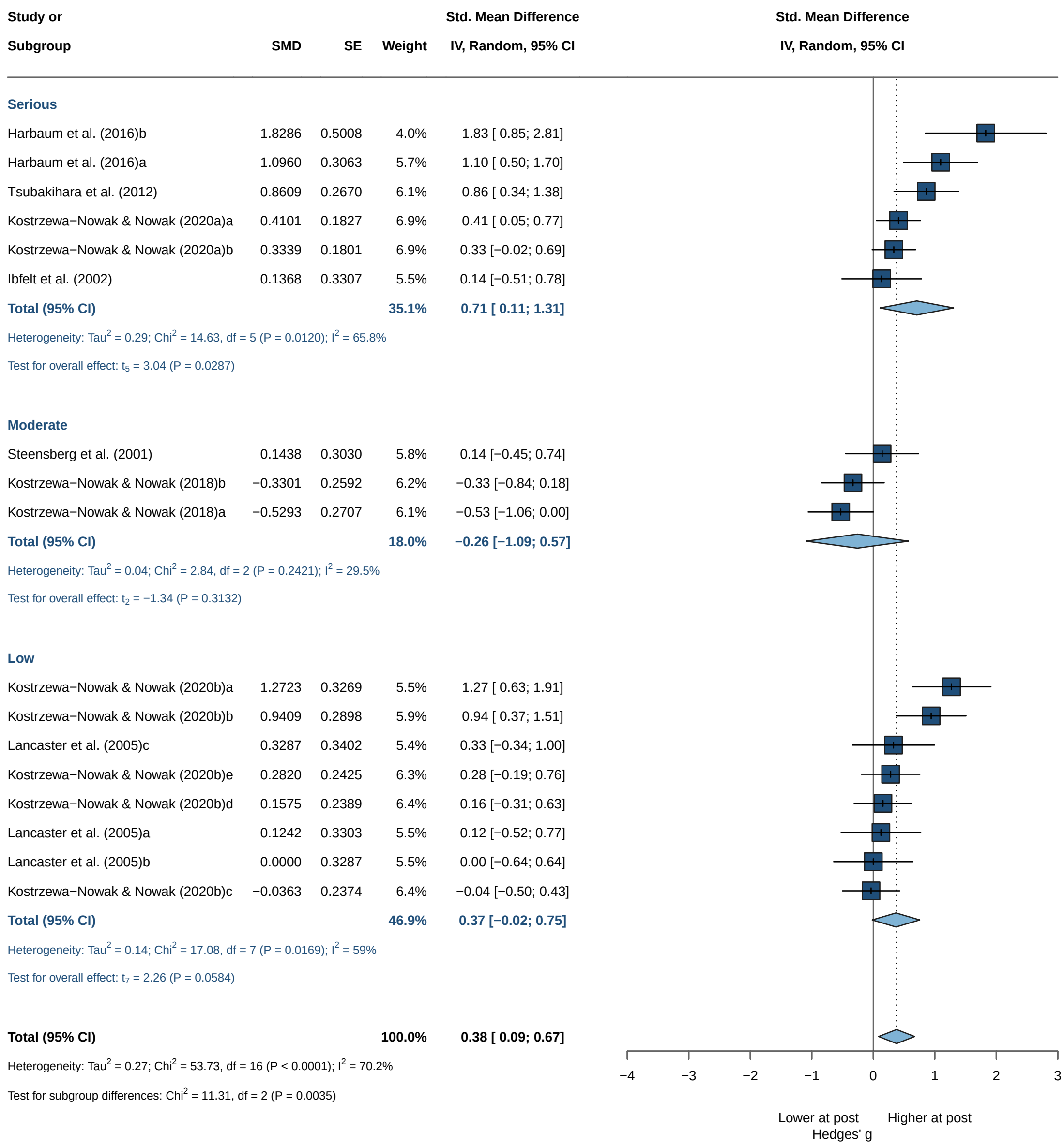
7.7.1.1. Model

```
Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.2749 [0.0825; 0.7087]; tau = 0.5243 [0.2873; 0.8418]
 I^2 = 70.2% [51.2%; 81.8%]; H = 1.83 [1.43; 2.35]

Test of heterogeneity:
     Q d.f.  p-value
 53.73   16 < 0.0001

Results for subgroups (random effects model (HK)):
                 k     SMD            95% CI  tau^2    tau
RoB = Serious    6  0.7080 [ 0.1099; 1.3062] 0.2926 0.5409
RoB = Moderate   3 -0.2584 [-1.0905; 0.5736] 0.0421 0.2053
RoB = Low        8  0.3689 [-0.0172; 0.7550] 0.1352 0.3677
                   Q   I^2
RoB = Serious   14.63 65.8%
RoB = Moderate   2.84 29.5%
RoB = Low       17.08 59.0%

Test for subgroup differences (random effects model (HK)):
                   Q d.f. p-value
```

```
Between groups 11.31    2  0.0035

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **! Important**
>
> Significant results

7.7.1.2. Forrest Plot



7.7.1.3. Head-to-Head

Head-to-head RoB comparisons. Because the overall RoB subgroup test is significant, the individual rating levels are compared pairwise.

7.7.1.4. Moderate vs Serious

```
Number of studies: k = 9

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.5204 [0.2137; 1.9229]; tau = 0.7214 [0.4622; 1.3867]
 I^2 = 98.0% [97.3%; 98.6%]; H = 7.11 [6.08; 8.31]

Test of heterogeneity:
      Q d.f.  p-value
 403.91    8 < 0.0001

Results for subgroups (random effects model (HK)):
                 k     SMD            95% CI  tau^2    tau
RoB = Serious    6  0.7540 [ 0.1167; 1.3914] 0.3759 0.6131
RoB = Moderate   3 -0.2420 [-1.0969; 0.6129] 0.1135 0.3368
                    Q   I^2
RoB = Serious  133.26 96.2%
RoB = Moderate  33.35 94.0%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 9.83    1  0.0017

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **! Important**
>
> Significant

7.7.1.5. Moderate vs Low

```
Number of studies: k = 11

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.2563 [0.1206; 0.8046]; tau = 0.5063 [0.3473; 0.8970]
 I^2 = 97.2% [96.1%; 97.9%]; H = 5.96 [5.10; 6.97]

Test of heterogeneity:
      Q d.f.  p-value
 355.37   10 < 0.0001

Results for subgroups (random effects model (HK)):
                 k     SMD            95% CI  tau^2    tau
```

```
RoB = Moderate   3 -0.2420 [-1.0969; 0.6129] 0.1135 0.3368
RoB = Low        8  0.3817 [-0.0113; 0.7748] 0.2141 0.4627
                    Q    I^2
RoB = Moderate  33.35 94.0%
RoB = Low      193.09 96.4%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 5.80    1  0.0161

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **! Important**
>
> Significant

7.7.1.6. Low vs Serious

```
Number of studies: k = 14

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.2999 [0.1367; 0.7776]; tau = 0.5476 [0.3697; 0.8818]
 I^2 = 96.3% [95.0%; 97.2%]; H = 5.18 [4.46; 6.03]

Test of heterogeneity:
      Q d.f.  p-value
 349.48   13 < 0.0001

Results for subgroups (random effects model (HK)):
                k    SMD            95% CI  tau^2    tau
RoB = Serious   6 0.7540 [ 0.1167; 1.3914] 0.3759 0.6131
RoB = Low       8 0.3817 [-0.0113; 0.7748] 0.2141 0.4627
                   Q    I^2
RoB = Serious 133.26 96.2%
RoB = Low     193.09 96.4%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 1.56    1  0.2123

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
```

```
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **Note**
>
> Not significant

7.7.2. Sex

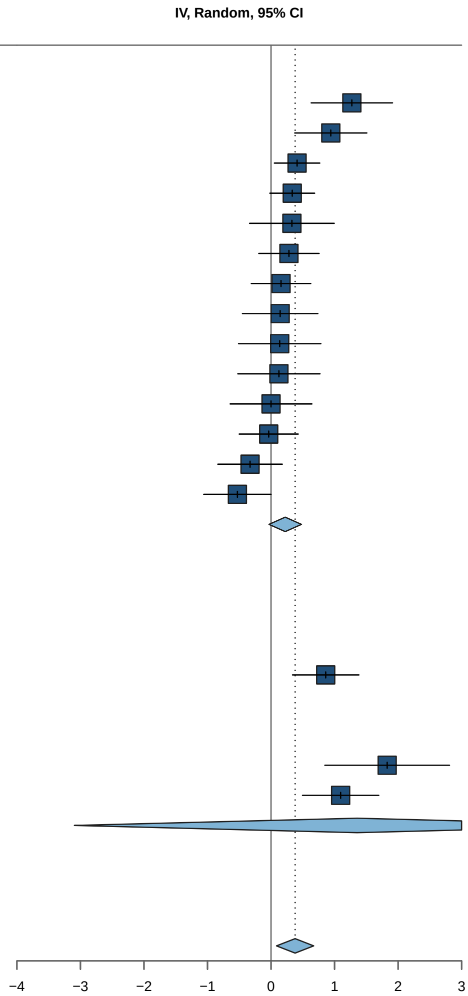
7.7.2.1. Model

```
Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.2749 [0.0825; 0.7087]; tau = 0.5243 [0.2873; 0.8418]
 I^2 = 70.2% [51.2%; 81.8%]; H = 1.83 [1.43; 2.35]

Test of heterogeneity:
     Q d.f.  p-value
 53.73   16 < 0.0001

Results for subgroups (random effects model (HK)):
               k    SMD            95% CI  tau^2    tau
sex = male    14 0.2235 [-0.0321; 0.4792] 0.1284 0.3584
sex = female   1 0.8609 [ 0.3377; 1.3842]     --     --
sex = mixed    2 1.3552 [-3.0956; 5.8059] 0.0960 0.3099
                 Q   I^2
sex = male   32.25 59.7%
sex = female  0.00    --
sex = mixed   1.56 35.8%

Test for subgroup differences (random effects model (HK)):
                   Q d.f. p-value
Between groups 12.61    2  0.0018

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **Important**
>
> Significant results but 15 vs 1 vs 2

7.7.2.2. Forrest Plot

Subgroup	SMD	SE	Weight	IV, Random, 95% CI
male				
Kostrzewa−Nowak & Nowak (2020b)a	1.2723	0.3269	5.5%	1.27 [0.63; 1.91]
Kostrzewa−Nowak & Nowak (2020b)b	0.9409	0.2898	5.9%	0.94 [0.37; 1.51]
Kostrzewa−Nowak & Nowak (2020a)a	0.4101	0.1827	6.9%	0.41 [0.05; 0.77]
Kostrzewa−Nowak & Nowak (2020a)b	0.3339	0.1801	6.9%	0.33 [−0.02; 0.69]
Lancaster et al. (2005)c	0.3287	0.3402	5.4%	0.33 [−0.34; 1.00]
Kostrzewa−Nowak & Nowak (2020b)e	0.2820	0.2425	6.3%	0.28 [−0.19; 0.76]
Kostrzewa−Nowak & Nowak (2020b)d	0.1575	0.2389	6.4%	0.16 [−0.31; 0.63]
Steensberg et al. (2001)	0.1438	0.3030	5.8%	0.14 [−0.45; 0.74]
Ibfelt et al. (2002)	0.1368	0.3307	5.5%	0.14 [−0.51; 0.78]
Lancaster et al. (2005)a	0.1242	0.3303	5.5%	0.12 [−0.52; 0.77]
Lancaster et al. (2005)b	0.0000	0.3287	5.5%	0.00 [−0.64; 0.64]
Kostrzewa−Nowak & Nowak (2020b)c	−0.0363	0.2374	6.4%	−0.04 [−0.50; 0.43]
Kostrzewa−Nowak & Nowak (2018)b	−0.3301	0.2592	6.2%	−0.33 [−0.84; 0.18]
Kostrzewa−Nowak & Nowak (2018)a	−0.5293	0.2707	6.1%	−0.53 [−1.06; 0.00]
Total (95% CI)			**84.1%**	**0.22 [−0.03; 0.48]**
Heterogeneity: $Tau^2 = 0.13$; $Chi^2 = 32.25$, df = 13 (P = 0.0022); $I^2 = 59.7\%$				
Test for overall effect: $t_{13} = 1.89$ (P = 0.0814)				
female				
Tsubakihara et al. (2012)	0.8609	0.2670	6.1%	0.86 [0.34; 1.38]
mixed				
Harbaum et al. (2016)b	1.8286	0.5008	4.0%	1.83 [0.85; 2.81]
Harbaum et al. (2016)a	1.0960	0.3063	5.7%	1.10 [0.50; 1.70]
Total (95% CI)			**9.7%**	**1.36 [−3.10; 5.81]**
Heterogeneity: $Tau^2 = 0.10$; $Chi^2 = 1.56$, df = 1 (P = 0.2121); $I^2 = 35.8\%$				
Test for overall effect: $t_1 = 3.87$ (P = 0.1610)				
Total (95% CI)			**100.0%**	**0.38 [0.09; 0.67]**
Heterogeneity: $Tau^2 = 0.27$; $Chi^2 = 53.73$, df = 16 (P < 0.0001); $I^2 = 70.2\%$				



7.7.3. Intensity

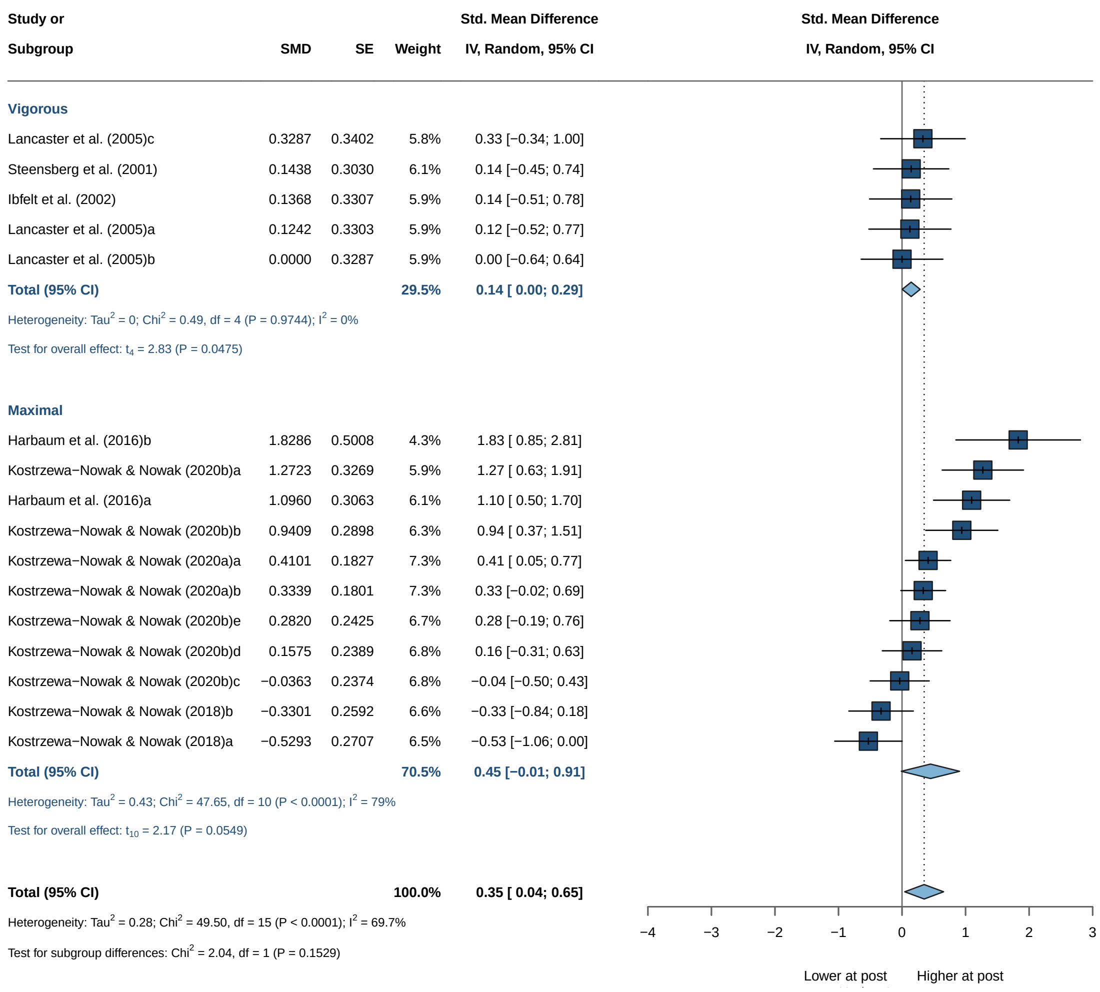
7.7.3.1. Model

```
Number of studies: k = 16

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.2839 [0.0797; 0.7558]; tau = 0.5328 [0.2822; 0.8694]
 I^2 = 69.7% [49.4%; 81.8%]; H = 1.82 [1.41; 2.35]

Test of heterogeneity:
     Q d.f.  p-value
 49.50   15 < 0.0001

Results for subgroups (random effects model (HK)):
                       k    SMD            95% CI  tau^2
intensity = Vigorous   5 0.1443 [ 0.0026; 0.2860]      0
intensity = Maximal   11 0.4478 [-0.0115; 0.9070] 0.4316
                        tau     Q   I^2
intensity = Vigorous      0  0.49  0.0%
```

```
intensity = Maximal  0.6570 47.65 79.0%

Test for subgroup differences (random effects model (HK)):
                    Q d.f. p-value
Between groups 2.04    1  0.1529

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **Note**
>
> Not significant

7.7.3.2. Forrest Plot



7.7.4. Staining

```
Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.2749 [0.0825; 0.7087]; tau = 0.5243 [0.2873; 0.8418]
 I^2 = 70.2% [51.2%; 81.8%]; H = 1.83 [1.43; 2.35]

Test of heterogeneity:
     Q d.f.  p-value
 53.73   16 < 0.0001

Results for subgroups (random effects model (HK)):
                   k    SMD            95% CI  tau^2    tau
staining = intra  14 0.2235 [-0.0321; 0.4792] 0.1284 0.3584
staining = extra   3 1.1539 [ 0.0415; 2.2663] 0.1161 0.3407
                     Q   I^2
staining = intra 32.25 59.7%
staining = extra  2.91 31.3%

Test for subgroup differences (random effects model (HK)):
                   Q d.f. p-value
Between groups 10.71    1  0.0011

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **! Important**
>
> Significant results

7.7.5. Continuous Moderators (Meta-Regression)

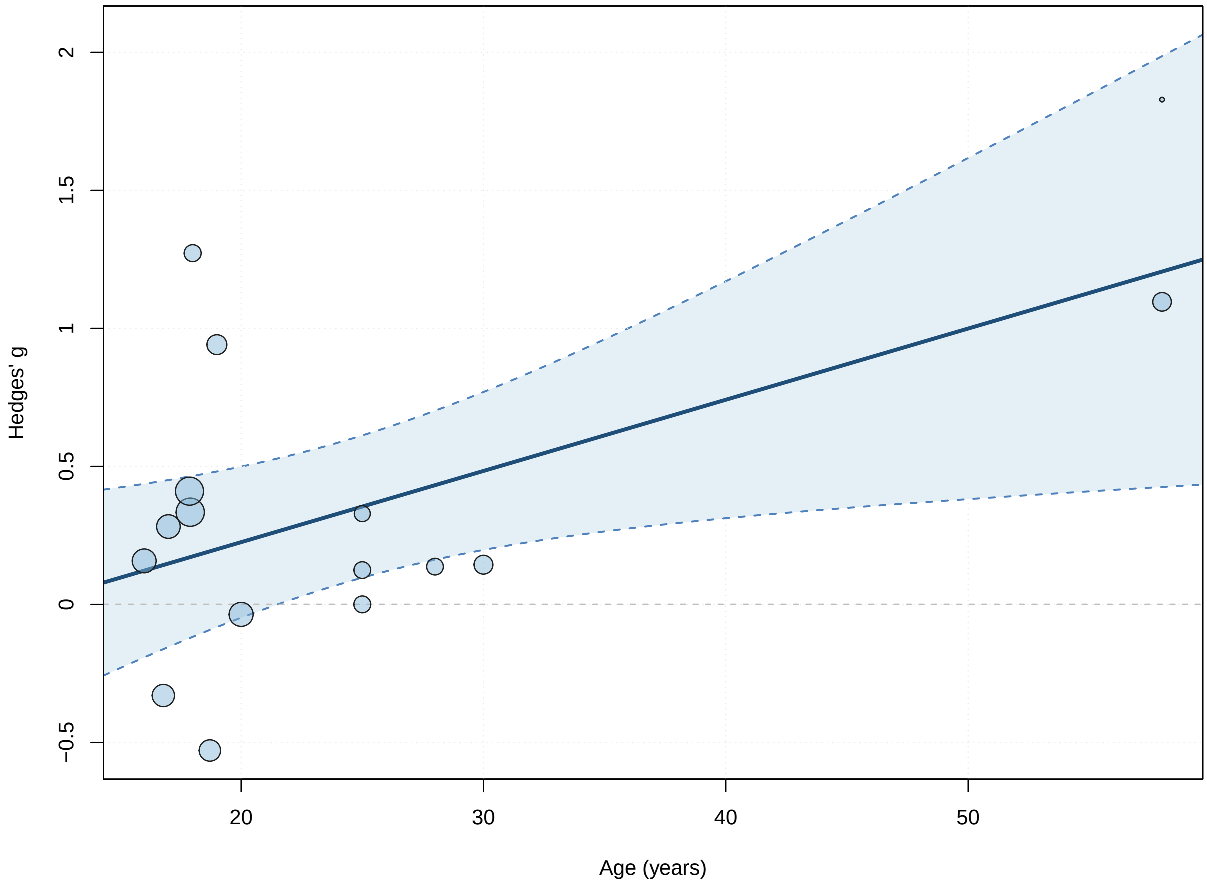
Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity (^{2}); bubble plots show the fitted line with 95% CI.

7.7.5.1. Age

```
Mixed-Effects Model (k = 16; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-10.1238   31.2182   26.2475   28.5653   28.2475
```

```
tau^2 (estimated amount of residual heterogeneity):     0.1586 (SE = 0.0936)
tau (square root of estimated tau^2 value):             0.3982
I^2 (residual heterogeneity / unaccounted variability): 68.97%
H^2 (unaccounted variability / sampling variability):   3.22
R^2 (amount of heterogeneity accounted for):            44.13%

Test for Residual Heterogeneity:
QE(df = 14) = 37.4912, p-val = 0.0006

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 14) = 6.4163, p-val = 0.0239

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt   -0.2905  0.2747  -1.0574  14  0.3082  -0.8798
age        0.0258  0.0102   2.5330  14  0.0239   0.0040
          ci.ub
intrcpt  0.2988
age      0.0476  *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **!** Important
>
> Significant



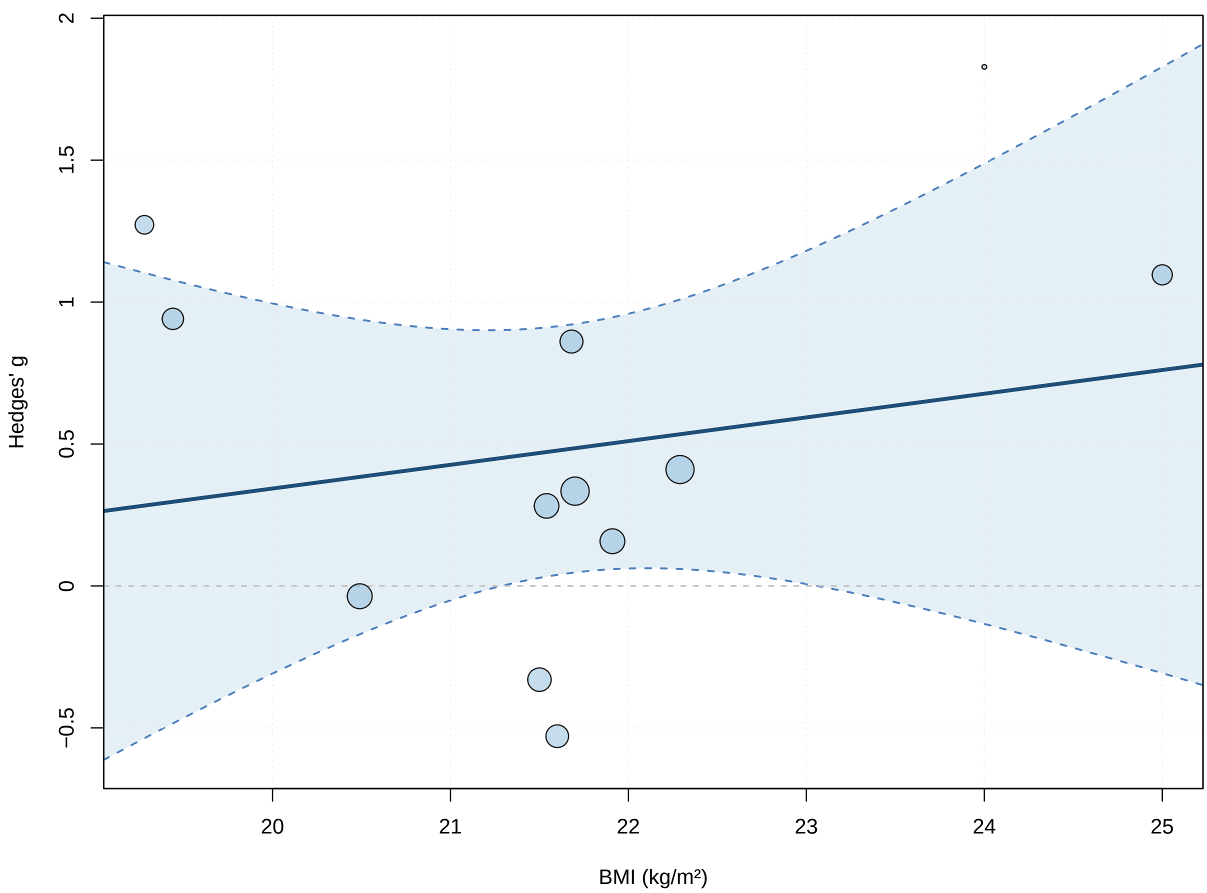
7.7.5.2. BMI

```
Mixed-Effects Model (k = 12; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-11.5110   32.7869   29.0221   30.4768   32.0221

tau^2 (estimated amount of residual heterogeneity):     0.4184 (SE = 0.2231)
tau (square root of estimated tau^2 value):             0.6468
I^2 (residual heterogeneity / unaccounted variability): 87.04%
H^2 (unaccounted variability / sampling variability):   7.71
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 10) = 50.4537, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 10) = 0.4072, p-val = 0.5377

Model Results:
```

```
         estimate      se     tval  df    pval    ci.lb
intrcpt   -1.3265  2.8404  -0.4670  10  0.6505  -7.6554
bmi        0.0835  0.1308   0.6381  10  0.5377  -0.2080
          ci.ub
intrcpt  5.0023
bmi      0.3750

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant



7.7.5.3. Duration

```
Mixed-Effects Model (k = 15; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-13.4344   35.4902   32.8688   34.9929   35.0506
```

```
tau^2 (estimated amount of residual heterogeneity):     0.3252 (SE = 0.1663)
tau (square root of estimated tau^2 value):             0.5702
I^2 (residual heterogeneity / unaccounted variability): 79.32%
H^2 (unaccounted variability / sampling variability):   4.84
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 13) = 53.3069, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 13) = 0.5003, p-val = 0.4919

Model Results:

          estimate      se     tval  df    pval    ci.lb
intrcpt     0.5088  0.2399   2.1212  13  0.0537  -0.0094
duration   -0.0020  0.0028  -0.7073  13  0.4919  -0.0079
           ci.ub
intrcpt   1.0270  .
duration  0.0040

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

7.7.5.4. Dose

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-12.4175   32.6531   30.8349   32.7521   33.2349

tau^2 (estimated amount of residual heterogeneity):     0.3219 (SE = 0.1725)
tau (square root of estimated tau^2 value):             0.5673
I^2 (residual heterogeneity / unaccounted variability): 78.97%
H^2 (unaccounted variability / sampling variability):   4.76
R^2 (amount of heterogeneity accounted for):            3.33%

Test for Residual Heterogeneity:
QE(df = 12) = 47.3298, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 12) = 1.0162, p-val = 0.3333

Model Results:
```

```
         estimate      se     tval  df    pval    ci.lb
intrcpt    0.5419  0.2552   2.1233  12  0.0552  -0.0142
dose      -0.0005  0.0005  -1.0080  12  0.3333  -0.0015
          ci.ub
intrcpt  1.0980  .
dose     0.0005

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

7.7.5.5. IFN

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -3.8105   14.8706   13.6209   13.4587   21.6209

tau^2 (estimated amount of residual heterogeneity):     0.1830 (SE = 0.1575)
tau (square root of estimated tau^2 value):             0.4277
I^2 (residual heterogeneity / unaccounted variability): 75.67%
H^2 (unaccounted variability / sampling variability):   4.11
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 15.1720, p-val = 0.0097

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.1169, p-val = 0.7463

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.7247  0.8225   0.8811   5  0.4186  -1.3896
IFN       -0.1511  0.4421  -0.3419   5  0.7463  -1.2876
          ci.ub
intrcpt  2.8389
IFN      0.9853

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

7.7.5.6. IL_4

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -7.3445   22.0122   20.6889   21.2806   25.4889

tau^2 (estimated amount of residual heterogeneity):     0.3176 (SE = 0.2182)
tau (square root of estimated tau^2 value):             0.5636
I^2 (residual heterogeneity / unaccounted variability): 82.89%
H^2 (unaccounted variability / sampling variability):   5.84
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 7) = 27.0308, p-val = 0.0003

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 0.3834, p-val = 0.5554

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.5539  0.2387  2.3206   7  0.0534  -0.0105
IL_4       0.1255  0.2026  0.6192   7  0.5554  -0.3536
          ci.ub
intrcpt  1.1183  .
IL_4     0.6045

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

7.7.5.7. IL_6

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-11.6641   33.3733   29.3282   31.2454   31.7282

tau^2 (estimated amount of residual heterogeneity):     0.3033 (SE = 0.1621)
```

```
tau (square root of estimated tau^2 value):             0.5507
I^2 (residual heterogeneity / unaccounted variability): 81.28%
H^2 (unaccounted variability / sampling variability):   5.34
R^2 (amount of heterogeneity accounted for):            8.75%

Test for Residual Heterogeneity:
QE(df = 12) = 41.9420, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 12) = 1.9487, p-val = 0.1880

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.5182  0.1867   2.7751  12  0.0168   0.1114
IL_6      -0.1428  0.1023  -1.3960  12  0.1880  -0.3656
          ci.ub
intrcpt  0.9251  *
IL_6     0.0801

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

7.7.5.8. IL_10

```
Mixed-Effects Model (k = 11; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-10.9326   30.8268   27.8652   29.0589   31.2937

tau^2 (estimated amount of residual heterogeneity):     0.4665 (SE = 0.2620)
tau (square root of estimated tau^2 value):             0.6830
I^2 (residual heterogeneity / unaccounted variability): 87.31%
H^2 (unaccounted variability / sampling variability):   7.88
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 9) = 46.2625, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 9) = 0.3720, p-val = 0.5570

Model Results:
```

```
         estimate      se     tval  df    pval    ci.lb
intrcpt    0.4614  0.2143   2.1530   9  0.0597  -0.0234
IL_10     -0.3677  0.6029  -0.6099   9  0.5570  -1.7315
          ci.ub
intrcpt  0.9461  .
IL_10    0.9961

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

7.7.5.9. IL_2

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -3.9212   15.0921   13.8425   13.6802   21.8425

tau^2 (estimated amount of residual heterogeneity):     0.1931 (SE = 0.1621)
tau (square root of estimated tau^2 value):             0.4394
I^2 (residual heterogeneity / unaccounted variability): 78.08%
H^2 (unaccounted variability / sampling variability):   4.56
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 15.3694, p-val = 0.0089

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.0000, p-val = 0.9986

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.4500  0.7121  0.6319   5  0.5552  -1.3805
IL_2       0.0007  0.3851  0.0019   5  0.9986  -0.9892
          ci.ub
intrcpt  2.2806
IL_2     0.9906

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

7.8. Multivariate Models

7.8.1. Intensity * Duration

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.2579 (SE = 0.1595)
tau (square root of estimated tau^2 value):             0.5079
I^2 (residual heterogeneity / unaccounted variability): 75.59%
H^2 (unaccounted variability / sampling variability):   4.10
R^2 (amount of heterogeneity accounted for):            22.54%

Test for Residual Heterogeneity:
QE(df = 10) = 34.9688, p-val = 0.0001

Test of Moderators (coefficients 2:4):
F(df1 = 3, df2 = 10) = 1.6514, p-val = 0.2396

Model Results:

                            estimate      se     tval  df
intrcpt                       2.1195  0.8475   2.5008  10
intensityVigorous            -1.9999  1.7640  -1.1337  10
duration                     -0.0926  0.0456  -2.0281  10
intensityVigorous:duration    0.0928  0.0470   1.9753  10
                              pval    ci.lb   ci.ub
intrcpt                     0.0314   0.2311  4.0079  *
intensityVigorous           0.2834  -5.9302  1.9305
duration                    0.0700  -0.1943  0.0091  .
intensityVigorous:duration  0.0765  -0.0119  0.1974  .

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

7.8.2. IL_10 * IL_4

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.3863 (SE = 0.3079)
tau (square root of estimated tau^2 value):             0.6215
I^2 (residual heterogeneity / unaccounted variability): 83.42%
H^2 (unaccounted variability / sampling variability):   6.03
R^2 (amount of heterogeneity accounted for):            0.00%
```

```
Test for Residual Heterogeneity:
QE(df = 5) = 23.7108, p-val = 0.0002

Test of Moderators (coefficients 2:4):
F(df1 = 3, df2 = 5) = 0.3848, p-val = 0.7691

Model Results:

            estimate      se     tval  df    pval    ci.lb
intrcpt       0.3992  0.3244   1.2307   5  0.2732  -0.4346
IL_10        -0.5850  0.7088  -0.8253   5  0.4468  -2.4069
IL_4          0.1813  0.2347   0.7725   5  0.4747  -0.4220
IL_10:IL_4    1.1806  1.5451   0.7641   5  0.4793  -2.7911
             ci.ub
intrcpt     1.2330
IL_10       1.2370
IL_4        0.7845
IL_10:IL_4  5.1524

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

8. Th2 — ~2 h Post-Exercise

8.1. Data Import

The ~2 h window (`effect_size/Th2/pre_post+2h/Th2_pre_post+2h_r_050.csv`) is reported for the main analysis, outlier check, funnel, and forest only; the limited data do not support moderator modelling. No study is excluded a priori.

8.2. Primary Meta-Analysis

8.2.1. {meta}

```
Number of studies: k = 5

                             SMD            95% CI    t
Random effects model (HK) 0.0744 [-0.0938; 0.2426] 1.23
Prediction interval              [-0.3288; 0.4775]
                          p-value
Random effects model (HK)  0.2869
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0 [0.0000; 0.0495]; tau = 0 [0.0000; 0.2226]
 I^2 = 0.0% [0.0%; 79.2%]; H = 1.00 [1.00; 2.19]

Test of heterogeneity:
    Q d.f. p-value
 0.70    4  0.9518

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 4)
- Prediction interval based on t-distribution (df = 4)
```

8.2.2. {metafor}

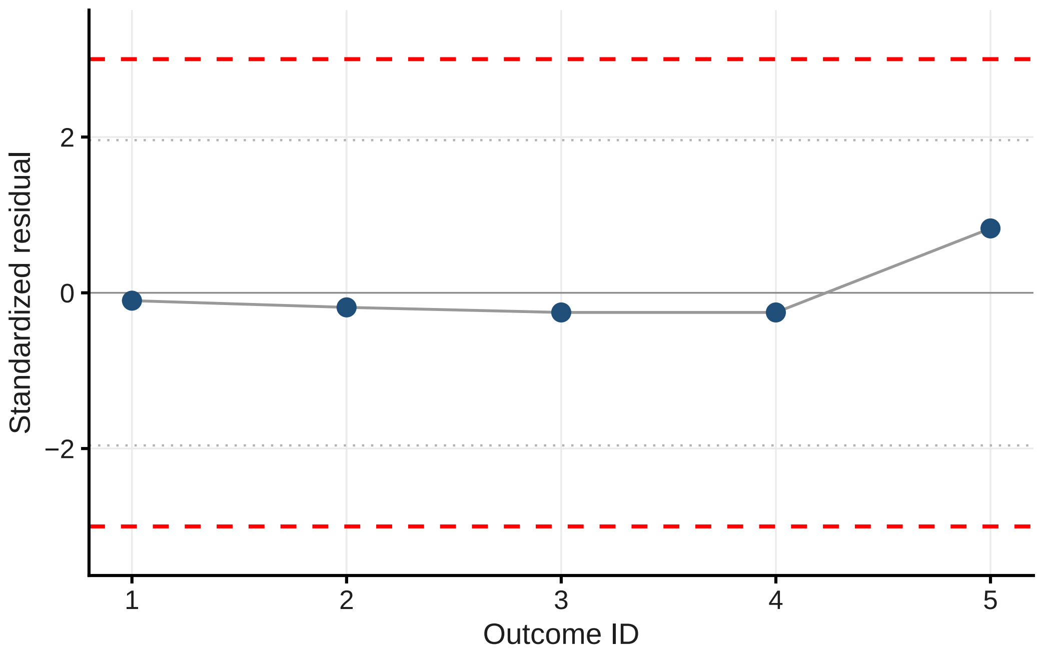
```
Random-Effects Model (k = 5; tau^2 estimator: HE)
```

```
tau^2 (estimated amount of total heterogeneity): 0 (SE = 0.0752)
tau (square root of estimated tau^2 value):      0
I^2 (total heterogeneity / total variability):   0.00%
H^2 (total variability / sampling variability):  1.00

Test for Heterogeneity:
Q(df = 4) = 0.6963, p-val = 0.9518

Model Results:

estimate      se    tval  df    pval    ci.lb   ci.ub
  0.0744  0.0606  1.2277   4  0.2869  -0.0938  0.2426

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

8.3. Outlier Detection

Table 8.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	0.045	-0.101
Steensberg et al. (2001)	0.025	-0.187
Lancaster et al. (2005)a	0.000	-0.252
Lancaster et al. (2005)b	0.000	-0.252
Lancaster et al. (2005)c	0.329	0.827



Note

One outcome has standardized residuals around -2 (Kostrzewa-Nowak et al. (2019)).

8.4. Funnel Plot & Influential Cases

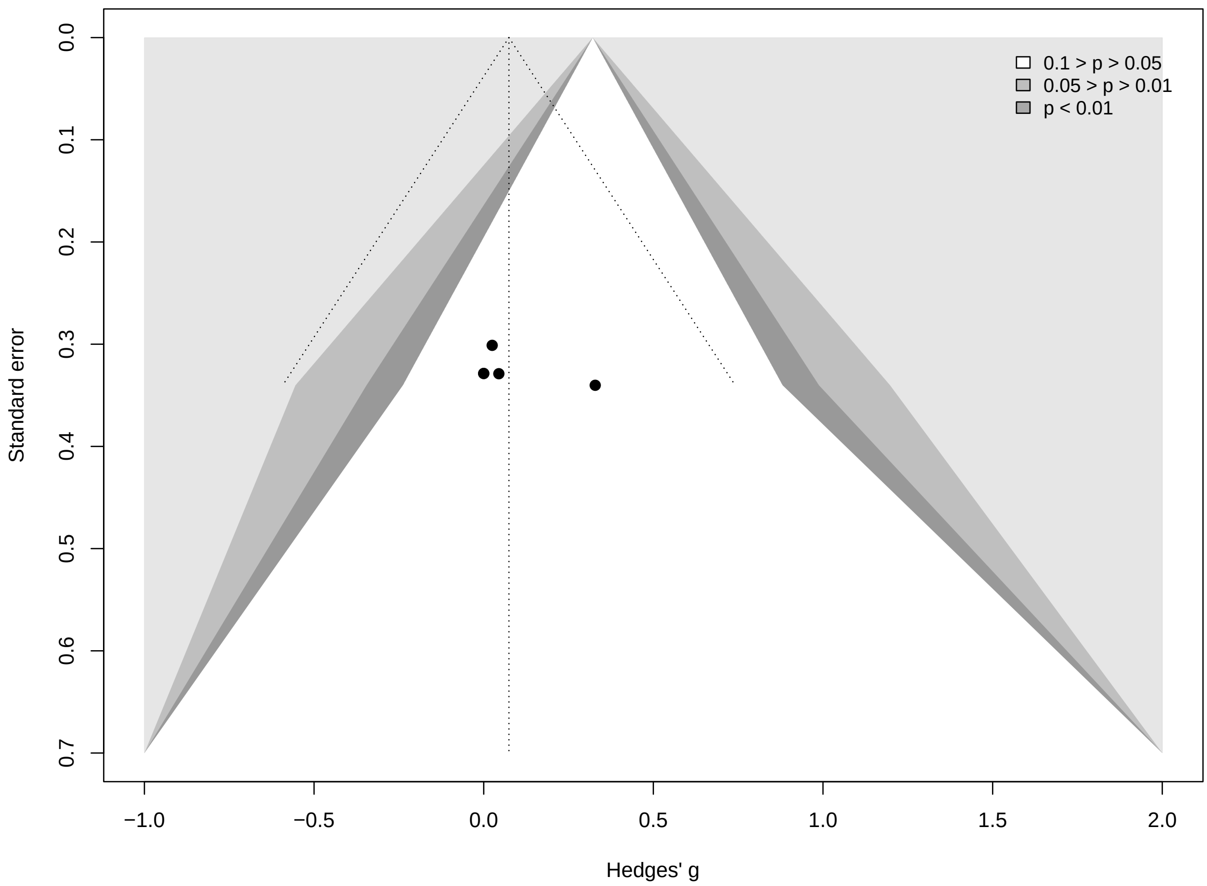


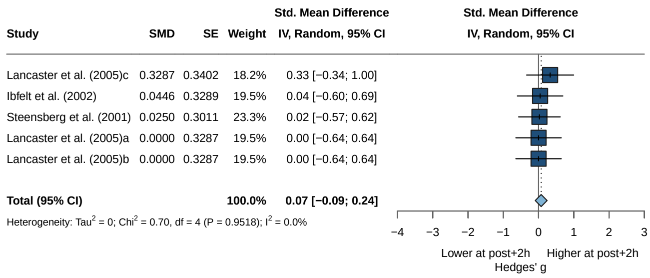
Table 8.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Ibfelt et al. (2002)	0.045	-0.04966493
Steensberg et al. (2001)	0.025	-0.10307622
Lancaster et al. (2005)a	0.000	-0.12421976
Lancaster et al. (2005)b	0.000	-0.12421976
Lancaster et al. (2005)c	0.329	0.39010883

Note

No influential case.

8.5. Forest Plot



8.6. Asymmetry Test

Note: the number of studies in this window is too small to reliably test for small-study effects (minimum $k = 10$). No Egger's test or trim-and-fill is performed.

9. Th2 — 17 h Post-Exercise

9.1. Data Import

9.2. Primary Meta-Analysis

Kostrzewa-Nowak et al. (2019) is removed before pooling and the model re-fitted.

9.2.1. {meta}

```
Number of studies: k = 11

                              SMD            95% CI    t
Random effects model (HK) 0.2468 [-0.4359; 0.9295] 0.81
Prediction interval              [-2.8310; 3.3247]
                          p-value
Random effects model (HK)  0.4392
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.7347 [0.0804; 4.1148]; tau = 1.3171 [0.2835; 2.0285]
 I^2 = 72.2% [48.9%; 84.9%]; H = 1.90 [1.40; 2.57]

Test of heterogeneity:
     Q d.f.  p-value
 35.94   10 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 10)
- Prediction interval based on t-distribution (df = 10)
```

9.2.2. {metafor}

```
Random-Effects Model (k = 11; tau^2 estimator: HE)

tau^2 (estimated amount of total heterogeneity): 1.7347 (SE = 0.9828)
tau (square root of estimated tau^2 value):      1.3171
```

```
I^2 (total heterogeneity / total variability):   96.28%
H^2 (total variability / sampling variability):  26.85

Test for Heterogeneity:
Q(df = 10) = 35.9359, p-val < .0001

Model Results:

estimate      se    tval  df    pval    ci.lb   ci.ub
  0.2468  0.3064  0.8056  10  0.4392  -0.4359  0.9295

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

9.2.3. Adapted Model {meta}

```
Number of studies: k = 10

                             SMD            95% CI    t
Random effects model (HK) 0.4009 [ 0.0582; 0.7436] 2.65
Prediction interval              [-0.6056; 1.4073]
                          p-value
Random effects model (HK)  0.0266
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.1740 [0.0376; 0.7263]; tau = 0.4171 [0.1939; 0.8522]
 I^2 = 69.1% [40.3%; 83.9%]; H = 1.80 [1.29; 2.50]

Test of heterogeneity:
     Q d.f. p-value
 29.08    9  0.0006

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 9)
- Prediction interval based on t-distribution (df = 9)
```

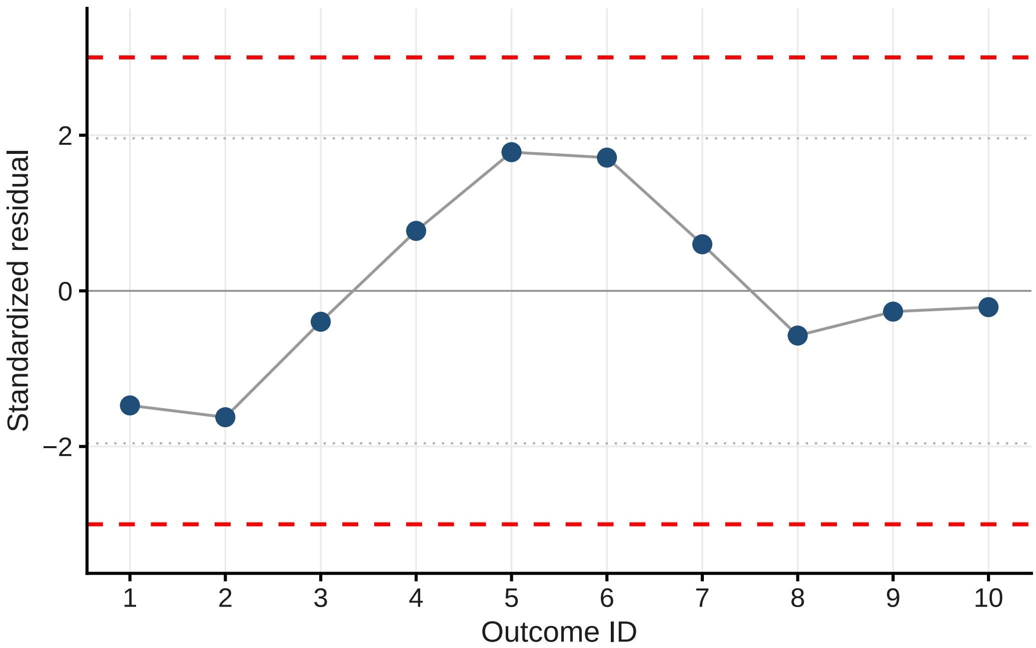
9.3. Outlier Detection

Table 9.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2018)a	-0.199	-1.472
Kostrzewa-Nowak & Nowak (2018)b	-0.250	-1.624

Table 9.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2020a)a	0.233	-0.397
Kostrzewa-Nowak & Nowak (2020a)b	0.707	0.771
Kostrzewa-Nowak & Nowak (2020b)a	1.162	1.782
Kostrzewa-Nowak & Nowak (2020b)b	1.133	1.712
Kostrzewa-Nowak & Nowak (2020b)c	0.664	0.599
Kostrzewa-Nowak & Nowak (2020b)d	0.150	-0.574
Kostrzewa-Nowak & Nowak (2020b)e	0.282	-0.267
Steensberg et al. (2001)	0.300	-0.210



Note

No outlier.

9.4. Funnel Plot & Influential Cases

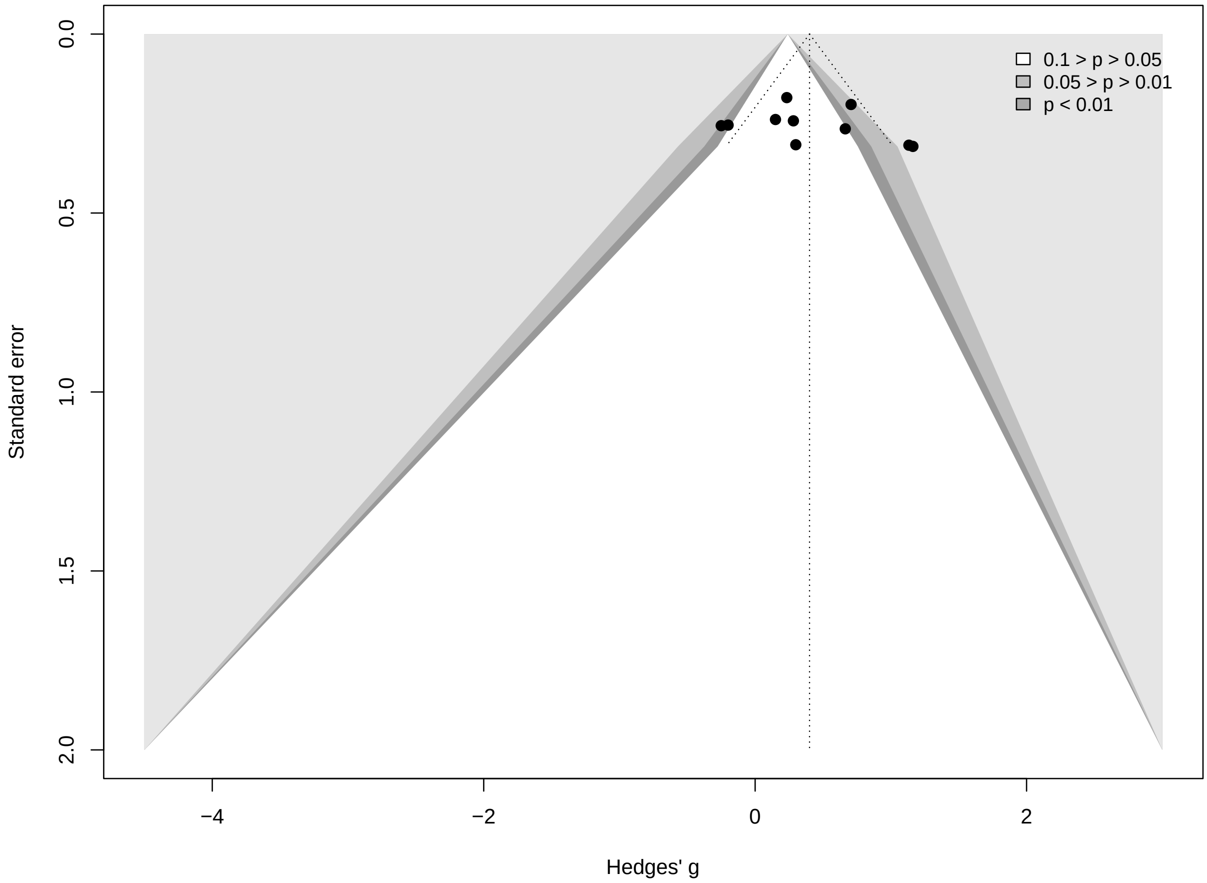


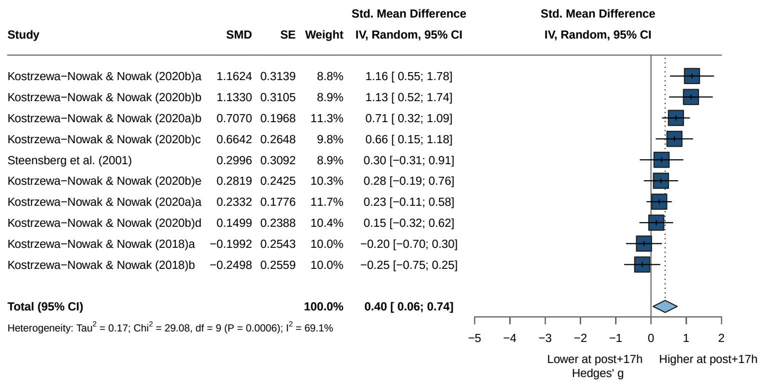
Table 9.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Kostrzewa-Nowak & Nowak (2018)a	-0.199	-0.7487610
Kostrzewa-Nowak & Nowak (2018)b	-0.250	-0.8040583
Kostrzewa-Nowak & Nowak (2020a)a	0.233	-0.4154425
Kostrzewa-Nowak & Nowak (2020a)b	0.707	0.7911110
Kostrzewa-Nowak & Nowak (2020b)a	1.162	0.6594062
Kostrzewa-Nowak & Nowak (2020b)b	1.133	0.6497221
Kostrzewa-Nowak & Nowak (2020b)c	0.664	0.3526656
Kostrzewa-Nowak & Nowak (2020b)d	0.150	-0.3340006
Kostrzewa-Nowak & Nowak (2020b)e	0.282	-0.1299465
Steensberg et al. (2001)	0.300	-0.0613988

Note

No influential case.

9.5. Forest Plot



9.6. Asymmetry Test

```
Linear regression test of funnel plot asymmetry

Test result: t = 0.76, df = 8, p-value = 0.4679
Bias estimate: 2.3883 (SE = 3.1343)

Details:
- multiplicative residual heterogeneity variance (tau^2 = 3.3889)
- predictor: standard error
- weight:    inverse variance
- reference: Egger et al. (1997), BMJ
```

Note

No asymmetry -> no trim-and-fill

9.7. Moderator Analyses

9.7.1. Risk of Bias

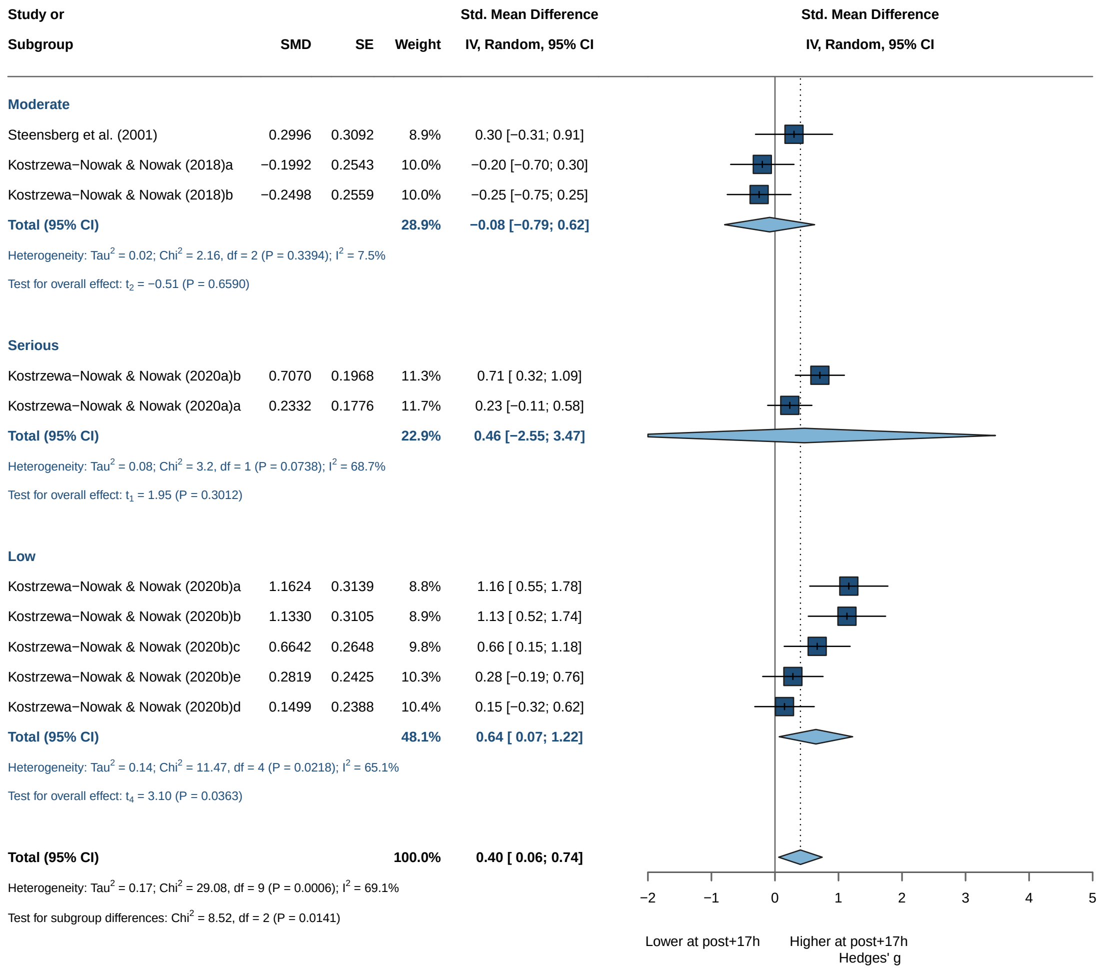
9.7.1.1. Model

```
Number of studies: k = 10

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.1740 [0.0376; 0.7263]; tau = 0.4171 [0.1939; 0.8522]
 I^2 = 69.1% [40.3%; 83.9%]; H = 1.80 [1.29; 2.50]

Test of heterogeneity:
     Q d.f. p-value
 29.08    9  0.0006

Results for subgroups (random effects model (HK)):
                 k     SMD            95% CI  tau^2    tau
RoB = Moderate   3 -0.0845 [-0.7928; 0.6239] 0.0169 0.1301
RoB = Serious    2  0.4625 [-2.5460; 3.4710] 0.0771 0.2777
RoB = Low        5  0.6449 [ 0.0667; 1.2230] 0.1432 0.3785
                   Q   I^2
RoB = Moderate  2.16  7.5%
RoB = Serious   3.20 68.7%
RoB = Low      11.47 65.1%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 8.52    2  0.0141

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

Note

Not significant results

9.7.1.2. Forrest Plot



Sex: all retained arms are male, so no sex subgroup analysis is performed.

9.7.2. Intensity

```
Number of studies: k = 10

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.1740 [0.0376; 0.7263]; tau = 0.4171 [0.1939; 0.8522]
 I^2 = 69.1% [40.3%; 83.9%]; H = 1.80 [1.29; 2.50]

Test of heterogeneity:
     Q d.f. p-value
 29.08    9  0.0006

Results for subgroups (random effects model (HK)):
                       k    SMD            95% CI  tau^2
intensity = Maximal    9 0.4129 [ 0.0242; 0.8016] 0.2053
intensity = Vigorous   1 0.2996 [-0.3065; 0.9057]     --
                        tau     Q   I^2
intensity = Maximal  0.4531 29.02 72.4%
intensity = Vigorous     --  0.00    --
```

```
Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 0.10    1  0.7477

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

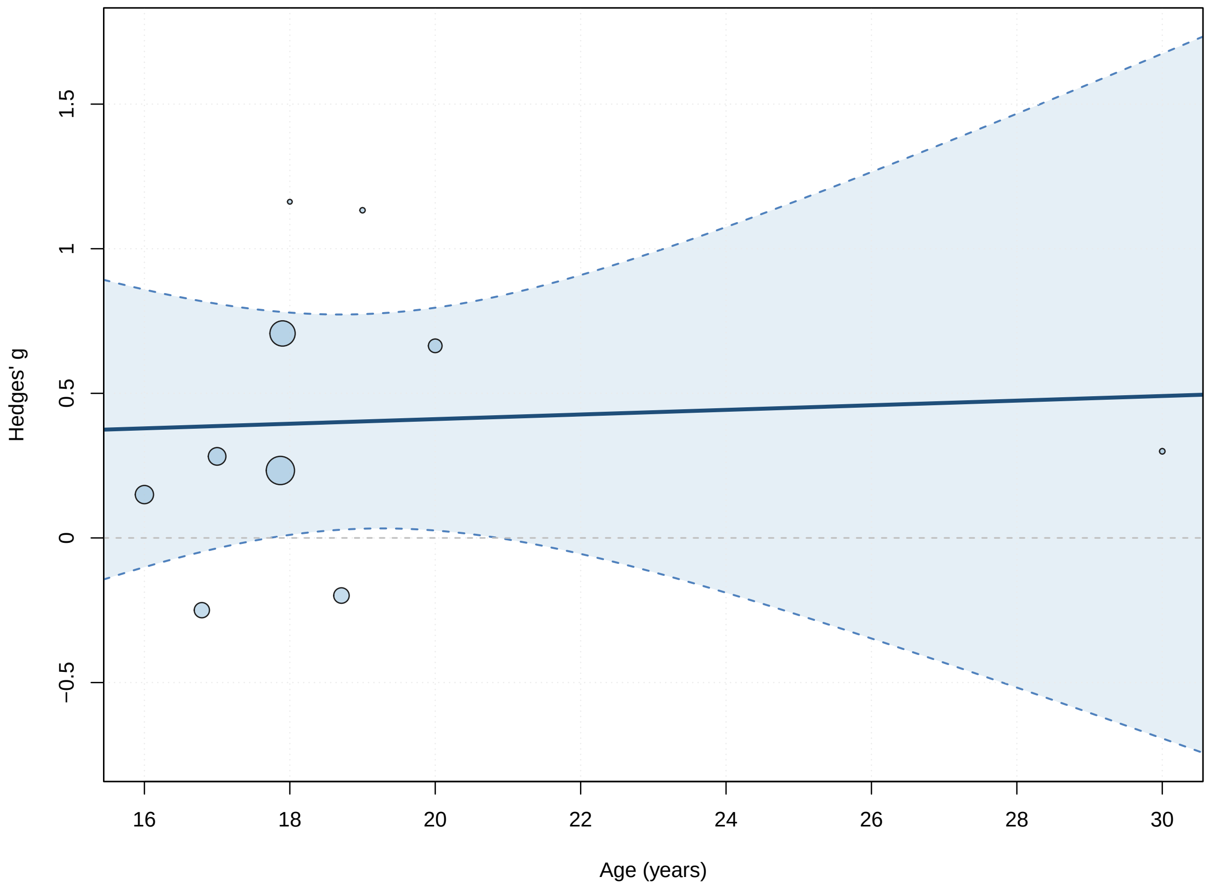
> **Note**
>
> Not significant but 10 vs 1

9.7.3. Continuous Moderators (Meta-Regression)

9.7.3.1. Age

```
Mixed-Effects Model (k = 10; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -6.5029   22.1591   19.0057   19.9135   23.0057

tau^2 (estimated amount of residual heterogeneity):     0.2063 (SE = 0.1360)
tau (square root of estimated tau^2 value):             0.4542
I^2 (residual heterogeneity / unaccounted variability): 77.93%
H^2 (unaccounted variability / sampling variability):   4.53
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 8) = 28.7965, p-val = 0.0003

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 8) = 0.0326, p-val = 0.8613

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.2511  0.8554  0.2936   8  0.7766  -1.7214
age        0.0080  0.0443  0.1805   8  0.8613  -0.0941
          ci.ub
intrcpt  2.2236
age      0.1100

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

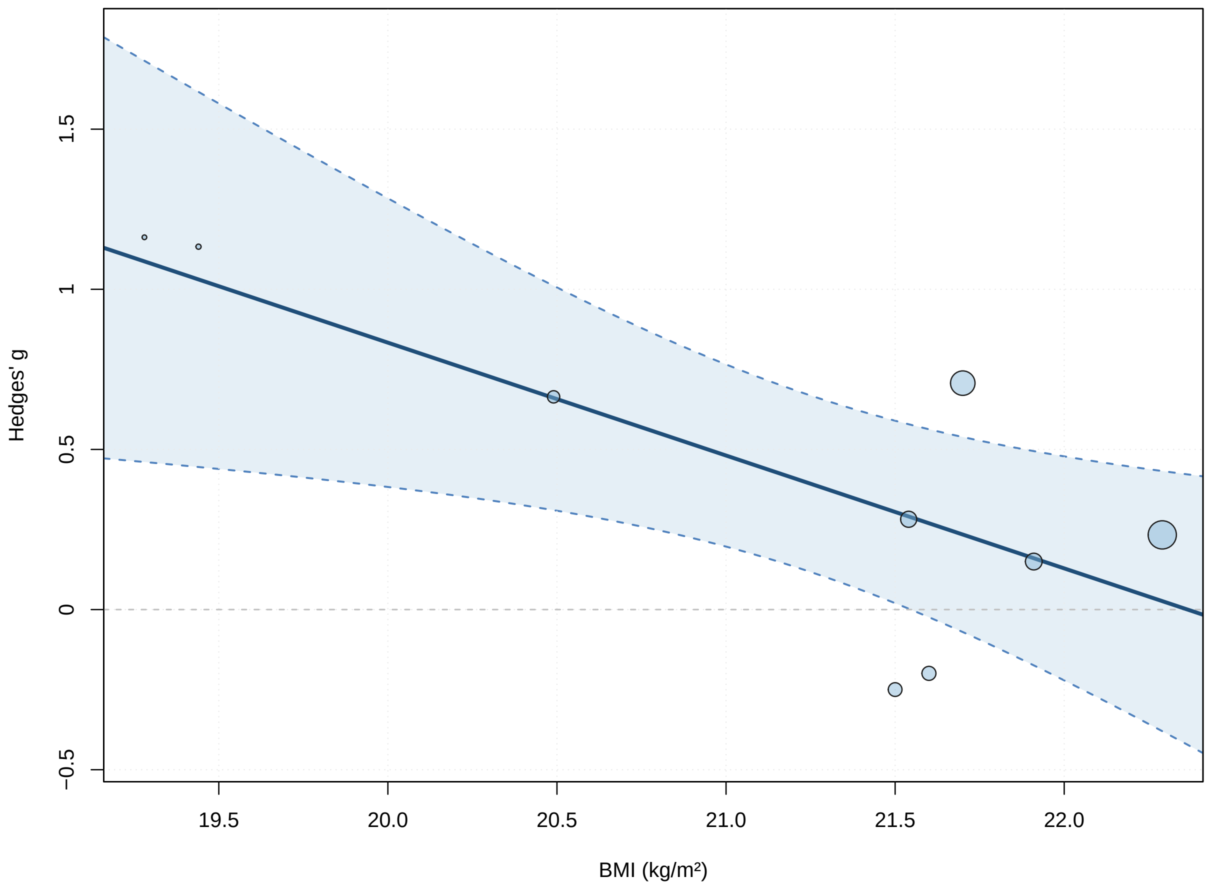


9.7.3.2. BMI

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -2.3155   13.2751   10.6311   11.2227   15.4311

tau^2 (estimated amount of residual heterogeneity):     0.0517 (SE = 0.0618)
tau (square root of estimated tau^2 value):             0.2273
I^2 (residual heterogeneity / unaccounted variability): 47.38%
H^2 (unaccounted variability / sampling variability):   1.90
R^2 (amount of heterogeneity accounted for):            74.84%

Test for Residual Heterogeneity:
QE(df = 7) = 15.1650, p-val = 0.0339

Test of Moderators (coefficient 2):
```

```
F(df1 = 1, df2 = 7) = 8.4875, p-val = 0.0226

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    7.8813  2.5727   3.0635   7  0.0182   1.7980
bmi       -0.3524  0.1210  -2.9133   7  0.0226  -0.6384
           ci.ub
intrcpt  13.9647  *
bmi      -0.0664  *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **! Important**
>
> Significant



9.7.3.3. Duration

```
Mixed-Effects Model (k = 8; tau^2 estimator: HE)
```

```
  logLik  deviance       AIC       BIC      AICc
 -5.9984   18.1177   17.9969   18.2352   23.9969

tau^2 (estimated amount of residual heterogeneity):     0.2675 (SE = 0.1968)
tau (square root of estimated tau^2 value):             0.5172
I^2 (residual heterogeneity / unaccounted variability): 79.17%
H^2 (unaccounted variability / sampling variability):   4.80
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 6) = 25.3804, p-val = 0.0003

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 6) = 0.0249, p-val = 0.8798

Model Results:

          estimate      se     tval  df    pval    ci.lb
intrcpt     0.4150  0.2673   1.5525   6  0.1715  -0.2391
duration   -0.0008  0.0049  -0.1578   6  0.8798  -0.0127
           ci.ub
intrcpt   1.0691
duration  0.0112

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

9.7.3.4. Dose

```
Mixed-Effects Model (k = 8; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -5.9984   18.1177   17.9969   18.2352   23.9969

tau^2 (estimated amount of residual heterogeneity):     0.2675 (SE = 0.1968)
tau (square root of estimated tau^2 value):             0.5172
I^2 (residual heterogeneity / unaccounted variability): 79.17%
H^2 (unaccounted variability / sampling variability):   4.80
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 6) = 25.3804, p-val = 0.0003
```

```
Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 6) = 0.0249, p-val = 0.8798

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.4198  0.2876   1.4593   6  0.1948  -0.2841
dose      -0.0001  0.0007  -0.1578   6  0.8798  -0.0019
           ci.ub
intrcpt   1.1236
dose      0.0017

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

9.7.3.5. IFN

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -2.4686   11.7933   10.9373   10.7750   18.9373

tau^2 (estimated amount of residual heterogeneity):     0.1048 (SE = 0.1101)
tau (square root of estimated tau^2 value):             0.3238
I^2 (residual heterogeneity / unaccounted variability): 62.72%
H^2 (unaccounted variability / sampling variability):   2.68
R^2 (amount of heterogeneity accounted for):            5.44%

Test for Residual Heterogeneity:
QE(df = 5) = 10.3245, p-val = 0.0665

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 1.4551, p-val = 0.2817

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.0498  0.4627  0.1077   5  0.9184  -1.1396
IFN        0.1801  0.1493  1.2063   5  0.2817  -0.2037
           ci.ub
intrcpt   1.2392
IFN       0.5640

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

9.7.3.6. IL_4

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -2.0026   10.8611   10.0052    9.8429   18.0052

tau^2 (estimated amount of residual heterogeneity):     0.0779 (SE = 0.0935)
tau (square root of estimated tau^2 value):             0.2790
I^2 (residual heterogeneity / unaccounted variability): 55.08%
H^2 (unaccounted variability / sampling variability):   2.23
R^2 (amount of heterogeneity accounted for):            29.77%

Test for Residual Heterogeneity:
QE(df = 5) = 10.3727, p-val = 0.0653

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 2.1405, p-val = 0.2033

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.7575  0.1881   4.0262   5  0.0101   0.2739
IL_4      -0.3349  0.2289  -1.4630   5  0.2033  -0.9234
          ci.ub
intrcpt  1.2411  *
IL_4     0.2535

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

9.7.3.7. IL_6

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -4.7669   18.1779   15.5338   16.1255   20.3338

tau^2 (estimated amount of residual heterogeneity):     0.1548 (SE = 0.1178)
```

```
tau (square root of estimated tau^2 value):             0.3934
I^2 (residual heterogeneity / unaccounted variability): 72.75%
H^2 (unaccounted variability / sampling variability):   3.67
R^2 (amount of heterogeneity accounted for):            24.62%

Test for Residual Heterogeneity:
QE(df = 7) = 21.5167, p-val = 0.0031

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 2.7235, p-val = 0.1429

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.6354  0.2046   3.1055   7  0.0172   0.1516
IL_6      -0.1053  0.0638  -1.6503   7  0.1429  -0.2562
          ci.ub
intrcpt  1.1193  *
IL_6     0.0456

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> ❗ Important
>
> Significant

9.7.3.8. IL_10

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -5.8893   20.4226   17.7785   18.3702   22.5785

tau^2 (estimated amount of residual heterogeneity):     0.2170 (SE = 0.1510)
tau (square root of estimated tau^2 value):             0.4658
I^2 (residual heterogeneity / unaccounted variability): 79.03%
H^2 (unaccounted variability / sampling variability):   4.77
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 7) = 26.1288, p-val = 0.0005

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 0.6854, p-val = 0.4351

Model Results:
```

```
         estimate      se     tval  df    pval    ci.lb
intrcpt    0.5383  0.2288   2.3529   7  0.0509  -0.0027
IL_10     -0.1244  0.1503  -0.8279   7  0.4351  -0.4798
          ci.ub
intrcpt  1.0793  .
IL_10    0.2309

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

9.7.3.9. IL_2

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -3.2017   13.2593   12.4033   12.2410   20.4033

tau^2 (estimated amount of residual heterogeneity):     0.1367 (SE = 0.1309)
tau (square root of estimated tau^2 value):             0.3697
I^2 (residual heterogeneity / unaccounted variability): 68.35%
H^2 (unaccounted variability / sampling variability):   3.16
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 13.5802, p-val = 0.0185

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.1962, p-val = 0.6763

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.4754  0.2972  1.5994   5  0.1706  -0.2887
IL_2       0.0608  0.1372  0.4430   5  0.6763  -0.2919
          ci.ub
intrcpt  1.2395
IL_2     0.4134

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

9.8. Multivariate Models

```
Mixed-Effects Model (k = 8; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.2675 (SE = 0.1968)
tau (square root of estimated tau^2 value):             0.5172
I^2 (residual heterogeneity / unaccounted variability): 79.17%
H^2 (unaccounted variability / sampling variability):   4.80
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 6) = 25.3804, p-val = 0.0003

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 6) = 0.0249, p-val = 0.8798

Model Results:

                   estimate      se     tval  df    pval
intrcpt              0.3996  0.2176   1.8360   6  0.1160
intensityVigorous   -0.1000  0.6338  -0.1578   6  0.8798
                     ci.lb   ci.ub
intrcpt            -0.1330  0.9322
intensityVigorous  -1.6510  1.4509

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

Part III.

Th1/Th2 Ratio

10. Th1/Th2 Ratio — Immediate Post-Exercise

10.1. Data Import

The base analysis uses the pre-computed Hedges' g file for the Th1/Th2 ratio at $r = 0.5$ (`effect_size/Ratio/pre_post/Ratio`
Sensitivity analyses can substitute the $r = 0.3$ / 0.4 / 0.6 / 0.7 files in the same folder. The pipeline column names are mapped to the analysis conventions (`VE_g`, `Sex`, `Age`, `BMI`) by `load_es()`.

10.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes (g, SE_g) using the **Hedges–Olkin (HE)** ^{2} estimator and the **Knapp–Hartung** small-sample correction (`hakn = TRUE`). **Kostrzewa-Nowak et al. (2019)** is removed before pooling and the model re-fitted; `metafor::rma()` cross-validates.

10.2.1. Full Model {meta}

```
Number of studies: k = 18

                             SMD            95% CI    t
Random effects model (HK) 0.3544 [-0.0597; 0.7685] 1.81
Prediction interval              [-1.7206; 2.4294]
                          p-value
Random effects model (HK)  0.0887
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.9096 [0.2029; 1.5996]; tau = 0.9537 [0.4504; 1.2647]
 I^2 = 80.5% [70.0%; 87.3%]; H = 2.26 [1.83; 2.81]

Test of heterogeneity:
     Q d.f.  p-value
 87.11   17 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 17)
- Prediction interval based on t-distribution (df = 17)
```

10.2.2. Full Model {metafor}

```
Random-Effects Model (k = 18; tau^2 estimator: HE)

tau^2 (estimated amount of total heterogeneity): 0.9096 (SE = 0.4257)
tau (square root of estimated tau^2 value):      0.9537
I^2 (total heterogeneity / total variability):   91.77%
H^2 (total variability / sampling variability):  12.15

Test for Heterogeneity:
Q(df = 17) = 87.1136, p-val < .0001

Model Results:

estimate      se    tval  df    pval    ci.lb   ci.ub
  0.3544  0.1963  1.8055  17  0.0887  -0.0597  0.7685  .

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

10.2.3. Adapted Model {meta}

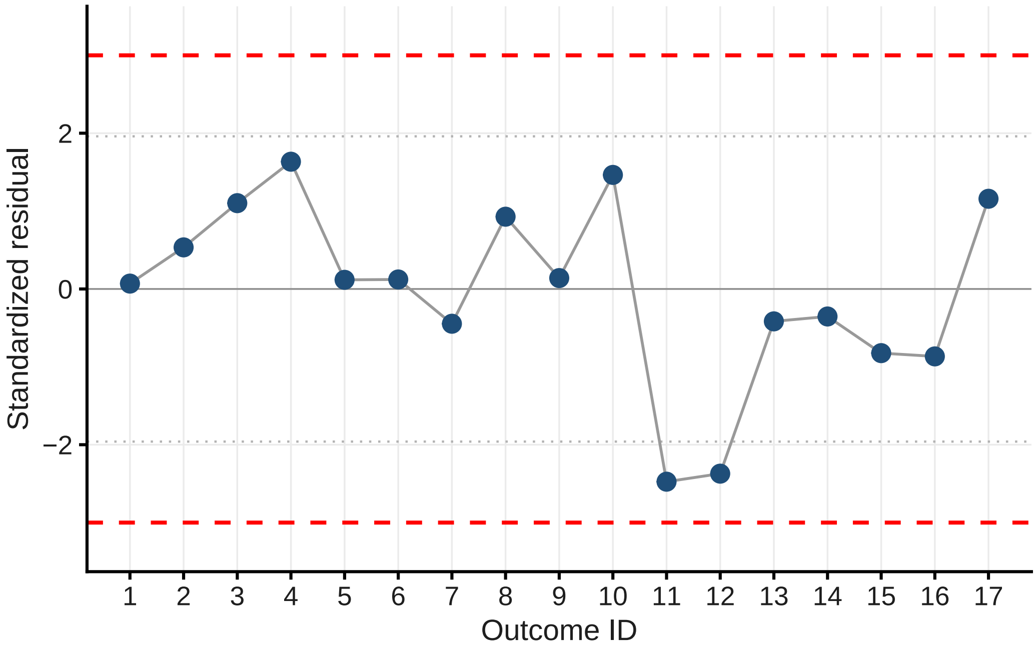
```
Number of studies: k = 17

                             SMD            95% CI    t
Random effects model (HK) 0.3005 [-0.0558; 0.6568] 1.79
Prediction interval              [-1.1139; 1.7149]
                          p-value
Random effects model (HK)  0.0928
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.4156 [0.1703; 1.0546]; tau = 0.6447 [0.4126; 1.0270]
 I^2 = 80.6% [69.8%; 87.5%]; H = 2.27 [1.82; 2.83]

Test of heterogeneity:
     Q d.f.  p-value
 82.37   16 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)
```

10.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m5$seTE^2`). Studies with $|z| > 3$ are flagged as outliers; $|z| > 1.96$ warrants inspection.

Table 10.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	0.347	0.069
Kostrzewa-Nowak & Nowak (2018)a	0.637	0.534
Kostrzewa-Nowak & Nowak (2018)b	0.993	1.102
Kostrzewa-Nowak & Nowak (2020a)a	1.209	1.634
Kostrzewa-Nowak & Nowak (2020a)b	0.374	0.117
Kostrzewa-Nowak & Nowak (2020b)a	0.378	0.122
Kostrzewa-Nowak & Nowak (2020b)b	0.028	-0.447
Kostrzewa-Nowak & Nowak (2020b)c	0.875	0.928
Kostrzewa-Nowak & Nowak (2020b)d	0.389	0.140
Kostrzewa-Nowak & Nowak (2020b)e	1.206	1.465
Steensberg et al. (2001)	-1.309	-2.475
Tsubakihara et al. (2012)	-0.959	-2.372
Lancaster et al. (2005)a	0.033	-0.416
Lancaster et al. (2005)b	0.073	-0.353
Lancaster et al. (2005)c	-0.226	-0.824
Harbaum et al. (2016)a	-0.215	-0.866
Harbaum et al. (2016)b	1.065	1.158



Note

No outlier.

10.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones (= 0.10, 0.05, 0.01). The dfbetas diagnostic from `metafor::rma.mv()` identifies **influential cases** — distinct from statistical outliers and not necessarily excluded.

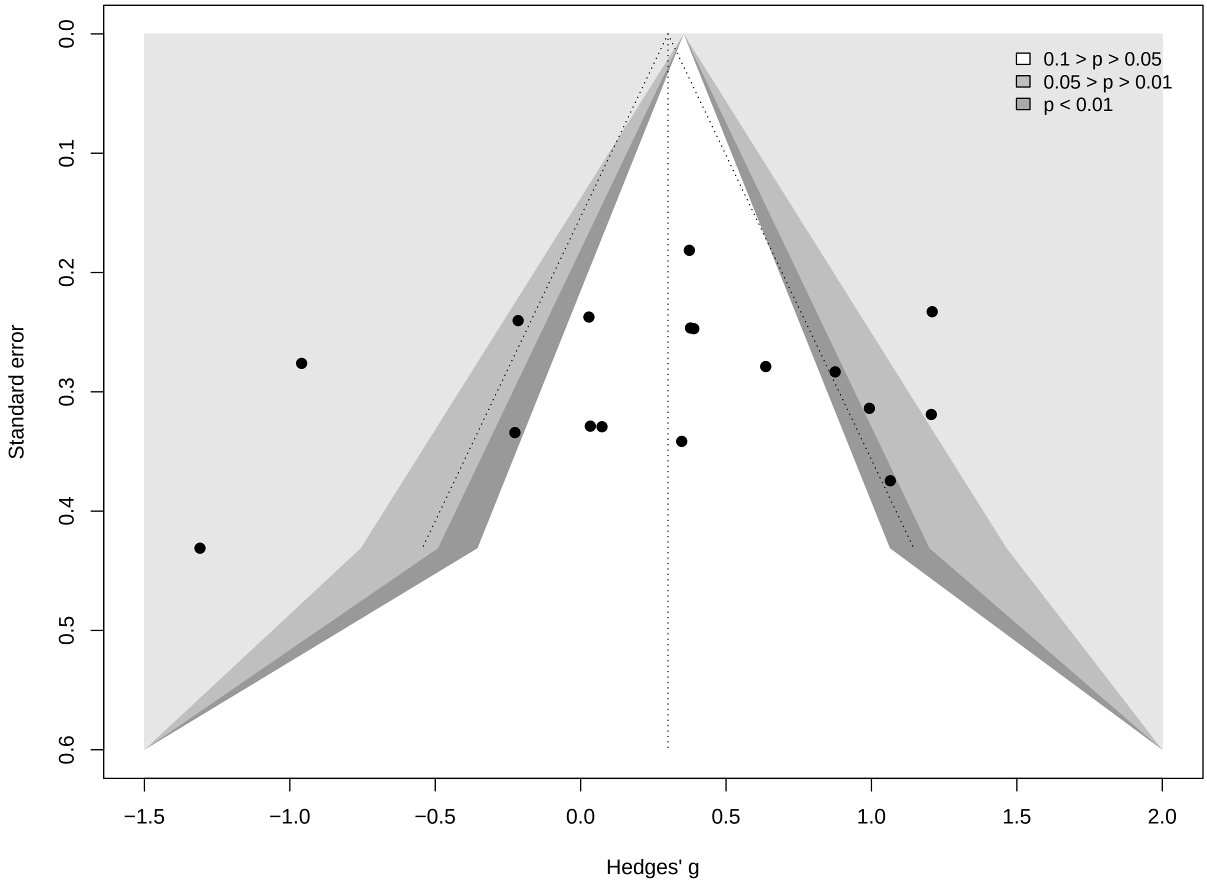


Table 10.2.: DFBETAS influence diagnostics

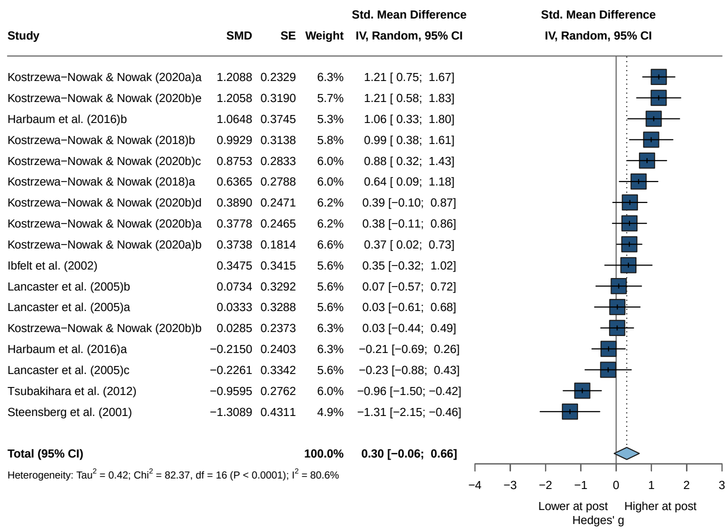
Short_reference	g	dfbetas
Ibfelt et al. (2002)	0.347	0.01004021
Kostrzewa-Nowak & Nowak (2018)a	0.637	0.27911537
Kostrzewa-Nowak & Nowak (2018)b	0.993	0.47102525
Kostrzewa-Nowak & Nowak (2020a)a	1.209	1.18025273
Kostrzewa-Nowak & Nowak (2020a)b	0.374	0.10134265
Kostrzewa-Nowak & Nowak (2020b)a	0.378	0.05596990
Kostrzewa-Nowak & Nowak (2020b)b	0.028	-0.38958693
Kostrzewa-Nowak & Nowak (2020b)c	0.875	0.48056733
Kostrzewa-Nowak & Nowak (2020b)d	0.389	0.06893600
Kostrzewa-Nowak & Nowak (2020b)e	1.206	0.60156479
Steensberg et al. (2001)	-1.309	-0.60449272
Tsubakihara et al. (2012)	-0.959	-1.20154886
Lancaster et al. (2005)a	0.033	-0.19186890
Lancaster et al. (2005)b	0.073	-0.16552634

Table 10.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Lancaster et al. (2005)c	-0.226	-0.34724912
Harbaum et al. (2016)a	-0.215	-0.68475128
Harbaum et al. (2016)b	1.065	0.36143534

One value was above 1 and one below −1, so both were identified as influential cases. Neither is an outlier; all outcomes are retained and the sources of heterogeneity are examined in the moderator analysis.

10.5. Forest Plot



10.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed, $p < .10$). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially missing studies.

```
Linear regression test of funnel plot asymmetry

Test result: t = -0.53, df = 15, p-value = 0.6026
Bias estimate: -1.4500 (SE = 2.7260)

Details:
```

```
- multiplicative residual heterogeneity variance (tau^2 = 5.3895)
- predictor: standard error
- weight:    inverse variance
- reference: Egger et al. (1997), BMJ
```

> **Note**
>
> No asymmetry -> no trim-and-fill

10.7. Moderator Analyses

10.7.1. Risk of Bias

Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic (Q_b) indicates whether subgroup means differ.

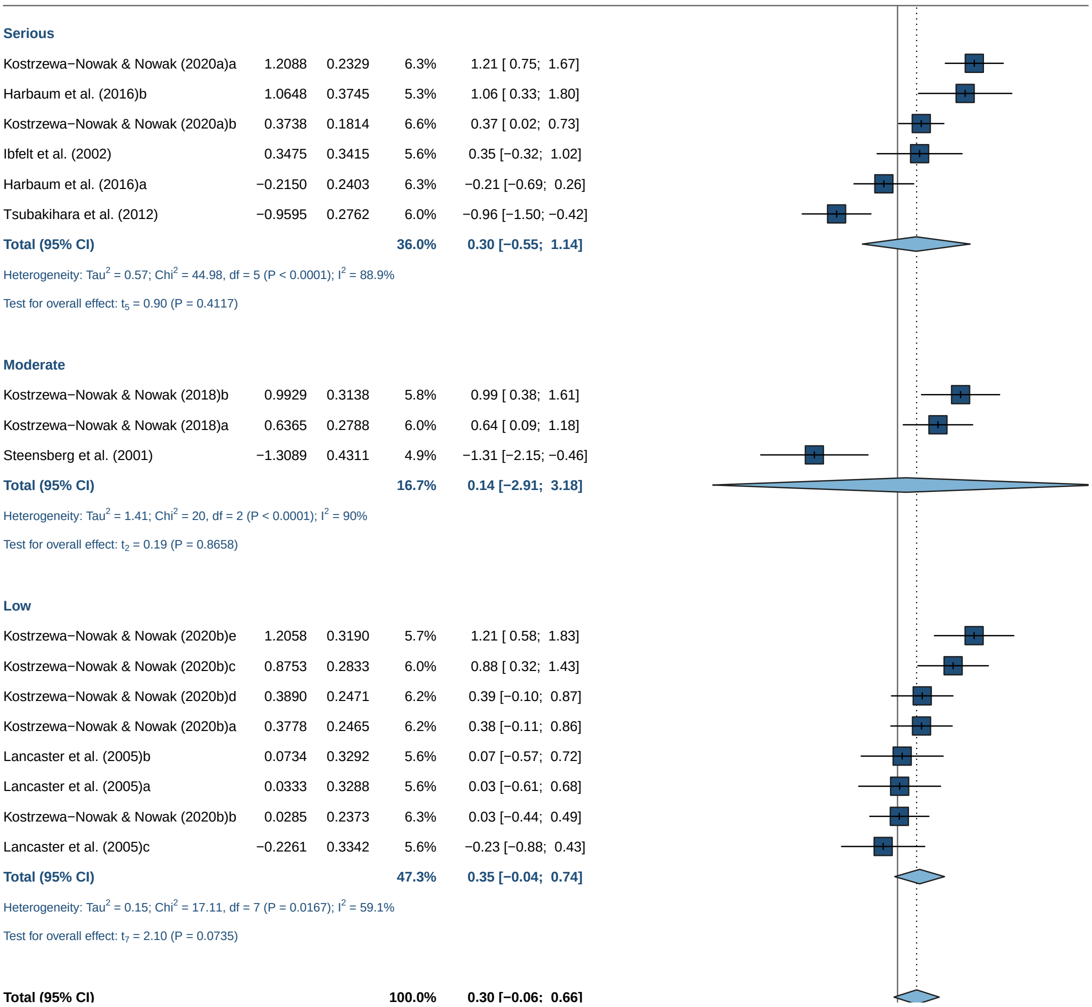
10.7.1.1. Model

```
Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.4156 [0.1703; 1.0546]; tau = 0.6447 [0.4126; 1.0270]
 I^2 = 80.6% [69.8%; 87.5%]; H = 2.27 [1.82; 2.83]

Test of heterogeneity:
     Q d.f.  p-value
 82.37   16 < 0.0001

Results for subgroups (random effects model (HK)):
                 k    SMD             95% CI  tau^2    tau
RoB = Serious    6 0.2952 [-0.5524; 1.1427] 0.5743 0.7578
RoB = Moderate   3 0.1355 [-2.9084; 3.1793] 1.4143 1.1892
RoB = Low        8 0.3483 [-0.0432; 0.7398] 0.1457 0.3818
                   Q   I^2
RoB = Serious  44.98 88.9%
RoB = Moderate 20.00 90.0%
RoB = Low      17.11 59.1%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 0.10    2  0.9517

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
```

```
- Prediction interval based on t-distribution
```

> **Note**
>
> Not significant results

10.7.1.2. Forrest Plot



10.7.2. Sex

10.7.2.1. Model

```
Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.4156 [0.1703; 1.0546]; tau = 0.6447 [0.4126; 1.0270]
 I^2 = 80.6% [69.8%; 87.5%]; H = 2.27 [1.82; 2.83]

Test of heterogeneity:
     Q d.f.  p-value
 82.37   16 < 0.0001
```

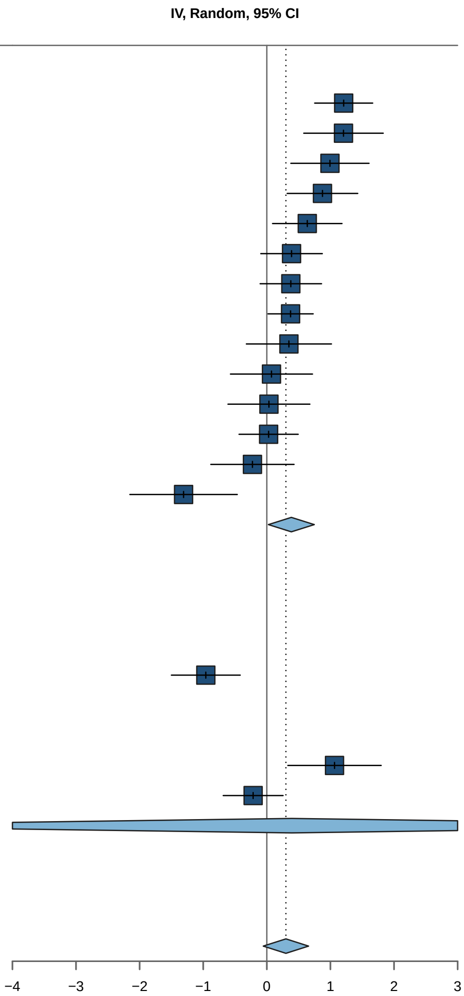
```
Results for subgroups (random effects model (HK)):
               k     SMD             95% CI  tau^2    tau
sex = male    14  0.3858 [ 0.0247;  0.7469] 0.3417 0.5846
sex = female   1 -0.9595 [-1.5008; -0.4182]     --     --
sex = mixed    2  0.3927 [-7.7274;  8.5127] 0.7199 0.8484
                 Q   I^2
sex = male   49.15 73.5%
sex = female  0.00    --
sex = mixed   8.27 87.9%

Test for subgroup differences (random effects model (HK)):
                   Q d.f. p-value
Between groups 17.68    2  0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

! Important

Significant results but 15 vs 1 vs 2

10.7.2.2. Forrest Plot

Subgroup	SMD	SE	Weight	IV, Random, 95% CI
male				
Kostrzewa−Nowak & Nowak (2020a)a	1.2088	0.2329	6.3%	1.21 [0.75; 1.67]
Kostrzewa−Nowak & Nowak (2020b)e	1.2058	0.3190	5.7%	1.21 [0.58; 1.83]
Kostrzewa−Nowak & Nowak (2018)b	0.9929	0.3138	5.8%	0.99 [0.38; 1.61]
Kostrzewa−Nowak & Nowak (2020b)c	0.8753	0.2833	6.0%	0.88 [0.32; 1.43]
Kostrzewa−Nowak & Nowak (2018)a	0.6365	0.2788	6.0%	0.64 [0.09; 1.18]
Kostrzewa−Nowak & Nowak (2020b)d	0.3890	0.2471	6.2%	0.39 [−0.10; 0.87]
Kostrzewa−Nowak & Nowak (2020b)a	0.3778	0.2465	6.2%	0.38 [−0.11; 0.86]
Kostrzewa−Nowak & Nowak (2020a)b	0.3738	0.1814	6.6%	0.37 [0.02; 0.73]
Ibfelt et al. (2002)	0.3475	0.3415	5.6%	0.35 [−0.32; 1.02]
Lancaster et al. (2005)b	0.0734	0.3292	5.6%	0.07 [−0.57; 0.72]
Lancaster et al. (2005)a	0.0333	0.3288	5.6%	0.03 [−0.61; 0.68]
Kostrzewa−Nowak & Nowak (2020b)b	0.0285	0.2373	6.3%	0.03 [−0.44; 0.49]
Lancaster et al. (2005)c	−0.2261	0.3342	5.6%	−0.23 [−0.88; 0.43]
Steensberg et al. (2001)	−1.3089	0.4311	4.9%	−1.31 [−2.15; −0.46]
Total (95% CI)			**82.4%**	**0.39 [0.02; 0.75]**
Heterogeneity: $Tau^2 = 0.34$; $Chi^2 = 49.15$, df = 13 ($P < 0.0001$); $I^2 = 73.5\%$				
Test for overall effect: $t_{13} = 2.31$ ($P = 0.0381$)				
female				
Tsubakihara et al. (2012)	−0.9595	0.2762	6.0%	−0.96 [−1.50; −0.42]
mixed				
Harbaum et al. (2016)b	1.0648	0.3745	5.3%	1.06 [0.33; 1.80]
Harbaum et al. (2016)a	−0.2150	0.2403	6.3%	−0.21 [−0.69; 0.26]
Total (95% CI)			**11.6%**	**0.39 [−7.73; 8.51]**
Heterogeneity: $Tau^2 = 0.72$; $Chi^2 = 8.27$, df = 1 ($P = 0.0040$); $I^2 = 87.9\%$				
Test for overall effect: $t_1 = 0.61$ ($P = 0.6492$)				
Total (95% CI)			**100.0%**	**0.30 [−0.06; 0.66]**
Heterogeneity: $Tau^2 = 0.42$; $Chi^2 = 82.37$, df = 16 ($P < 0.0001$); $I^2 = 80.6\%$				



10.7.3. Intensity

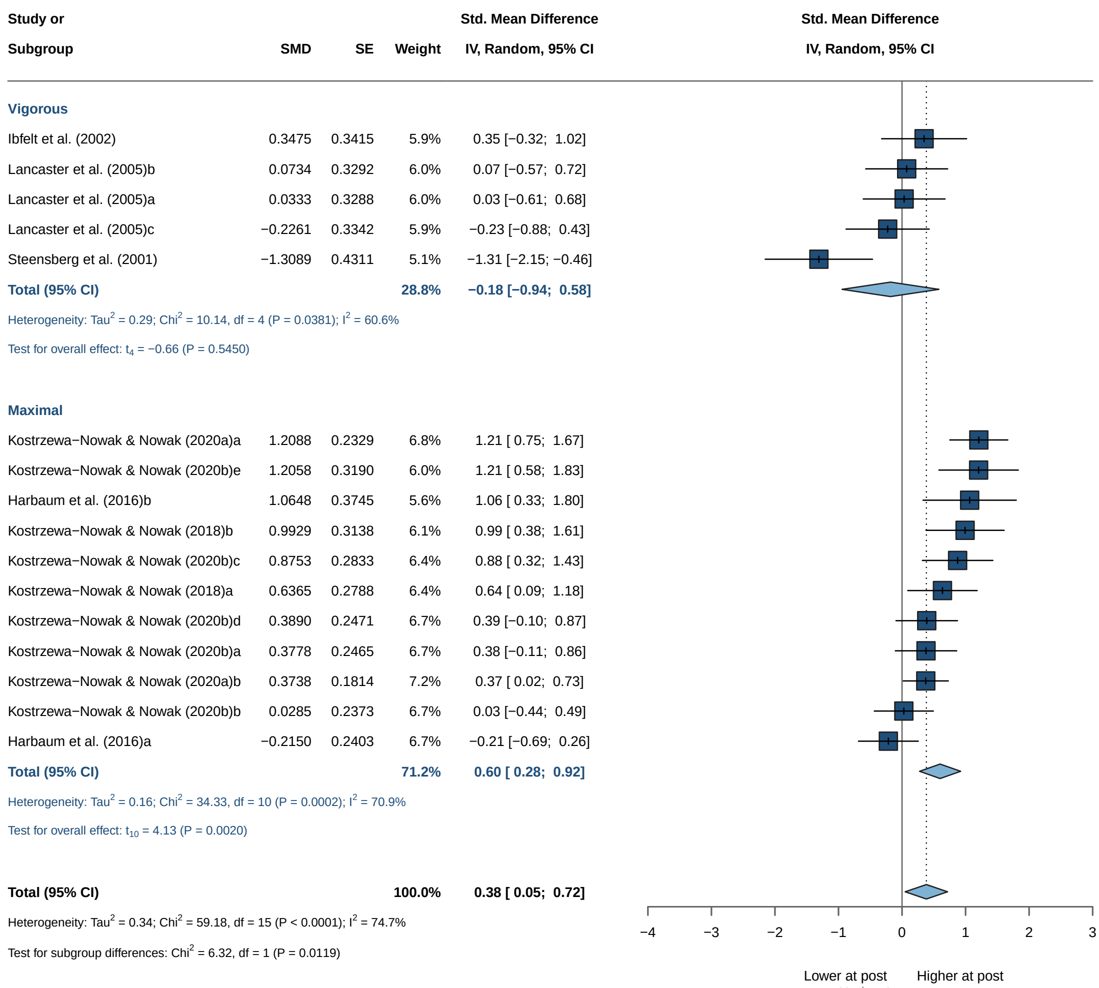
10.7.3.1. Model

```
Number of studies: k = 16

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.3382 [0.1128; 0.8975]; tau = 0.5815 [0.3358; 0.9474]
 I^2 = 74.7% [58.6%; 84.5%]; H = 1.99 [1.56; 2.54]

Test of heterogeneity:
     Q d.f.  p-value
 59.18   15 < 0.0001

Results for subgroups (random effects model (HK)):
                       k     SMD            95% CI  tau^2
intensity = Vigorous   5 -0.1814 [-0.9439; 0.5811] 0.2883
intensity = Maximal   11  0.5997 [ 0.2761; 0.9232] 0.1564
                        tau     Q   I^2
intensity = Vigorous 0.5370 10.14 60.6%
```

```
intensity = Maximal  0.3955 34.33 70.9%

Test for subgroup differences (random effects model (HK)):
                    Q d.f. p-value
Between groups 6.32    1  0.0119

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **! Important**
>
> Significant

10.7.3.2. Forrest Plot



10.7.4. Staining

```
Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.4156 [0.1703; 1.0546]; tau = 0.6447 [0.4126; 1.0270]
 I^2 = 80.6% [69.8%; 87.5%]; H = 2.27 [1.82; 2.83]

Test of heterogeneity:
     Q d.f.  p-value
 82.37   16 < 0.0001

Results for subgroups (random effects model (HK)):
                   k     SMD            95% CI  tau^2
staining = intra  14  0.3858 [ 0.0247; 0.7469] 0.3417
staining = extra   3 -0.0594 [-2.5758; 2.4570] 0.9569
                    tau     Q   I^2
staining = intra 0.5846 49.15 73.5%
staining = extra 0.9782 18.93 89.4%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 0.54    1  0.4643

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

Note

Not significant results

10.7.5. Continuous Moderators (Meta-Regression)

Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity (^{2}); bubble plots show the fitted line with 95% CI.

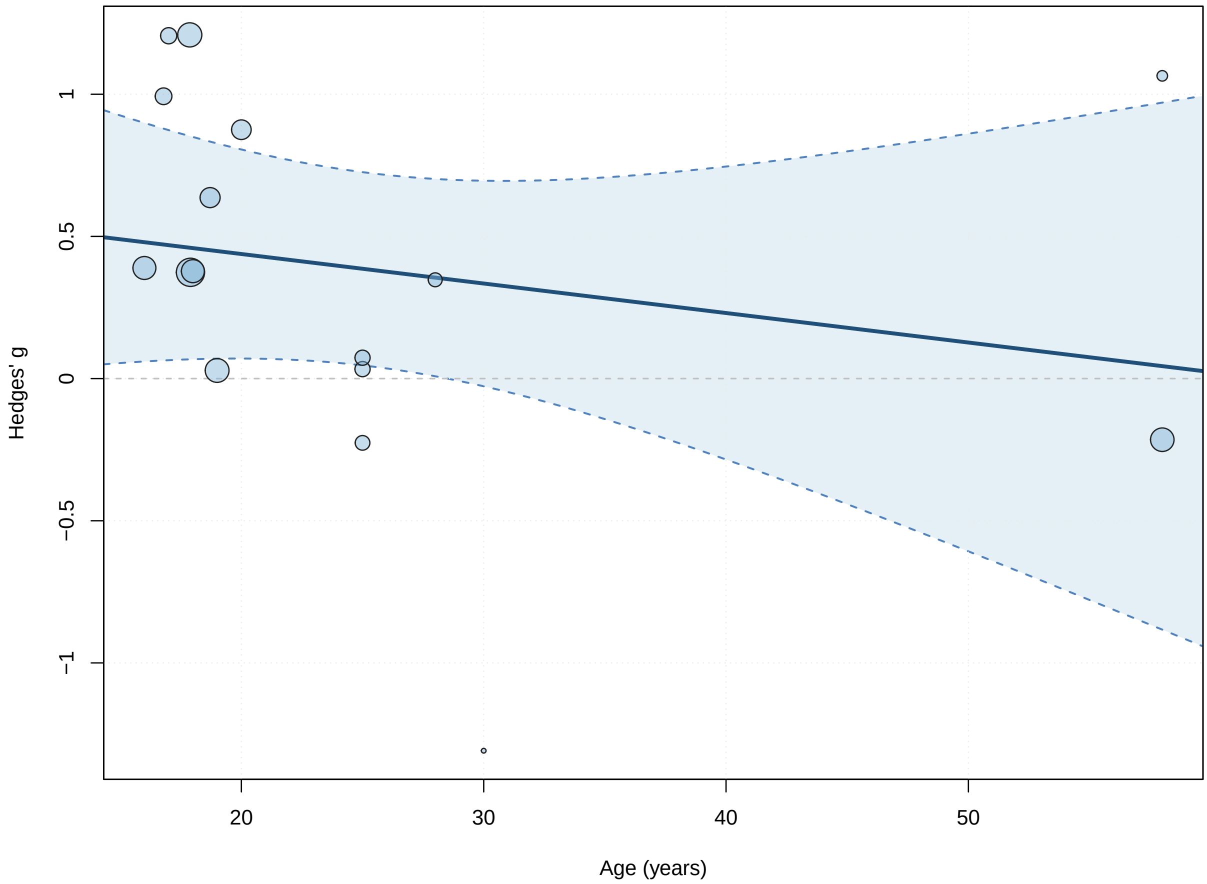
10.7.5.1. Age

```
Mixed-Effects Model (k = 16; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-14.5497   39.4622   35.0995   37.4173   37.0995
```

```
tau^2 (estimated amount of residual heterogeneity):     0.3523 (SE = 0.1679)
tau (square root of estimated tau^2 value):             0.5936
I^2 (residual heterogeneity / unaccounted variability): 81.90%
H^2 (unaccounted variability / sampling variability):   5.52
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 14) = 53.0007, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 14) = 0.7114, p-val = 0.4131

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.6457  0.3495   1.8473  14  0.0859  -0.1040
age       -0.0104  0.0123  -0.8435  14  0.4131  -0.0368
          ci.ub
intrcpt  1.3953  .
age      0.0160

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

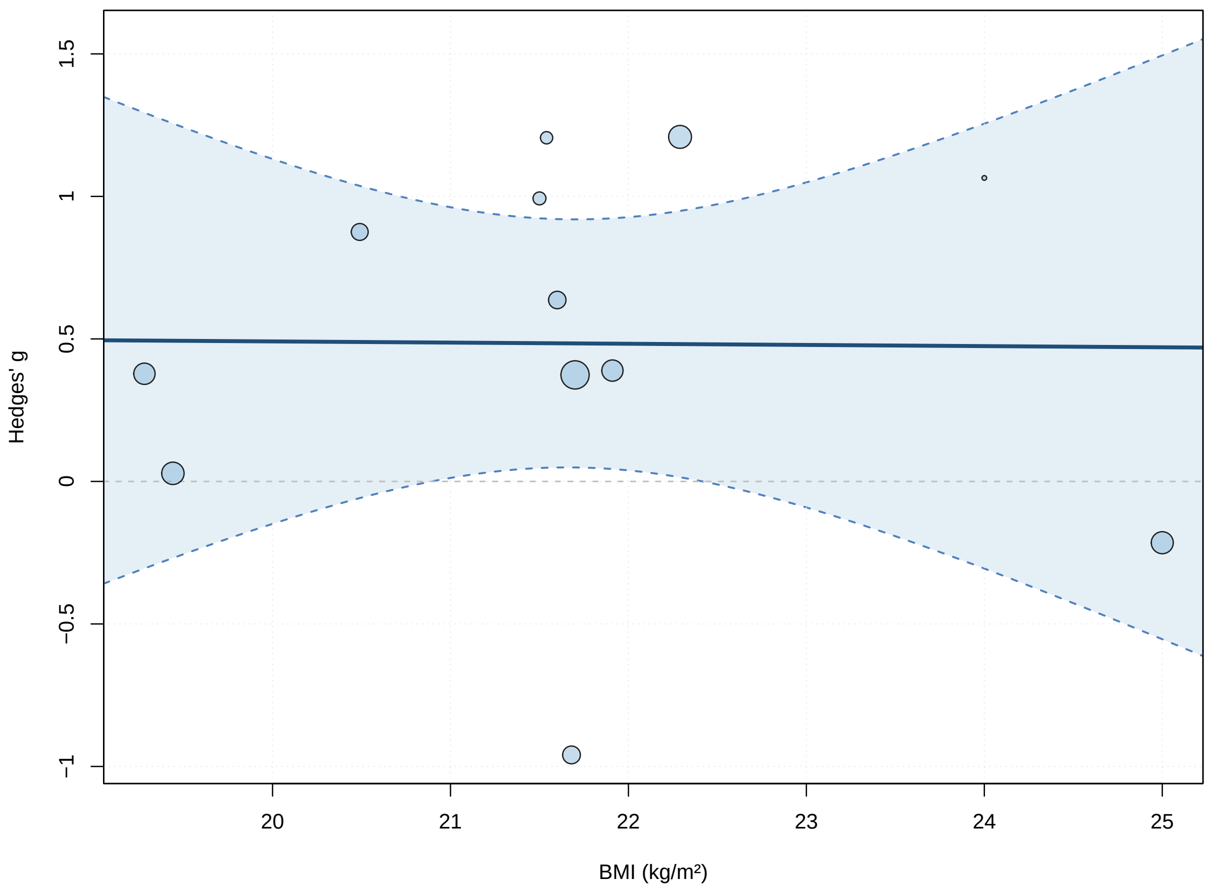


10.7.5.2. BMI

```
Mixed-Effects Model (k = 12; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-11.3566   32.5305   28.7132   30.1679   31.7132

tau^2 (estimated amount of residual heterogeneity):     0.3880 (SE = 0.2073)
tau (square root of estimated tau^2 value):             0.6229
I^2 (residual heterogeneity / unaccounted variability): 85.24%
H^2 (unaccounted variability / sampling variability):   6.77
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 10) = 61.5799, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 10) = 0.0011, p-val = 0.9744

Model Results:
```

```
         estimate      se     tval  df    pval    ci.lb
intrcpt    0.5737  2.7249   0.2106  10  0.8375  -5.4977
bmi       -0.0041  0.1254  -0.0329  10  0.9744  -0.2834
          ci.ub
intrcpt  6.6452
bmi      0.2752

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant



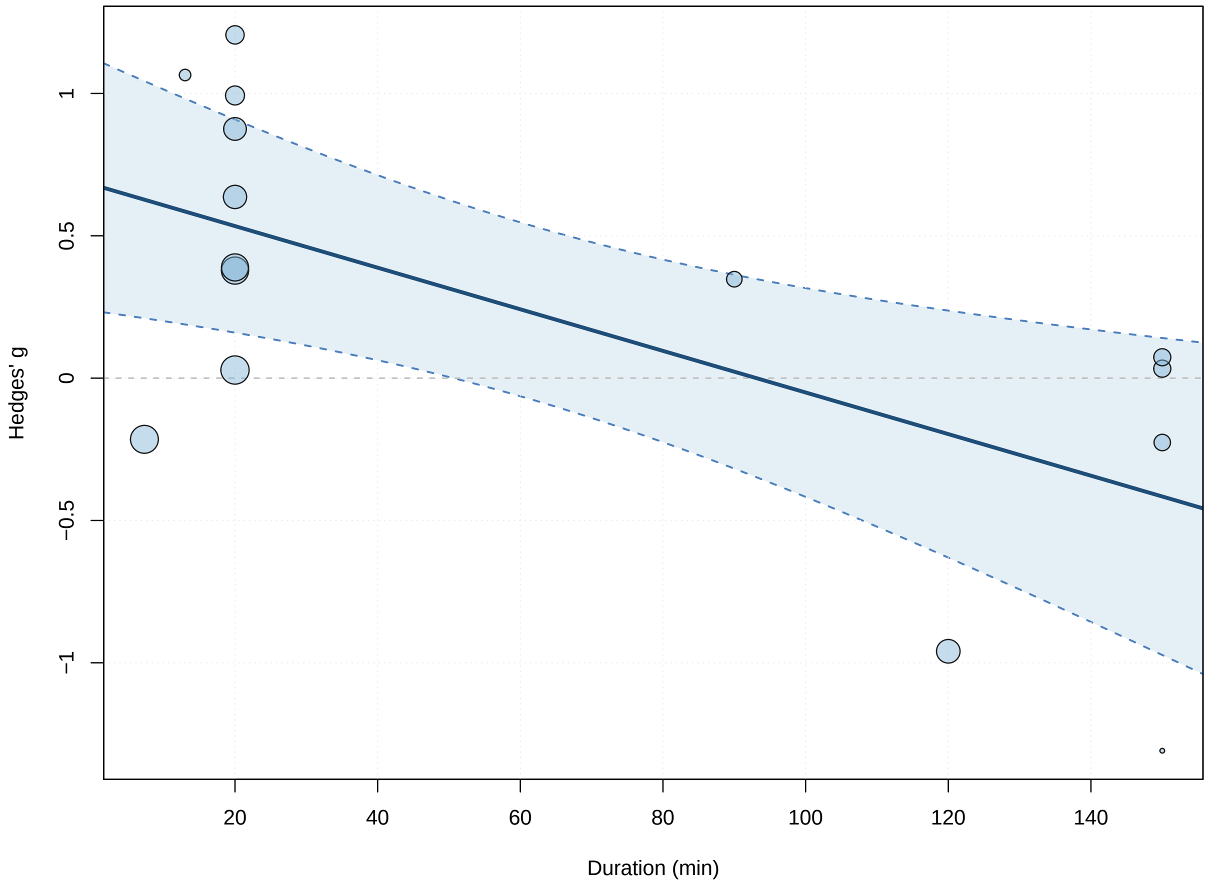
10.7.5.3. Duration

```
Mixed-Effects Model (k = 15; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
-11.2968   31.0389   28.5936   30.7177   30.7754
```

```
tau^2 (estimated amount of residual heterogeneity):     0.2107 (SE = 0.1206)
tau (square root of estimated tau^2 value):             0.4590
I^2 (residual heterogeneity / unaccounted variability): 71.01%
H^2 (unaccounted variability / sampling variability):   3.45
R^2 (amount of heterogeneity accounted for):            49.32%

Test for Residual Heterogeneity:
QE(df = 13) = 43.6946, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 13) = 9.0460, p-val = 0.0101

Model Results:

          estimate      se     tval  df    pval    ci.lb
intrcpt     0.6802  0.2053   3.3142  13  0.0056   0.2368
duration   -0.0073  0.0024  -3.0077  13  0.0101  -0.0126
            ci.ub
intrcpt    1.1237  **
duration  -0.0021   *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

! Important

Significant



10.7.5.4. Dose

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -9.3620   26.4336   24.7240   26.6412   27.1240

tau^2 (estimated amount of residual heterogeneity):     0.1656 (SE = 0.1075)
tau (square root of estimated tau^2 value):             0.4070
I^2 (residual heterogeneity / unaccounted variability): 65.73%
H^2 (unaccounted variability / sampling variability):   2.92
R^2 (amount of heterogeneity accounted for):            51.12%

Test for Residual Heterogeneity:
QE(df = 12) = 34.1013, p-val = 0.0007

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 12) = 7.2942, p-val = 0.0193

Model Results:
```

```
         estimate      se     tval  df    pval    ci.lb
intrcpt    0.7315  0.2022   3.6180  12  0.0035   0.2910
dose      -0.0010  0.0004  -2.7008  12  0.0193  -0.0019
            ci.ub
intrcpt   1.1721  **
dose     -0.0002   *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

10.7.5.5. IFN

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -4.0987   14.9426   14.1973   14.0350   22.1973

tau^2 (estimated amount of residual heterogeneity):     0.1868 (SE = 0.1615)
tau (square root of estimated tau^2 value):             0.4322
I^2 (residual heterogeneity / unaccounted variability): 74.03%
H^2 (unaccounted variability / sampling variability):   3.85
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 19.9174, p-val = 0.0013

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.0002, p-val = 0.9881

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.6319  0.8621   0.7330   5  0.4965  -1.5842
IFN       -0.0073  0.4649  -0.0156   5  0.9881  -1.2024
           ci.ub
intrcpt  2.8480
IFN      1.1878

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

10.7.5.6. IL_4

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -6.2496   20.3845   18.4992   19.0909   23.2992

tau^2 (estimated amount of residual heterogeneity):     0.2223 (SE = 0.1585)
tau (square root of estimated tau^2 value):             0.4715
I^2 (residual heterogeneity / unaccounted variability): 77.77%
H^2 (unaccounted variability / sampling variability):   4.50
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 7) = 30.1193, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 0.3341, p-val = 0.5814

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.4967  0.2145  2.3160   7  0.0537  -0.0104
IL_4       0.1043  0.1804  0.5780   7  0.5814  -0.3224
          ci.ub
intrcpt  1.0038  .
IL_4     0.5310

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

10.7.5.7. IL_6

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -8.9389   28.0846   23.8778   25.7949   26.2778

tau^2 (estimated amount of residual heterogeneity):     0.1675 (SE = 0.1022)
tau (square root of estimated tau^2 value):             0.4092
I^2 (residual heterogeneity / unaccounted variability): 70.18%
H^2 (unaccounted variability / sampling variability):   3.35
R^2 (amount of heterogeneity accounted for):            7.58%
```

```
Test for Residual Heterogeneity:
QE(df = 12) = 36.6447, p-val = 0.0003

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 12) = 1.8665, p-val = 0.1969

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.3688  0.1537  2.4001  12  0.0335   0.0340
IL_6       0.1207  0.0884  1.3662  12  0.1969  -0.0718
          ci.ub
intrcpt  0.7036  *
IL_6     0.3132

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

10.7.5.8. IL_10

```
Mixed-Effects Model (k = 11; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -7.0987   23.2792   20.1975   21.3911   23.6260

tau^2 (estimated amount of residual heterogeneity):     0.1760 (SE = 0.1201)
tau (square root of estimated tau^2 value):             0.4195
I^2 (residual heterogeneity / unaccounted variability): 71.18%
H^2 (unaccounted variability / sampling variability):   3.47
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 9) = 34.2151, p-val < .0001

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 9) = 0.0965, p-val = 0.7631

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.5975  0.1530  3.9061   9  0.0036   0.2515
IL_10      0.1301  0.4189  0.3107   9  0.7631  -0.8174
          ci.ub
```

```
intrcpt  0.9435  **
IL_10    1.0777

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

10.7.5.9. IL_2

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -3.8027   14.3505   13.6053   13.4430   21.6053

tau^2 (estimated amount of residual heterogeneity):     0.1669 (SE = 0.1434)
tau (square root of estimated tau^2 value):             0.4086
I^2 (residual heterogeneity / unaccounted variability): 74.80%
H^2 (unaccounted variability / sampling variability):   3.97
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 19.7007, p-val = 0.0014

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.4392, p-val = 0.5368

Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.1414  0.7407  0.1909   5  0.8561  -1.7626
IL_2       0.2684  0.4049  0.6627   5  0.5368  -0.7725
          ci.ub
intrcpt  2.0453
IL_2     1.3093

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

10.8. Multivariate Models

Multivariate meta-regression tests interactions between candidate moderators. None of the models below yields independent predictors.

10.8.1. Intensity * Duration

```
Mixed-Effects Model (k = 14; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.1901 (SE = 0.1290)
tau (square root of estimated tau^2 value):             0.4360
I^2 (residual heterogeneity / unaccounted variability): 68.47%
H^2 (unaccounted variability / sampling variability):   3.17
R^2 (amount of heterogeneity accounted for):            43.88%

Test for Residual Heterogeneity:
QE(df = 10) = 26.5082, p-val = 0.0031

Test of Moderators (coefficients 2:4):
F(df1 = 3, df2 = 10) = 2.9355, p-val = 0.0857

Model Results:

                            estimate      se     tval  df
intrcpt                      -0.2459  0.6993  -0.3516  10
intensityVigorous             1.5730  1.5618   1.0071  10
duration                      0.0452  0.0381   1.1868  10
intensityVigorous:duration   -0.0561  0.0394  -1.4245  10
                              pval    ci.lb   ci.ub
intrcpt                     0.7324  -1.8040  1.3122
intensityVigorous           0.3376  -1.9070  5.0529
duration                    0.2627  -0.0397  0.1300
intensityVigorous:duration  0.1848  -0.1438  0.0316

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

10.8.2. IL_10 * IL_4

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.3297 (SE = 0.2594)
tau (square root of estimated tau^2 value):             0.5742
```

```
I^2 (residual heterogeneity / unaccounted variability): 81.69%
H^2 (unaccounted variability / sampling variability):   5.46
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 30.1034, p-val < .0001

Test of Moderators (coefficients 2:4):
F(df1 = 3, df2 = 5) = 0.0923, p-val = 0.9611

Model Results:

            estimate      se     tval  df    pval    ci.lb
intrcpt       0.5401  0.3177   1.6997   5  0.1499  -0.2767
IL_10         0.0951  0.6813   0.1395   5  0.8945  -1.6563
IL_4          0.0953  0.2254   0.4229   5  0.6899  -0.4842
IL_10:IL_4   -0.2817  1.4854  -0.1897   5  0.8570  -4.1000
             ci.ub
intrcpt     1.3568
IL_10       1.8464
IL_4        0.6748
IL_10:IL_4  3.5365

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

10.8.3. IL_2 * IFN

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.2707 (SE = 0.2724)
tau (square root of estimated tau^2 value):             0.5203
I^2 (residual heterogeneity / unaccounted variability): 81.40%
H^2 (unaccounted variability / sampling variability):   5.38
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 3) = 17.0604, p-val = 0.0007

Test of Moderators (coefficients 2:4):
F(df1 = 3, df2 = 3) = 0.2470, p-val = 0.8596

Model Results:
```

```
          estimate      se     tval  df    pval     ci.lb
intrcpt     5.7989  9.5568   0.6068   3  0.5868  -24.6150
IL_2       -2.8869  5.5986  -0.5156   3  0.6417  -20.7043
IFN        -3.1991  5.2880  -0.6050   3  0.5879  -20.0280
IL_2:IFN    1.7440  3.0132   0.5788   3  0.6033   -7.8454
             ci.ub
intrcpt   36.2129
IL_2      14.9304
IFN       13.6298
IL_2:IFN  11.3335

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

11. Th1/Th2 Ratio — ~2 h Post-Exercise

11.1. Data Import

The ~2 h window (`effect_size/Ratio/pre_post+2h/Ratio_pre_post+2h_r_050.csv`) is reported for the main analysis, outlier check, funnel, and forest only; the limited data do not support moderator modelling. No study is excluded a priori.

11.2. Primary Meta-Analysis

11.2.1. {meta}

```
Number of studies: k = 5

                              SMD            95% CI     t
Random effects model (HK) -0.5071 [-1.1495; 0.1353] -2.19
Prediction interval               [-1.8150; 0.8007]
                          p-value
Random effects model (HK)  0.0935
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.1626 [0.0000; 2.3056]; tau = 0.4033 [0.0000; 1.5184]
 I^2 = 44.6% [0.0%; 79.7%]; H = 1.34 [1.00; 2.22]

Test of heterogeneity:
    Q d.f. p-value
 7.22    4  0.1245

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 4)
- Prediction interval based on t-distribution (df = 4)
```

11.2.2. {metafor}

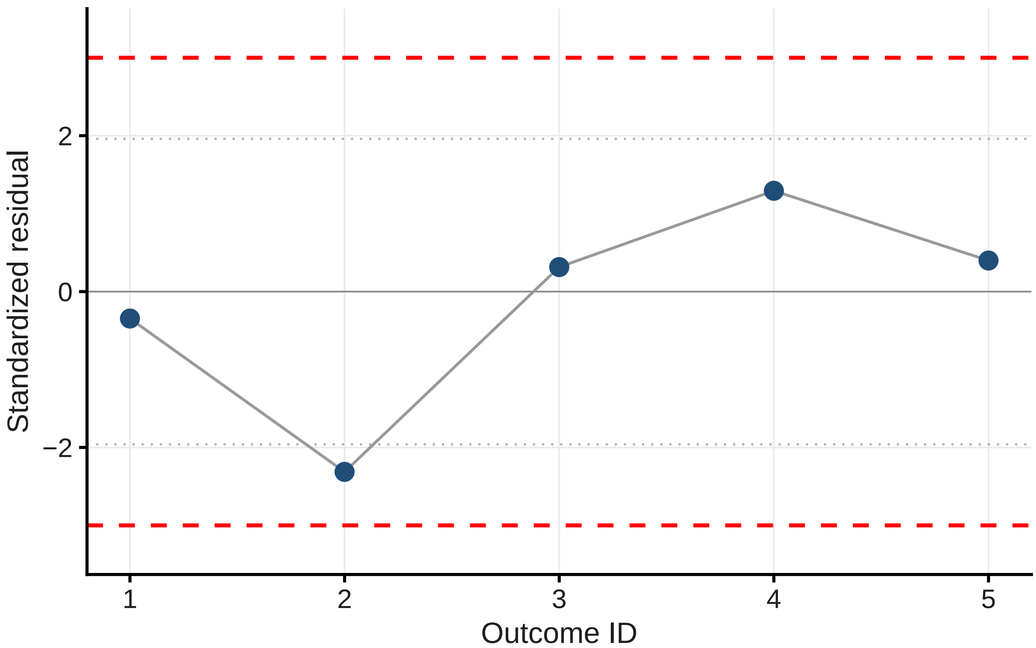
```
Random-Effects Model (k = 5; tau^2 estimator: HE)
```

```
tau^2 (estimated amount of total heterogeneity): 0.1626 (SE = 0.2133)
tau (square root of estimated tau^2 value):      0.4033
I^2 (total heterogeneity / total variability):   55.30%
H^2 (total variability / sampling variability):  2.24

Test for Heterogeneity:
Q(df = 4) = 7.2248, p-val = 0.1245

Model Results:

estimate      se     tval  df    pval    ci.lb   ci.ub
 -0.5071  0.2314  -2.1916   4  0.0935  -1.1495  0.1353  .

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

11.3. Outlier Detection

Table 11.1.: Standardised residuals

Short_reference	g	std_res
Ibfelt et al. (2002)	-0.679	-0.346
Steensberg et al. (2001)	-1.439	-2.313
Lancaster et al. (2005)a	-0.357	0.314
Lancaster et al. (2005)b	0.000	1.293
Lancaster et al. (2005)c	-0.317	0.398



Note

No outlier.

11.4. Funnel Plot & Influential Cases

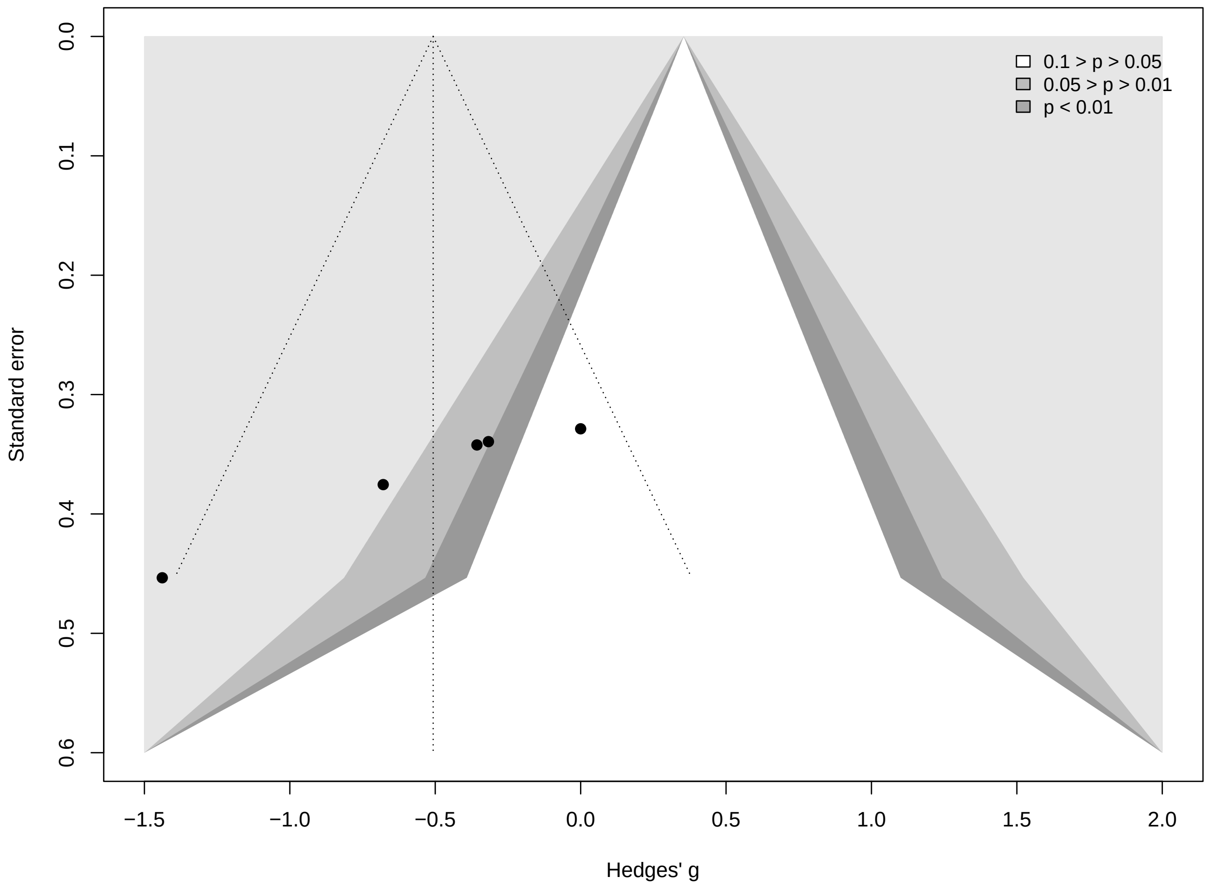
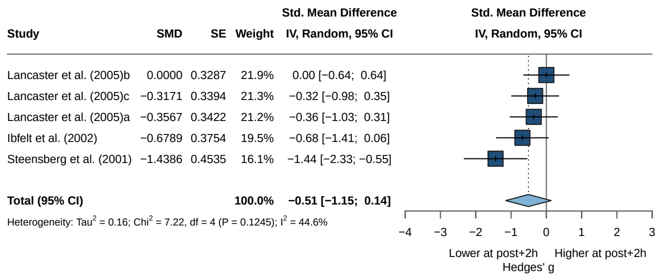


Table 11.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Ibfelt et al. (2002)	-0.679	-0.3097587
Steensberg et al. (2001)	-1.439	-0.8804841
Lancaster et al. (2005)a	-0.357	0.1798076
Lancaster et al. (2005)b	0.000	0.9012816
Lancaster et al. (2005)c	-0.317	0.2552408

One value above 1 and one below -1 were identified as influential cases; neither is an outlier, so all outcomes are retained.

11.5. Forest Plot



11.6. Asymmetry Test

Note: the number of studies in this window is too small to reliably test for small-study effects (minimum $k = 10$). No Egger's test or trim-and-fill is performed.

12. Th1/Th2 Ratio — 17 h Post-Exercise

12.1. Data Import

12.2. Primary Meta-Analysis

Kostrzewa-Nowak et al. (2019) is removed before pooling and the model re-fitted.

12.2.1. {meta}

```
Number of studies: k = 11

                             SMD            95% CI    t
Random effects model (HK) 0.6433 [-0.0265; 1.3131] 2.14
Prediction interval              [-2.4114; 3.6980]
                          p-value
Random effects model (HK)  0.0580
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 1.7065 [0.1105; 3.8540]; tau = 1.3063 [0.3324; 1.9632]
 I^2 = 73.5% [51.6%; 85.5%]; H = 1.94 [1.44; 2.62]

Test of heterogeneity:
     Q d.f.  p-value
 37.71   10 < 0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 10)
- Prediction interval based on t-distribution (df = 10)
```

12.2.2. {metafor}

```
Random-Effects Model (k = 11; tau^2 estimator: HE)

tau^2 (estimated amount of total heterogeneity): 1.7065 (SE = 1.0895)
tau (square root of estimated tau^2 value):      1.3063
```

```
I^2 (total heterogeneity / total variability):   95.92%
H^2 (total variability / sampling variability):  24.49

Test for Heterogeneity:
Q(df = 10) = 37.7068, p-val < .0001

Model Results:

estimate      se    tval  df    pval    ci.lb   ci.ub
  0.6433  0.3006  2.1401  10  0.0580  -0.0265  1.3131  .

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

12.2.3. Adapted Model {meta}

```
Number of studies: k = 10

                             SMD             95% CI    t
Random effects model (HK) 0.5129 [ 0.0830; 0.9427] 2.70
Prediction interval              [-0.8512; 1.8770]
                          p-value
Random effects model (HK)  0.0244
Prediction interval

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.3238 [0.0746; 1.2164]; tau = 0.5690 [0.2731; 1.1029]
 I^2 = 72.7% [48.5%; 85.6%]; H = 1.92 [1.39; 2.63]

Test of heterogeneity:
     Q d.f. p-value
 33.01    9  0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 9)
- Prediction interval based on t-distribution (df = 9)
```

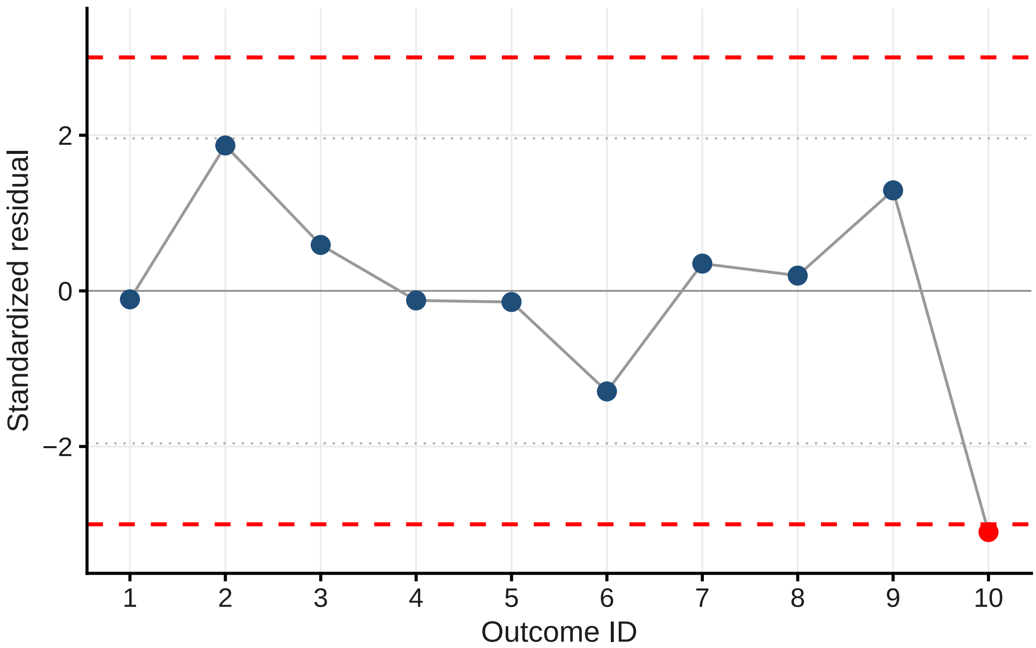
12.3. Outlier Detection

Table 12.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2018)a	0.458	-0.109
Kostrzewa-Nowak & Nowak (2018)b	1.463	1.868

Table 12.1.: Standardised residuals

Short_reference	g	std_res
Kostrzewa-Nowak & Nowak (2020a)a	0.785	0.590
Kostrzewa-Nowak & Nowak (2020a)b	0.454	-0.123
Kostrzewa-Nowak & Nowak (2020b)a	0.442	-0.144
Kostrzewa-Nowak & Nowak (2020b)b	-0.051	-1.292
Kostrzewa-Nowak & Nowak (2020b)c	0.685	0.350
Kostrzewa-Nowak & Nowak (2020b)d	0.609	0.196
Kostrzewa-Nowak & Nowak (2020b)e	1.141	1.291
Steensberg et al. (2001)	-0.854	-3.100



ℹ Note

No outlier.

12.4. Funnel Plot & Influential Cases

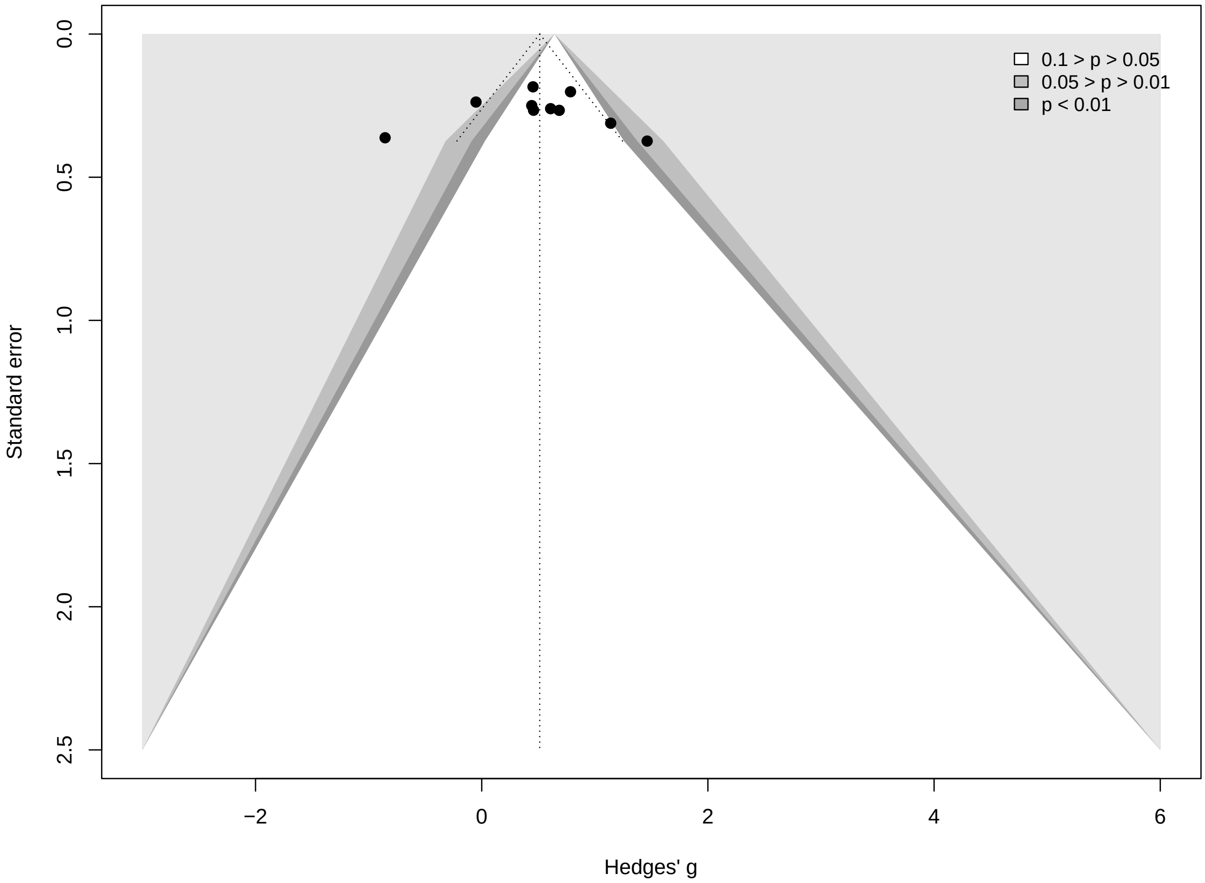


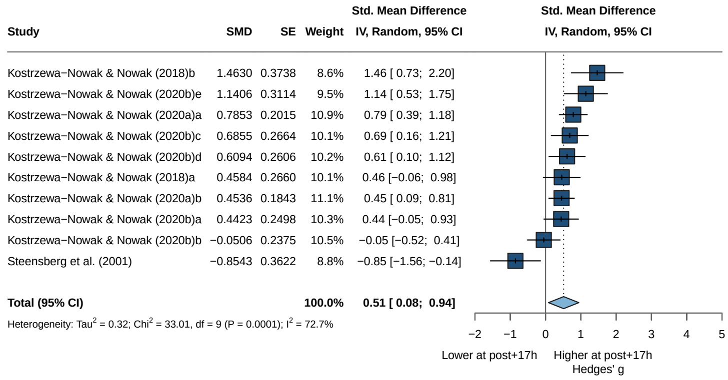
Table 12.2.: DFBETAS influence diagnostics

Short_reference	g	dfbetas
Kostrzewa-Nowak & Nowak (2018)a	0.458	-0.06601183
Kostrzewa-Nowak & Nowak (2018)b	1.463	0.57178772
Kostrzewa-Nowak & Nowak (2020a)a	0.785	0.64171703
Kostrzewa-Nowak & Nowak (2020a)b	0.454	-0.16812828
Kostrzewa-Nowak & Nowak (2020b)a	0.442	-0.09893333
Kostrzewa-Nowak & Nowak (2020b)b	-0.051	-0.90092632
Kostrzewa-Nowak & Nowak (2020b)c	0.685	0.21588205
Kostrzewa-Nowak & Nowak (2020b)d	0.609	0.12756806
Kostrzewa-Nowak & Nowak (2020b)e	1.141	0.55656301
Steensberg et al. (2001)	-0.854	-0.87675665

ℹ Note

No influential case.

12.5. Forest Plot



12.6. Asymmetry Test

```
Linear regression test of funnel plot asymmetry

Test result: t = 0.02, df = 8, p-value = 0.9847
Bias estimate: 0.0606 (SE = 3.0606)

Details:
- multiplicative residual heterogeneity variance (tau^2 = 4.1263)
- predictor: standard error
- weight:    inverse variance
- reference: Egger et al. (1997), BMJ
```

> **Note**
>
> No asymmetry -> no trim-and-fill

12.7. Moderator Analyses

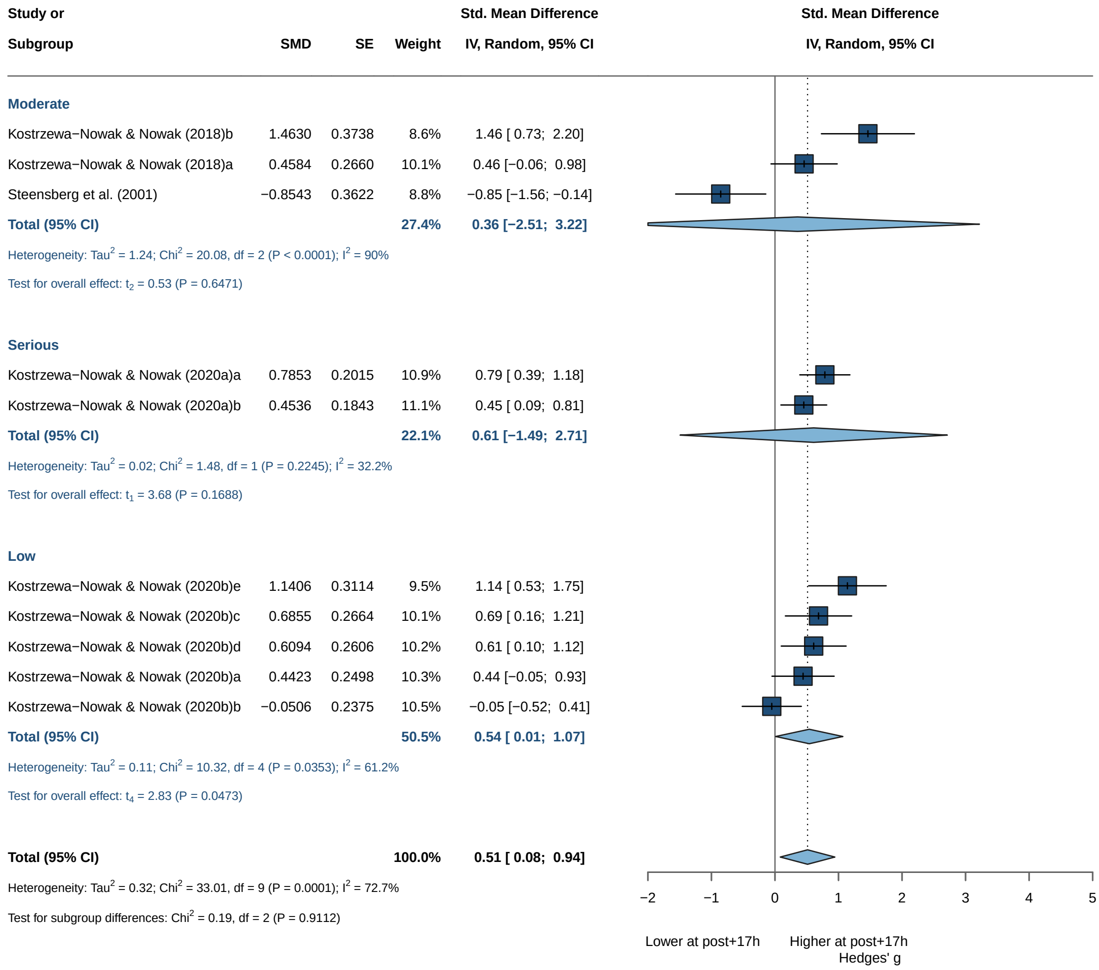
12.7.1. Risk of Bias

12.7.1.1. Model

```
Number of studies: k = 10

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.3238 [0.0746; 1.2164]; tau = 0.5690 [0.2731; 1.1029]
 I^2 = 72.7% [48.5%; 85.6%]; H = 1.92 [1.39; 2.63]

Test of heterogeneity:
     Q d.f. p-value
 33.01    9  0.0001

Results for subgroups (random effects model (HK)):
                 k    SMD            95% CI  tau^2    tau
RoB = Moderate   3 0.3550 [-2.5088; 3.2188] 1.2364 1.1119
RoB = Serious    2 0.6095 [-1.4940; 2.7129] 0.0177 0.1331
RoB = Low        5 0.5399 [ 0.0105; 1.0692] 0.1145 0.3384
                   Q   I^2
RoB = Moderate 20.08 90.0%
RoB = Serious   1.48 32.2%
RoB = Low      10.32 61.2%

Test for subgroup differences (random effects model (HK)):
                  Q d.f. p-value
Between groups 0.19    2  0.9112

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

> **Note**
>
> Not significant results

12.7.1.2. Forrest Plot



12.7.2. Intensity

12.7.2.1. Model

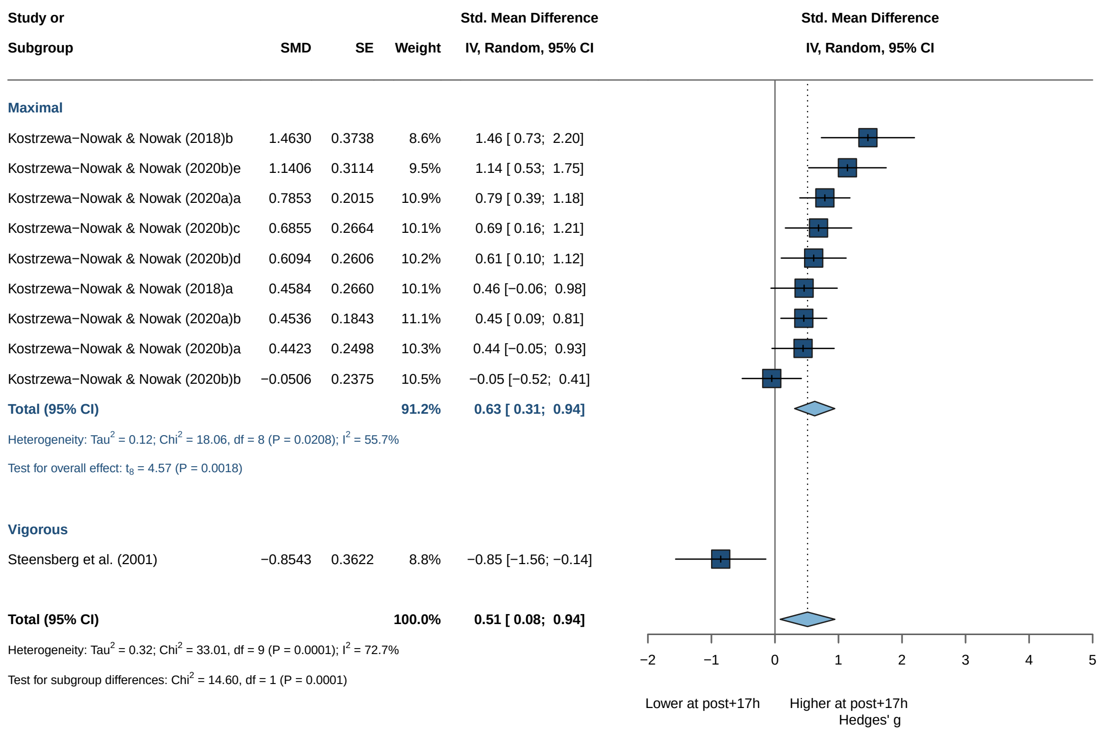
```
Number of studies: k = 10

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.3238 [0.0746; 1.2164]; tau = 0.5690 [0.2731; 1.1029]
 I^2 = 72.7% [48.5%; 85.6%]; H = 1.92 [1.39; 2.63]

Test of heterogeneity:
     Q d.f. p-value
 33.01    9  0.0001

Results for subgroups (random effects model (HK)):
                       k     SMD             95% CI  tau^2
intensity = Maximal    9  0.6254 [ 0.3097;  0.9411] 0.1202
intensity = Vigorous   1 -0.8543 [-1.5641; -0.1444]     --
                        tau     Q   I^2
intensity = Maximal  0.3466 18.06 55.7%
```

```
intensity = Vigorous     --  0.00    --

Test for subgroup differences (random effects model (HK)):
                   Q d.f. p-value
Between groups 14.60    1  0.0001

Details of meta-analysis methods:
- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution
```

! Important

Significant results but 10 vs 1

12.7.2.2. Forrest Plot



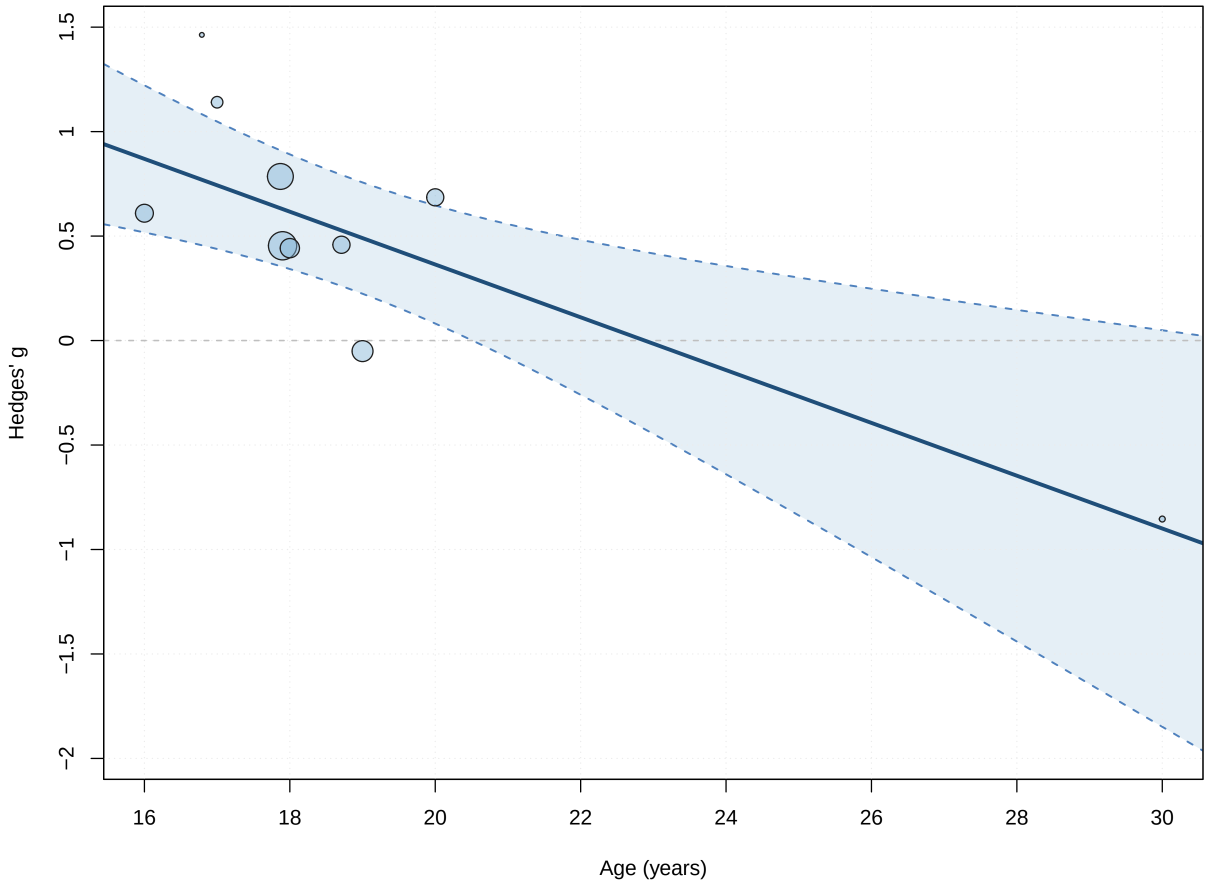
12.7.3. Continuous Moderators (Meta-Regression)

12.7.3.1. Age

```
Mixed-Effects Model (k = 10; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -3.3225   14.8176   12.6449   13.5527   16.6449

tau^2 (estimated amount of residual heterogeneity):     0.0757 (SE = 0.0748)
tau (square root of estimated tau^2 value):             0.2751
I^2 (residual heterogeneity / unaccounted variability): 54.86%
H^2 (unaccounted variability / sampling variability):   2.22
R^2 (amount of heterogeneity accounted for):            76.63%

Test for Residual Heterogeneity:
QE(df = 8) = 14.6115, p-val = 0.0672

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 8) = 12.7719, p-val = 0.0073

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    2.8912  0.6757   4.2791   8  0.0027   1.3331
age       -0.1264  0.0354  -3.5738   8  0.0073  -0.2079
           ci.ub
intrcpt   4.4493  **
age      -0.0448  **

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant



12.7.3.2. BMI

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -2.9847   13.9489   11.9695   12.5612   16.7695

tau^2 (estimated amount of residual heterogeneity):     0.0868 (SE = 0.0866)
tau (square root of estimated tau^2 value):             0.2947
I^2 (residual heterogeneity / unaccounted variability): 57.84%
H^2 (unaccounted variability / sampling variability):   2.37
R^2 (amount of heterogeneity accounted for):            27.72%

Test for Residual Heterogeneity:
QE(df = 7) = 12.7253, p-val = 0.0791

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 2.7310, p-val = 0.1424

Model Results:
```

```
         estimate      se     tval  df    pval    ci.lb
intrcpt   -3.4303  2.4532  -1.3983   7  0.2047  -9.2312
bmi        0.1920  0.1162   1.6526   7  0.1424  -0.0827
          ci.ub
intrcpt  2.3706
bmi      0.4668

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

12.7.3.3. Duration

```
Mixed-Effects Model (k = 8; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -4.6365   14.5351   15.2729   15.5113   21.2729

tau^2 (estimated amount of residual heterogeneity):     0.1642 (SE = 0.1421)
tau (square root of estimated tau^2 value):             0.4052
I^2 (residual heterogeneity / unaccounted variability): 68.62%
H^2 (unaccounted variability / sampling variability):   3.19
R^2 (amount of heterogeneity accounted for):            60.58%

Test for Residual Heterogeneity:
QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

          estimate      se     tval  df    pval    ci.lb
intrcpt     0.8663  0.2241   3.8659   6  0.0083   0.3180
duration   -0.0115  0.0043  -2.6689   6  0.0371  -0.0220
            ci.ub
intrcpt    1.4146   **
duration  -0.0010    *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

12.7.3.4. Dose

```
Mixed-Effects Model (k = 8; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -4.6365   14.5351   15.2729   15.5113   21.2729

tau^2 (estimated amount of residual heterogeneity):     0.1642 (SE = 0.1421)
tau (square root of estimated tau^2 value):             0.4052
I^2 (residual heterogeneity / unaccounted variability): 68.62%
H^2 (unaccounted variability / sampling variability):   3.19
R^2 (amount of heterogeneity accounted for):            60.58%

Test for Residual Heterogeneity:
QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.9372  0.2421   3.8715   6  0.0083   0.3448
dose      -0.0017  0.0006  -2.6689   6  0.0371  -0.0033
           ci.ub
intrcpt   1.5295  **
dose     -0.0001   *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

12.7.3.5. MET

> **Note**
>
> Only modelled when the MET column is present; the current $r = 0.5$ export does not include it, so this meta-regression is skipped.

12.7.3.6. IFN

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
```

```
 -2.0723   11.1835   10.1447    9.9824   18.1447

tau^2 (estimated amount of residual heterogeneity):     0.0848 (SE = 0.0943)
tau (square root of estimated tau^2 value):             0.2913
I^2 (residual heterogeneity / unaccounted variability): 58.34%
H^2 (unaccounted variability / sampling variability):   2.40
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 10.9775, p-val = 0.0518

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.4201, p-val = 0.5455

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.8400  0.4512   1.8617   5  0.1217  -0.3198
IFN       -0.0933  0.1440  -0.6481   5  0.5455  -0.4634
           ci.ub
intrcpt  1.9998
IFN      0.2768

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

12.7.3.7. IL_4

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -1.4940   10.0268    8.9880    8.8258   16.9880

tau^2 (estimated amount of residual heterogeneity):     0.0657 (SE = 0.0832)
tau (square root of estimated tau^2 value):             0.2563
I^2 (residual heterogeneity / unaccounted variability): 52.41%
H^2 (unaccounted variability / sampling variability):   2.10
R^2 (amount of heterogeneity accounted for):            8.99%

Test for Residual Heterogeneity:
QE(df = 5) = 8.9170, p-val = 0.1124

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 1.4107, p-val = 0.2883
```

```
Model Results:

         estimate      se    tval  df    pval    ci.lb
intrcpt    0.4280  0.1697  2.5220   5  0.0530  -0.0083
IL_4       0.2553  0.2149  1.1877   5  0.2883  -0.2972
          ci.ub
intrcpt  0.8643  .
IL_4     0.8077

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note

Not significant

12.7.3.8. IL_6

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -0.3049    8.5892    6.6098    7.2014   11.4098

tau^2 (estimated amount of residual heterogeneity):     0.0027 (SE = 0.0374)
tau (square root of estimated tau^2 value):             0.0522
I^2 (residual heterogeneity / unaccounted variability): 4.44%
H^2 (unaccounted variability / sampling variability):   1.05
R^2 (amount of heterogeneity accounted for):            97.74%

Test for Residual Heterogeneity:
QE(df = 7) = 8.6202, p-val = 0.2811

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 7.8742, p-val = 0.0263

Model Results:

         estimate      se    tval  df    pval   ci.lb
intrcpt    0.3614  0.1208  2.9922   7  0.0202  0.0758
IL_6       0.1441  0.0514  2.8061   7  0.0263  0.0227
          ci.ub
intrcpt  0.6469  *
IL_6     0.2655  *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **!** Important
>
> Significant

12.7.3.9. IL_10

```
Mixed-Effects Model (k = 9; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -4.3135   16.6064   14.6270   15.2186   19.4270

tau^2 (estimated amount of residual heterogeneity):     0.1340 (SE = 0.1107)
tau (square root of estimated tau^2 value):             0.3661
I^2 (residual heterogeneity / unaccounted variability): 68.68%
H^2 (unaccounted variability / sampling variability):   3.19
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 7) = 17.5453, p-val = 0.0142

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 7) = 0.3559, p-val = 0.5696

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.7042  0.1924   3.6597   7  0.0081   0.2492
IL_10     -0.0754  0.1263  -0.5966   7  0.5696  -0.3741
          ci.ub
intrcpt  1.1592  **
IL_10    0.2233

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **i** Note
>
> Not significant

12.7.3.10. IL_2

```
Mixed-Effects Model (k = 7; tau^2 estimator: HE)

  logLik  deviance       AIC       BIC      AICc
 -2.1646   11.3680   10.3292   10.1669   18.3292

tau^2 (estimated amount of residual heterogeneity):     0.0869 (SE = 0.0971)
```

```
tau (square root of estimated tau^2 value):             0.2947
I^2 (residual heterogeneity / unaccounted variability): 58.04%
H^2 (unaccounted variability / sampling variability):   2.38
R^2 (amount of heterogeneity accounted for):            0.00%

Test for Residual Heterogeneity:
QE(df = 5) = 11.4142, p-val = 0.0438

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 5) = 0.2537, p-val = 0.6359

Model Results:

         estimate      se     tval  df    pval    ci.lb
intrcpt    0.6698  0.2567   2.6093   5  0.0477   0.0099
IL_2      -0.0585  0.1162  -0.5037   5  0.6359  -0.3573
          ci.ub
intrcpt  1.3296  *
IL_2     0.2402

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **Note**
>
> Not significant

12.8. Multivariate Models

```
Mixed-Effects Model (k = 8; tau^2 estimator: HE)

tau^2 (estimated amount of residual heterogeneity):     0.1642 (SE = 0.1421)
tau (square root of estimated tau^2 value):             0.4052
I^2 (residual heterogeneity / unaccounted variability): 68.62%
H^2 (unaccounted variability / sampling variability):   3.19
R^2 (amount of heterogeneity accounted for):            60.58%

Test for Residual Heterogeneity:
QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):
F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

                  estimate      se     tval  df    pval
intrcpt             0.6369  0.1810   3.5193   6  0.0125
intensityVigorous  -1.4912  0.5587  -2.6689   6  0.0371
```

```
                     ci.lb    ci.ub
intrcpt             0.1941   1.0797  *
intensityVigorous  -2.8583  -0.1240  *

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

! Important

Model significant but no independent predictors

References

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